

STEPPERONLINE[®]

Modbus Series

Bus type driver product Features Manual

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Preface

Thank you for using our Modbus series bus type driver .

Before using this product, please be sure to read this manual carefully to understand the necessary safety information , precautions, and operating methods.

Incorrect operation may lead to extremely serious consequences.

statement

The design and manufacture of this product are not capable of protecting personal safety from threats from mechanical systems. Please consider safety protection measures during the design and manufacture of mechanical systems to prevent accidents caused by improper operation or product abnormalities.

Due to product improvements, the manual contents may be changed without prior notice.

Our company will not be responsible for any modification of the product by the user.

When reading, please pay attention to the following signs in the manual:



NOTE: Draws your attention to important points in the text.



CAUTION: Indicates that incorrect operation may result in personal injury and equipment damage.

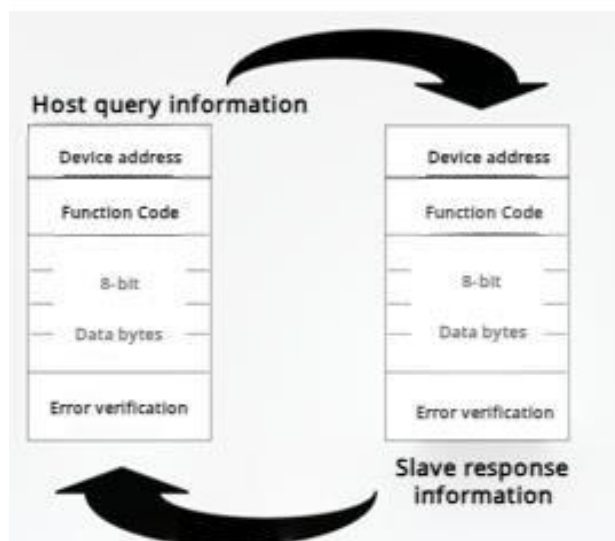
The contents described in this user manual are only applicable to the following models:

Model
ESS17-RSx1 Series
ESS23-RSx1 Series
DM556PR Series
DM860PR Series

1 Modbus/RTU Protocol Definition

Modbus is a serial communication protocol that was published by Modicon (now Schneider Electric) in 1979 for communication with programmable logic controllers (PLCs). Modbus has become the de facto industry standard for communication protocols and is now a common way to connect industrial electronic devices. The main reasons why Modbus is more widely used than other communication protocols are: it is publicly published and has no copyright requirements; it is easy to deploy and maintain; there are not many restrictions on suppliers to modify local bits or bytes;

The Modbus protocol is a master/slave protocol. There is one node that is the master node, and the other nodes that use the Modbus protocol to participate in the communication are slave nodes. Each slave device has a unique address. In the serial and MB+ network, only the node designated as the master node can initiate a command.



2 Modbus/RTU Hardware Interface

2.1 Wiring method

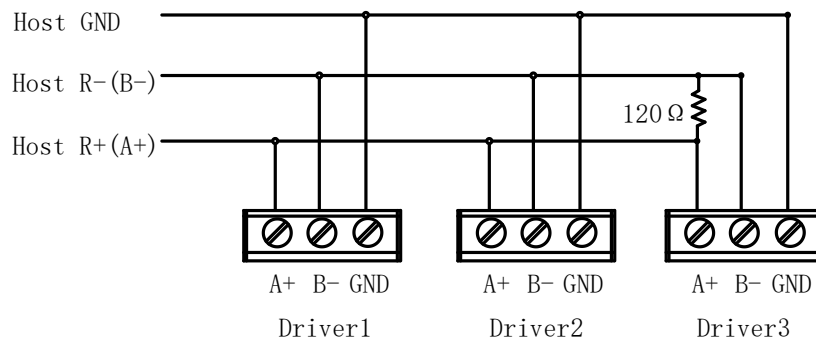
The Modbus series products communicate as Modbus devices through the serial RS485 physical layer, allowing multiple devices to be connected on the same network.

RS485 uses differential signals to complete communication transmission. Ordinary twisted pair cables can be used in low-speed, short-distance, and interference-free situations. In high-speed, long-line transmission, RS485 dedicated cables with impedance matching (generally 120Ω) must be used. In environments with harsh interference, armored twisted pair shielded cables should also be used.

When using the RS485 interface, for a specific transmission line, the maximum cable length allowed for data signal transmission from the RS485 interface to the load is inversely proportional to the baud rate of the signal transmission. This length data is mainly affected by factors such as signal distortion and noise. During the transmission process, the signal can be amplified by adding relays, and up to eight relays can be added.

RS485 has two interface types: two-wire and four-wire. This series of products all use two-wire interface.

The two-wire half-duplex connection method is shown in the figure:



When working in high-speed, long-line, and multi-node conditions, a 120-ohm impedance matching resistor must be connected to the differential communication line of the last slave in the network. The Modbus series driver products all have integrated 120-ohm resistors, and the validity can be selected by the dial code .

2.2 Twenty two Node Settings

Each slave node in the Modbus network needs to be set to a different address so that the master node can complete device addressing.

Modbus series products generally have 4-bit (up to 15 node addresses) or 5-bit (up to 31 node addresses) address dip switches specifically used for node settings.

At the same time, you can also customize the slave node configuration through Modbus: 0013h register. Note that 0013h is only effective when all address dip switches are OFF.

※ Modbus: 0013h register

Register Address	name	illustrate
0x0013	Custom drive node number	0 ~ 255 : custom slave address; This register takes effect when all the drive address DIP switches are OFF;

2.3 Twenty three Communication parameter settings

Modbus network communication parameters mainly include: baud rate, data bit, stop bit, and check bit.

Modbus network can set different communication parameters for communication. All nodes in the same network must be set to the same baud rate, data bit, stop bit, and check bit.

Modbus series products generally use the baud rate DIP switch to set the baud rate of the node, and the default configuration is 8 data bits, 1 stop bit, and no parity bit.

For products without baud rate dial, you can also customize the slave node communication parameters through Modbus: 0014h~0015h registers. After modifying the customized communication baud rate, it will take effect after powering on again.

※ Modbus: 0014h~0015h register

Register Address	name	illustrate
0x0014	Customize communication baud rate	0:115200 1 : 38400 2 : 19200 3 : 9600 Note: After modification, power must be turned on again to take effect;
0x0015	Serial port data format	0 : 8 -bit data, no parity, 1 stop bit; 1 : 8 -bit data, no parity, 2 stop bits; 2 : 8 -bit data, even parity, 1 stop bit; 3 : 8 -bit data, odd parity, 1 stop bit; Note: After modification, power must be turned on again to take effect;

3 Modbus/RTU protocol

3.1 Message Format

Modbus/RTU message is a data frame, and its format is shown in the following table:

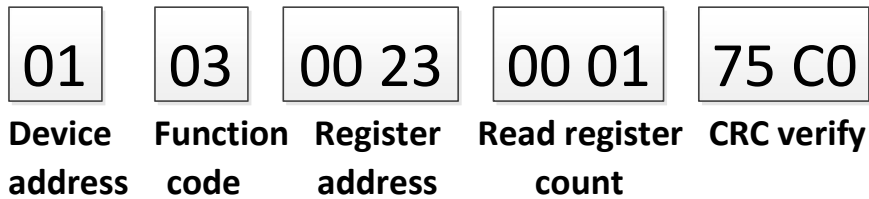
Device Address	Function code	Data Format	CRC Check
8 bits	8 bits	N * 8 bits	16 bit

For detailed message formats of each function code, see Section 3.2.

3.2 Function code introduction

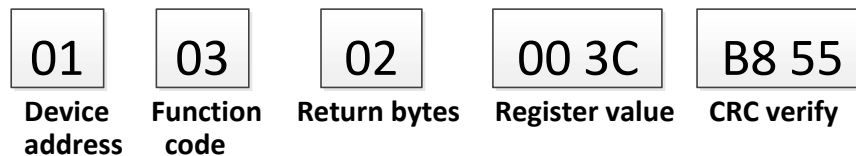
3.2.1 Read Holding Register Command 03

➤ Master->Slave Data



The host sends a command to query the positioning motion speed register to the slave.

Slave->Master data:



The slave returns the maximum speed register value of 60.

➤ read holding register commands are as follows :

Query the current position (0x00 0A~0x000B) and current speed (0x00 0C).

Master->Slave data: 01 03 00 0A 00 0 3 25 C9

Slave->Host data: 01 03 0 6 00 0 0 13 88 00 0 0 A5 D B (current position 5000, current speed 0)



Note: The maximum number of queries must not exceed 16 registers.

3.2.2 Write Single Register Command 06

➤ Master->Slave Data



address code address

The host writes a value of 60 to the positioning motion speed register of the slave .
After receiving the command, the slave returns the same command for confirmation.

Slave->Master data:

01	06	00 23	00 3C	78 11
Device address	Function code	Register address	Write data	CRC verify

➤ Other examples of writing a single register command are as follows :

Set the acceleration time register to 500ms:

Master->Slave data: 01 06 00 21 01 F4 D9 D7

Slave->Host data: 01 06 00 21 01 F4 D9 D7

3.2.3 Write multiple registers command 10

01	10	00 24	00 02	04	00 00	13 88	B9 56
address	Function code	Origin address	Number of writes	Number of bytes	Write content	Write content	CRC check

An example of a write multiple registers command is as follows:

The host writes two registers to the slave, setting the pulse number high register and the pulse number low register respectively.

Master->Slave data: 01 10 00 24 00 02 04 00 00 13 88 FD 12

Slave->Host data: 01 10 00 24 00 02 01 C3

3.3 CRC Checksum Algorithm

The CRC area is 2 bytes, containing a 16-bit binary data. The sending device calculates the CRC value and attaches the calculated value to the information. When receiving the information, the receiving device recalculates the CRC value and compares the calculated value with the actual value received in the CRC area. If the two are different, an error occurs.

At the beginning of CRC, all 16 bits of the register are set to "1", and then the data of two adjacent 8-bit bytes are placed in the current register. Only the 8-bit data of each character is used to generate CRC, and the start bit, stop bit and parity bit are not added to the CRC.

During CRC generation, every 8 bits of data are XORed with the value in the register, the result is shifted one position to the right (towards the LSB), and the MSB is filled with "0". The LSB is tested. If the LSB is "1", it is XORed with the preset fixed value . If the LSB is "0", no XOR operation is performed.

Repeat the above process until it shifts 8 times. After the 8th shift is completed, the next 8 bits of data are XORed with the current value of the register . After all information is processed, the final value in the register is the CRC value.

※ CRC check routine

```

Uint16 CRC ( Uint16 *tempMsg, Uint16 tempLength)
{
    //Here CRC check is a query table
    Uint16 i=0;
    Uint16 tempCrcHigh = 0xFF;
    Uint16 tempCrcLow = 0xFF;
    Uint16 tempIndex=0;
    for (i = 0; i < tempLength; i++)
    {
        tempIndex = tempCrcLow ^ (tempMsg[i]); tempCrcLow =
        tempCrcHigh ^ (CRCVALUE[tempIndex] >>8); tempCrcHigh
        =CRCVALUE[tempIndex]&0xff;
    }
    return (tempCrcHigh | (tempCrcLow<<8));
}

const Uint16 CRCVALUE[] =
{
    0x0000,0xC1C0,0x81C1,0x4001,0x01C3,0xC003,0x8002,0x41C2,0x01C6,0xC
    006,0x8007,0x41C7,
    0x0005,0xC1C5,0x81C4,0x4004,0x01CC,0xC00C,0x800D,0x41CD,0x000F,0xC
    1CF,0x81CE,0x400E,
    0x000A,0xC1CA,0x81CB,0x400B,0x01C9,0xC009,0x8008,0x41C8,0x01D8,0xC
    018,0x8019,0x41D9,
    0x001B,0xC1DB,0x81DA,0x401A,0x001E,0xC1DE,0x81DF,0x401F,0x01DD,0xC
    01D,0x801C,0x41DC,
    0x0014,0xC1D4,0x81D5,0x4015,0x01D7,0xC017,0x8016,0x41D6,0x01D2,0xC
    012,0x8013,0x41D3,
    0x0011,0xC1D1,0x81D0,0x4010,0x01F0,0xC030,0x8031,0x41F1,0x0033,0xC
    1F3,0x81F2,0x4032,
    0x0036,0xC1F6,0x81F7,0x4037,0x01F5,0xC035,0x8034,0x41F4,0x003C,0xC

```

```

1FC,0x81FD,0x403D,
  0x01FF,0xC03F,0x803E,0x41FE,0x01FA,0xC03A,0x803B,0x41FB,0x0039,0xC
1F9,0x81F8,0x4038,
  0x0028,0xC1E8,0x81E9,0x4029,0x01EB,0xC02B,0x802A,0x41EA,0x01EE,0xC
02E,0x802F,0x41EF,
  0x002D,0xC1ED,0x81EC,0x402C,0x01E4,0xC024,0x8025,0x41E5,0x0027,0xC
1E7,0x81E6,0x4026,
  0x0022,0xC1E2,0x81E3,0x4023,0x01E1,0xC021,0x8020,0x41E0,0x01A0,0xC
060,0x8061,0x41A1,
  0x0063,0xC1A3,0x81A2,0x4062,0x0066,0xC1A6,0x81A7,0x4067,0x01A5,0xC
065,0x8064,0x41A4,
  0x006C,0xC1AC,0x81AD,0x406D,0x01AF,0xC06F,0x806E,0x41AE,0x01AA,0xC
06A,0x806B,0x41AB,
  0x0069,0xC1A9,0x81A8,0x4068,0x0078,0xC1B8,0x81B9,0x4079,0x01BB,0xC
07B,0x807A,0x41BA,
  0x01BE,0xC07E,0x807F,0x41BF,0x007D,0xC1BD,0x81BC,0x407C,0x01B4,0xC
074,0x8075,0x41B5,
  0x0077,0xC1B7,0x81B6,0x4076,0x0072,0xC1B2,0x81B3,0x4073,0x01B1,0xC
071,0x8070,0x41B0,
  0x0050,0xC190,0x8191,0x4051,0x0193,0xC053,0x8052,0x4192,0x0196,0xC
056,0x8057,0x4197,
  0x0055,0xC195,0x8194,0x4054,0x019C,0xC05C,0x805D,0x419D,0x005F,0xC
19F,0x819E,0x405E,
  0x005A,0xC19A,0x819B,0x405B,0x0199,0xC059,0x8058,0x4198,0x0188,0xC
048,0x8049,0x4189,
  0x004B,0xC18B,0x818A,0x404A,0x004E,0xC18E,0x818F,0x404F,0x018D,0xC
04D,0x804C,0x418C,
  0x0044,0xC184,0x8185,0x4045,0x0187,0xC047,0x8046,0x4186,0x0182,0xC
042,0x8043,0x4183,
  0x0041,0xC181,0x8180,0x4040 };

```

3.4 Communication error code

➤ CRC check error

If an error occurs during data transmission, and the CRC check value calculated by the slave device for a frame of data is not **85 C0**, the slave device discards the frame of data and does not return any data.

Master->Slave data: 01 03 00 20 00 01 85 **C1**

Slave->Master data: 01 **83 01 80 F0**

➤ Instruction code error

If the function code requested by the host is not 03 , 06 or 10 , the device returns exception code 02 .

Master->Slave data: 01 02 00 00 00 04 79 C9

Slave->Master data: 01 82 02 61 C1

➤ Invalid data address

If the data address requested by the host is illegal, the device returns exception code 03.

Host->Slave data: 01 03 00 1 C 00 01 4 5 C C

Slave->Master data: 01 83 03 01 31

Register address 0x001 C is empty and the device returns exception code 03.

➤ Out of address range

Host- > Slave data: 01 06 FF 00 0B 580B

Slave- > Master data: 01 86 04 43 A3

Register address 0 Xff 00 is beyond the register address definition range. The device returns exception code 04.

➤ Read address overflow

If the data requested by the host exceeds the range of one read, the device returns the exception code 05. For details of the exception code 05, please refer to the table ✕

MODBUS exception code .

Master->Slave data: 01 03 00 20 00 20 45 D8

Slave->Master data: 01 83 05 81 33

Reading 32 data at a time exceeds the range and returns exception code 05

➤ Illegal read and write errors

Function code read and write attributes are divided into three types: read-only, writeonly, and read-write. For operations that do not conform to the function code attributes, exception code 06 will be reported.

Master->Slave data: 01 03 00 27 00 01 34 01

Slave->Host data: 01 83 06 C1 32

Function code 0x27 is a write-only function code, and its read operation reports exception code 06.

➤ Error writing content

The content of the written function code exceeds its specified range.

Master->Slave data: 01 06 00 2 7 FF FF 38 71

Slave->Master data: 01 86 07 03 A2

If the write function code is out of range, the exception code 07 is returned.

※ MODBUS abnormal code

Code	name	meaning
01	CRC check error	CRC check error.
02	Instruction code error	The slave receives a function code other than 03 and 06.
03	Function code address error	The received data address is not allowed by the slave.
04	Exceeded function code address	The received data address exceeds the function code range.
05	Read function code number overflow	A maximum of 16 function codes can be read at one time.
06	Illegal function code reading and writing error	Function code read and write attributes are divided into three types: read-only, write-only, and read-write. Operations that do not conform to the function code attributes will be abnormally incorrect.
07	Function code writing error	Data outside the specified range is written to the function code.

4 Motion control function

Modbus series bus drivers can realize 5 single-axis motion control functions, including:

- Positioning movement
- Speed Movement
- Zero return function
- Multi-stage position control
- Multi-speed control

there are two control modes for the three functions of positioning motion, speed control and zero return:

- 1) Control is achieved through Modbus: 0027h motion control command register;
- 2) Control is achieved through external input signals;

The multi-stage position control and multi-stage speed control functions can only be controlled by external input signals.

The driver integrates several input signals, and its functions can be configured through Modbus: 0041h~0047h registers.

This section describes in detail how the above five functions work.

※ Modbus: 0027h motion control command register

Register Address	name	illustrate
0x0027	Motion Control Command Register	Bit0: Position mode start command bit; 0: invalid; 1: Valid; Bit1: Speed mode start command bit; 0: invalid; 1: Valid; Bit2: Position mode positioning method; 0: relative positioning; 1: absolute positioning; Bit3: Sports mode switching method; 0: Ignore motion commands; 1: interrupt the current motion and execute it immediately; Bit4: Return to origin start command bit; 0: invalid; 1: Valid; Bit8: stop command bit; 0: invalid; 1: Valid;

		Bit9: Emergency stop command bit; 0; invalid; 1: Valid;
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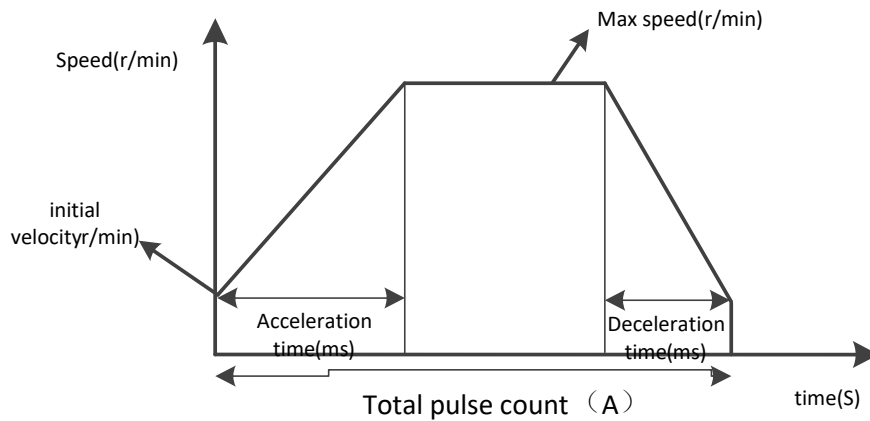
※ **Modbus: 0041h~0047h input terminal function configuration register**

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	0 : undefined; 1 : origin signal; 2 : Positive limit signal; 3 : Anti-limit signal; 4 : Motor MF signal; 5 : Stop signal; 6 : Emergency stop signal; 7 : Position mode movement; 8: Speed mode movement; 9 : JOG+ point motion; 10 : JOG -point movement; 11 : Return to origin enable signal; 12 : PT trigger signal ; 13: PV trigger signal; 14 : PIN0; 15 : PIN1; 16 : PIN2; 17 : PIN3;

4.1 Positioning movement

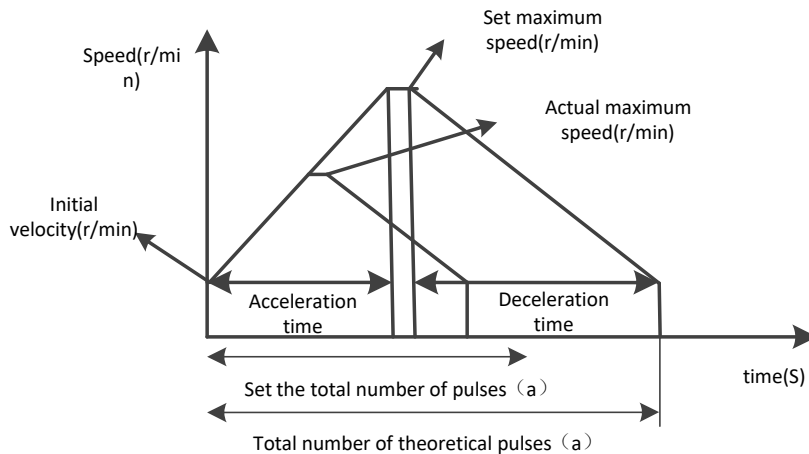
The positioning motion mode is implemented using a trapezoidal acceleration and deceleration curve. Users can achieve precise position control by setting several parameters, including acceleration time, deceleration time, running speed, and target total pulse number.

Note: The driver determines the rotation direction of the motor by judging the positive or negative of the total number of target pulses . When the total number of pulses is positive, the motor is defined as forward rotation, and when the total number of pulses is negative, the motor is defined as reverse rotation. The trapezoidal acceleration and deceleration curve is shown in the figure below.



Position mode acceleration and deceleration curve

When the total number of pulses set by the user is small, the motor may need to decelerate before accelerating to the maximum speed (that is, the motor does not accelerate to the maximum speed set by the user during actual operation). The speed curve is shown in the figure below. The solid line in the figure shows the actual operation curve of the motor, and the dotted line is the curve required to accelerate to the set maximum speed. The theoretical total number of pulses is the minimum total number of pulses calculated according to the user-set parameters (maximum speed, acceleration time, deceleration time). When the total number of pulses set by the user is less than the theoretical total number of pulses, the motor will run according to the solid line in the figure below.



Position mode acceleration and deceleration curve (not accelerated to the set maximum speed)

The relevant motion control parameter registers are:

Register Address	name	illustrate
0x0021	Acceleration time	Acceleration time , unit: ms ;
0x0022	Deceleration time	Deceleration time , unit: ms ;
0x0023	Positioning movement speed	Movement speed during positioning movement, unit: r/min;

0x0024	Target pulse value high	Target pulse value, unit: pulse; When the value is positive, the motor moves in the positive direction, and when the value is negative, the motor moves in the negative direction .
0x0025	Target pulse value low	

※ **Control is completed through Modbus: 0027h motion control command register**

Control is completed through 0027h : Bit0, Bit2, Bit3 .

0027h	Order	illustrate
Bit0	Position mode start command bit 0: invalid; 1: Start;	When this bit is written as 1, a start command is generated and the drive starts moving according to the given motion parameters;
Bit2	Position mode positioning method 0: relative positioning; 1: absolute positioning;	This bit indicates whether the given target position is a relative position or an absolute position;
Bit3	The current exercise is not completed, how to switch the exercise mode 0: Ignore the new given motion command; 1: interrupt the current motion and execute it immediately;	The current drive is in motion. This bit is used to set to ignore the new given motion command or interrupt the current motion and immediately execute the new given motion command. This bit is only valid in positioning mode ;

Example:

move forward relative to the specified parameters for 1000 pulses (starting speed 10r/min, acceleration time 100ms, deceleration time 100ms, maximum speed 60 /min) .

Step 1: Write parameters: acceleration time 100ms , deceleration time 100ms , running speed 60 /min , set total pulse number 1000.

Host->Slave: 01 10 00 2 1 00 0 5 0 A 00 64 00 64 00 3C 00 00 00 03 E8 98 EA

Slave->Host: 01 10 00 2 1 00 0 5 50 00

Step 2 : Relative position mode start command

Host -> Slave: 01 06 00 27 00 01 F8 01

Slave->Host: 01 06 00 27 00 01 F8 01

※ **Control is completed through external input terminals**

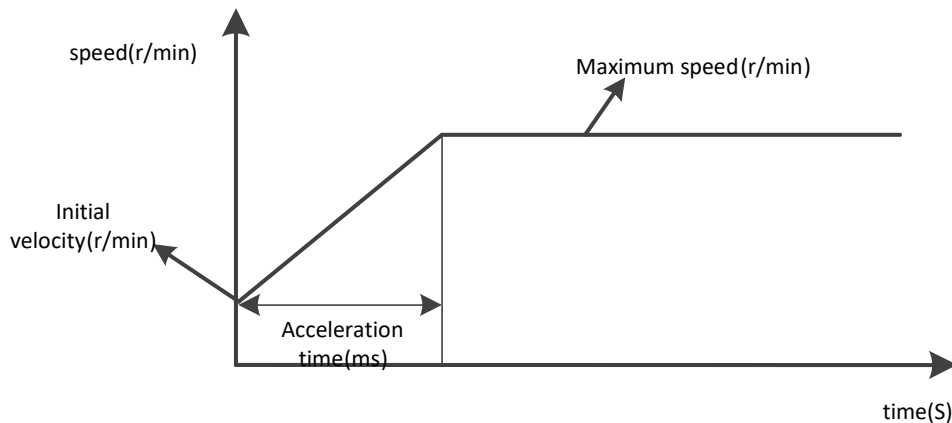
Any input terminal of the driver and set its function to **7: position motion mode through the corresponding function setting register** . When a valid level signal is given to this terminal, the driver completes the action according to the given motion parameters.

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	7: Position mode movement;

4.2 Speed Movement

The speed motion mode is shown in the figure below. Unlike the positioning motion mode, the speed mode only requires the setting of three parameters: running speed, acceleration time, and deceleration time. After the motor accelerates to the maximum speed according to these three parameter settings, it runs at a constant speed at the maximum speed.

Note: The positive and negative values of the running speed register are absolutely related to the forward and reverse rotation of the motor. When the maximum speed register is positive, the motor is defined as forward rotation. When the maximum speed register is negative, the motor is defined as reverse rotation.



Speed mode acceleration curve

The relevant motion control parameter registers are:

Register Address	name	illustrate
0x001D	Jog mode operation speed	Jog mode running speed, unit: r/min; When the value is positive, the motor moves in the positive direction, and when the value is negative, the motor moves in the negative direction .
0x001E	Jog mode acceleration time	Jog mode acceleration time, in ms;
0x001F	Jog mode deceleration time	Jog mode deceleration time, unit: ms;

※ **Control is completed through Modbus: 0027h motion control command register**

Through 0027h :Bit1 .

0027h	Order	illustrate
Bit1	Speed mode start command bit 0; invalid; 1: Start;	When this bit is written as 1, a start command is generated and the drive starts moving according to the given motion parameters;

Example:

Accelerate in the reverse direction according to the parameters (operating speed 60r/min, acceleration/ deceleration time 100ms) to 500r/min, and then run at a constant speed.

Before starting this example, be sure to set the driver device address to 1, that is, turn the DIP switches SW5-SW2 to OFF and SW1 to ON.

Step 1: Set the running speed to 60 r/min and the acceleration/deceleration time to 100ms.

Host->Slave: 01 10 00 1D 00 03 06 00 3C 00 64 00 64 66 DE

Slave->Host: 01 10 00 1D 00 03 10 0E

Step 2 : Speed Mode Start Command

Host -> Slave: 01 06 00 27 00 02 B8 00

Slave->Host: 01 06 00 27 00 02 B8 00

※ **Control is completed through external input terminals**

Any input terminal of the driver , and set its function to **8: speed mode motion, 9: JOG+ point motion, 10: JOG- point motion through the corresponding function setting register** . When the terminal is given an effective level signal, the driver completes the action according to the given motion parameters.

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	8: Speed mode movement; 9: JOG+point motion; 10: JOG-point movement;

The difference between setting it to 8, 9, and 10 is:

When set to mode 8, the forward and reverse directions of the motor are determined by the positive and negative values of the 001Dh register;

When set to mode 9, the drive always moves in the positive direction regardless of the positive or negative value of the 001Dh register parameter;

When set to mode 10, the drive always moves in the negative direction regardless of whether the 001Dh register parameter value is positive or negative.

4.3 Return to zero function

The selection of the homing mode is set by the Modbus: 0031h register. When the limit signal or origin signal is needed in the homing process, you need to select the limit signal or origin signal function of the input terminal according to the mode before using the homing function. At the same time, the homing function can be triggered by external I/O or Modbus: 0027h register. If an external I/O trigger is used, a certain input terminal function needs to be enabled as the "homing enable" function.

Before triggering the homing action, the relevant motion parameter registers need to be set, including the homing mode, homing speed, homing query speed, homing acceleration and deceleration time. If the origin needs to be offset, the origin offset value needs to be set.

The relevant motion control parameter registers are:

Register Address	name	illustrate
0x0031	Return to origin mode	1~14, 17~30, 33~35 modes;
0x0032	Return to origin speed	Running speed when querying the origin position;
0x0033	Return to origin query speed	The return speed after querying the origin;
0x0034	Acceleration and deceleration time when returning to origin	The acceleration and deceleration time when querying the origin position;
0x0035	Origin offset value high	Origin offset value: after finding the origin sensor, it moves to the correct position according to the value set in this register.
0x0036	Origin offset value low	

The driver supports 1~14, 17~30, 33, 34, and 35 return to origin methods. Among them, 1~14, 33, and 34 return to zero modes require the use of a closed-loop stepper motor with a Z signal. For detailed description, see Appendix 1.

※ **Control is completed through Modbus: 0027h motion control command register** through 0027h :Bit4.

0027h	Order	illustrate
Bit4	Return to origin start command position	When this bit is written as 1, a start command is generated and
	0: invalid; 1: Start;	the drive starts moving according to the given motion parameters;

Example:

Complete the homing action according to the parameters (homing mode 24 , homing speed 60r/min, homing query speed 30r/min , homing acceleration/ deceleration time 100ms, origin offset 0) .

Assume that before returning to the origin, the functions of the relevant input terminals have been set to the origin, positive limit, and negative limit;

Step 1: Write parameters: homing mode 24 , homing speed 60r/min, homing query speed 30r/min , homing acceleration/ deceleration time 100ms, origin offset 0.

Host->Slave: 01 10 00 31 00 0 6 0 C 00 18 00 3C 00 1E 00 64 00 0 0 00 00 2F 3B

Slave->Host: 01 10 00 31 00 06 11 C4

Step 2 : Return to origin and start command

Host -> Slave: 01 06 00 27 00 10 38 0D

Slave->Host: 01 06 00 27 00 10 38 0D

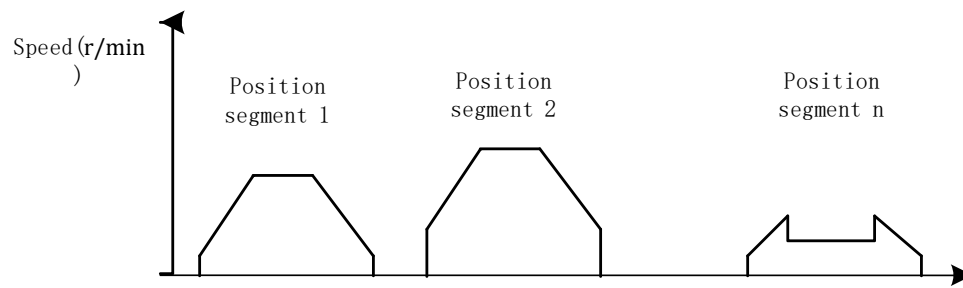
※ **Control is completed through external input terminals**

Any input terminal of the driver , and set its function to **11 through the corresponding function setting register: homing enable signal** . When a valid level signal is given to this terminal, the driver completes the action according to the given motion parameters.

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	1 1 : Return to origin enable signal;

4.4 Multi-stage position control

Multi-segment position control is a working mode that combines multiple position segments in a certain order, triggers motion through external IO signals, and completes a series of position segment actions . This function can also be regarded as a multi-segment combination of the positioning motion mode described in **4.1 Positioning motion** . The difference is that the user can store the motion parameters of several position segments such as acceleration and deceleration time, target pulse number , acceleration and deceleration in eeprom or local flash in advance . When these position segments need to be enabled, only a trigger signal is needed to complete the work. The working process is described as shown in the figure below.



Multi-position working mode

Currently, a maximum of 16 position segments are supported. Here are the relevant motion parameters describing the first position segment:

Register Address	name	illustrate
0x00 6 0	The first segment motion pulse command high	The first segment is the motion pulse instruction, the unit is pulse; When the value is positive, the motor moves in the positive direction, and when the value is negative, the motor moves in the negative direction .
0x0061	The first segment motion pulse command low	
0x00 62	Positioning speed of the first stage	the first position positioning, unit: r/min;
0x0063	1st stage positioning motion acceleration	The first segment position positioning motion acceleration, in ms;
0x0064	1st stage position positioning motion deceleration	The first stage of the positioning motion deceleration, unit: ms;

For the addresses of motion parameters related to the remaining position segments, **please refer to the Appendix: Modbus Register Parameter Table .**

Multi-stage position control uses external input terminals to select position segments and start motion, so the input terminal functions need to be set.

Among them, **1 2 : The PT trigger signal** function is the trigger signal, and **PIN0 ~PIN3** are the position segment selection signals .

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	1 2 : PT trigger signal ; 1 4 : PIN0 ; 1 5 : PIN1 ; 1 6 : PIN2 ; 1 7 : PIN3 ;

The position segment selection is completed according to the binary number composed of PIN0~PIN3. The corresponding relationship is as follows:

※ Input terminal selection position segment

PIN3	PIN2	PIN1	PIN0	Location segment
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
0	0	1	1	4
0	1	0	0	5

0	1	0	1	6
0	1	1	0	7
0	1	1	1	8
1	0	0	0	9
1	0	0	1	10
1	0	1	0	11
1	0	1	1	12
1	1	0	0	13
1	1	0	1	14
1	1	1	0	15
1	1	1	1	16


Notice:

- using the PIN terminal for segment selection , it must remain valid 5ms before and after the " PT trigger signal " .
- In multi-segment mode, the position segment is controlled by the relative/absolute position register 0x00 50 to determine whether it is relative position movement or absolute position movement.
- Some driver peripherals only support 4 input signals from X0 to X3, so the position segment selection supports a maximum of 8 segments;

4.5 Multi- speed modes

The multi- speed mode function is a working method that stores multiple speed segments in advance, triggers movement through external IO signals, and completes a series of actions at different speeds .

Currently, a maximum of 16 speed segments are supported. Here are the relevant motion parameters describing the first speed segment:

Register Address	name	illustrate
0x00 C0	1st speed segment running speed	running speed of the first speed segment is in r/min ; When the value is positive, the motor moves in the positive direction, and when the value is negative, the motor moves in the negative direction .
0x00C1	1st speed segment acceleration time	The first speed segment motion acceleration, unit: ms;
0x00C2	1st speed stage deceleration time	The first speed segment is the deceleration of the movement, in ms;

For the addresses of motion parameters related to other speed segments, **please refer to the Appendix: Modbus Register Parameter Table** .

Multi-speed control uses external input terminals to select speed segments and start motion, so the input terminal function needs to be set.

Among them, **1 3 : PV trigger signal** function is trigger signal, **PIN0 ~ PIN3** are speed segment selection signals .

Register Address	name	illustrate
0x004 1~0x0047	Input terminal function selection	1 3 : PV trigger signal ; 1 4 : PIN0 ; 1 5 : PIN1 ; 1 6 : PIN2 ; 1 7 : PIN3 ;

Speed segment selection is completed according to the binary number composed of PIN0~PIN3. The corresponding relationship is shown in the following table:

※ **Input terminal selects speed segment**

PIN3	PIN2	PIN1	PIN0	Speed range
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
0	0	1	1	4
0	1	0	0	5
0	1	0	1	6
0	1	1	0	7
0	1	1	1	8
1	0	0	0	9
1	0	0	1	10
1	0	1	0	11
1	0	1	1	12
1	1	0	0	13
1	1	0	1	14
1	1	1	0	15
1	1	1	1	16



Notice:

- Some driver peripherals only support 4 input signals from X0 to X3, so the position segment selection supports a maximum of 8 segments;

4.6 Stop and emergency stop commands

When the drive needs to stop the current moving state, it can be completed through Modbus: 0027h motion control command register.

through 0027 h: Bit8 and Bit9 .

0027h	Order	illustrate
Bit8	Stop command bit 0: invalid; 1: Start;	When this bit is written as 1, a start command is generated and the drive starts moving according to the given motion parameters;
Bit9	Emergency stop command position 0: relative positioning; 1: absolute positioning;	This bit indicates whether the given target position is a relative position or an absolute position;

Example:

Send a stop command to the slave:

Host -> Slave: 01 06 00 27 01 00 38 51

Slave->Host: 01 06 00 27 01 00 38 51

Send an emergency stop command to the slave:

Master->Slave: 01 06 00 27 02 00 38 A1 Slave->Host: 01 06 00

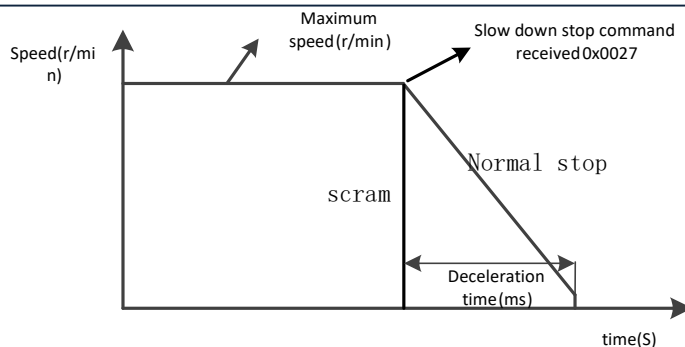
27 02 00 38 A1

the motor is running in positioning mode or speed mode, if it receives a normal stop command, the motor will decelerate and stop according to the set deceleration time. If it receives an emergency stop command, it will stop directly without deceleration.



Notice:

- The deceleration time parameter needs to be set before the motor runs. If the driver receives the command after the motor starts running, it will execute the command according to the deceleration time set before the motor runs.



Normal stop and emergency stop

5 Other functions

In addition to the main motion control function, the driver also has a series of other functions to achieve functions such as register parameter saving, factory reset, terminal function configuration, etc., for the convenience of customers.

5.1 Auxiliary control instructions

Modbus : 002Dh auxiliary control command register is used to realize a series of functions such as motor enable release, alarm clearing, parameter recovery and storage.

※ **Modbus: 002Dh auxiliary control register**

Register Address	name	illustrate
0x002D	Auxiliary control instructions	0x0000: invalid; 0x0011: Motor release; 0x0012: Motor enable; 0x0021: drive alarm cleared; 0x0031: Clear the current position of the motor; 0x0041: Restore parameters to factory settings; 0x0042: save all parameters; Note: When performing factory recovery and parameter saving operations on the drive, it is necessary to ensure that the motor is in the stopped state, otherwise the relevant instructions will be ignored;

Example:1) Driver controls the motor to release:

Host->Slave: 01 06 00 2D 00 11 D9 CF

Slave->Host: 01 06 00 2D 00 11 D9 CF

2) Driver controls the motor to enable:

Host->Slave: 01 06 00 2D 00 12 99 CE

Slave->Host: 01 06 00 2D 00 12 99 CE

3) When the driver generates a resettable alarm, reset the alarm:

Host->Slave: 01 06 00 2D 00 21 D9 DB

Slave->Host: 01 06 00 2D 00 21 D9 DB

4) Restore the drive factory parameters

Host->Slave: 01 06 00 2D 00 41 D9 F3

Slave->Host: 01 06 00 2D 00 41 D9 F3

5) Save the current parameters of the drive

Host->Slave: 01 06 00 2D 00 42 99 F2

Slave->Host: 01 06 00 2D 00 42 99 F2

5.2 Input and output terminals

The driver integrates several input and output terminals, and its functions can be configured according to the actual use function. If the return to origin function is required, several terminals can be selected and their functions can be set to origin, positive and negative limit; if multi-stage position/multi-stage speed control is required, several terminals can be selected and their functions can be set to PT enable, PV enable, PIN0~3, etc.

All functions of the input and output terminals are as follows:

※ Input terminal function table

Input terminal function	illustrate
0 : undefined;	The terminal has no function;
1 : origin signal;	The input signal is the origin sensor signal;
2 : Positive limit signal;	The input signal is the positive limit sensor signal;
3 : Anti-limit signal;	The input signal is the negative limit sensor signal;
4 : Motor MF signal;	The motor is released when a valid level is given;
5 : Stop signal;	When a valid level is given, the motor movement is forced to stop;
6 : Emergency stop signal;	When a valid level is given, the motor is forced to stop urgently;
7 : Position mode movement;	When a valid level is given, the movement is completed according to the movement parameters and the edge is valid;
8: Speed mode movement;	When a valid level is given, the movement is completed according to the movement parameters and the level is valid;
9 : JOG+ point motion;	When a valid level is given, the movement is completed according to the movement parameters and the level is valid;
10 : JOG -point movement;	When a valid level is given, the movement is completed according to the movement parameters and the level is valid;
1 1 : Return to origin enable signal;	When a valid level is given, the movement is completed according to the movement parameters and the edge is valid;
1 2: PT trigger signal ;	When a valid level is given, the movement is completed according to the movement parameters and the edge is valid;
13: PV trigger signal;	When a valid level is given, the movement is completed according to the movement parameters and the level is valid;
1 4 : PIN0 ;	Multiple position/speed segment binary selection bits, level effective;
1 5 : PIN1 ;	
1 6 : PIN2 ;	
1 7 : PIN3 ;	

※ Output terminal function table

Output terminal function	illustrate
0 : undefined;	The terminal has no function;
1: Alarm signal;	When the driver is in alarm state, it outputs a valid level;
2: Driver status signal;	When the driver is in motion, it outputs a valid level;
3: Return to origin completion signal;	When the driver returns to the origin, it outputs a valid level;
4: Arrival signal;	When the driver starts and reaches the correct position, it outputs a valid level.
5: Braking signal;	When the driver locks the motor, it outputs a valid level; When the driver releases the motor, it outputs an invalid level;
9: User defined 0;	The driver outputs a valid or invalid level according to the 004F h register setting;
10: User defined 1;	
11: User defined 2;	

※ Modbus: 0040h~004Fh input and output terminal function configuration register

Register Address	name	illustrate
0x0040	Input terminal effective level selection	Used to set the input terminal voltage to be normally open or normally closed;
0x0041~0x0047	Input terminal function selection	Used to set the input terminal function
0x004B	Output terminal effective level selection	Used to set the output terminal voltage to be normally open or normally closed;
0x004C~0x004E	Output terminal function selection	Used to set the output terminal function
0x004F	Y group terminal custom output	When the output terminal function is set to custom, this register is used to control the corresponding output terminal to output a valid level or an invalid level;

5.3 Software limit function

In some special applications, the limit sensor cannot be installed in the stroke due to structural space limitations, and at the same time it is necessary to ensure that the stroke is limited within a safe range. In this case, the internal soft limit function of the driver can be used.

The soft limit function can be started through Modbus: 0019h register, and the soft limit range can be set through Modbus: 0037h~003Ah register.

※ **Modbus: 0018h, 0037h~003Ah soft limit configuration register**

Register Address	name	illustrate
0x0018	Internal software limit switches	0: invalid; 1: Take effect after returning to zero;
0x0037	Soft limit positive high position	Software positive limit setting point, this limit point will take effect only after zero return is completed ;
0x0038	Soft limit positive low position	
0x0039	Soft limit negative high position	Software negative limit setting point, this limit point will take effect only after zero return is completed ;
0x003A	Soft limit negative low position	

Note: The soft limit function will only take effect after the origin position is determined.

5.4 Interrupt fixed length function

The interrupt fixed length function means that when the motor is currently running in speed mode, the running mode can be switched to positioning mode by triggering an external signal, and the positioning length can be set in advance.

The interrupt fixed length function can be implemented in many ways.

Method 1 :

Start the motor into speed operation mode through Modbus: 0027h motion control command register;
Select any input signal and set its function to **7 : Position mode motion** function. When the input signal receives a valid level, the motor enters fixed-length positioning. The fixed-length pulse value can be set in Modbus: 0024h~0025h; **Method 2:**

This function is achieved by mixing the multi-speed control mode and the multiposition mode. Note that the multi-speed mode and multi-position mode functions can only be mixed after the Modbus: 0051h register is set to 1.

5.5 Setting the endian mode of high and low registers

In the Modbus register parameter table, some parameters extend the register working range by high and low bits to meet the actual application requirements. The following table lists all high and low bit parameters:

※ **All high and low register tables**

Register Address	name
0x000A	Current position high
0x000B	Current position low
0x0024	Total pulse count high
0x0025	Total pulse count low
0x0037	Soft limit positive high position
0x0038	Soft limit positive low position
0x0039	Soft limit negative high position
0x003A	Soft limit negative low position
0x0060	The first segment motion pulse command high
0x0061	The first segment motion pulse command low
...	...
0x00BA	16th segment motion pulse command high
0x00BB	16th segment motion pulse command low

The Modbus series bus driver can adapt to mainstream touch screens, PLCs, control cards and other control devices that support the Modbus protocol. However, there are differences in the big-endian and small-endian modes for 32-bit variables in different devices. In the big-endian mode, the high byte of the data is stored in the low address of the memory, and the low byte of the data is stored in the high address of the memory; in the little-endian mode, the high byte of the data is stored in the high address of the memory, and the low byte of the data is stored in the low address of the memory. For different control devices, the Modbus: 0019h register can be set to adapt the big-endian and small-endian modes of the control device.

※ **Modbus: 0019h big and small end configuration register**

Register Address	name	illustrate
0x0019	32-bit register endianness configuration	0: high position first, low position last; 1: High position at the back, low position at the front; Adapt to different PLC or touch screen usage habits;

6 Revision History

Version	describe	time	Remark
V1.0	First edition released	2025/03/05	

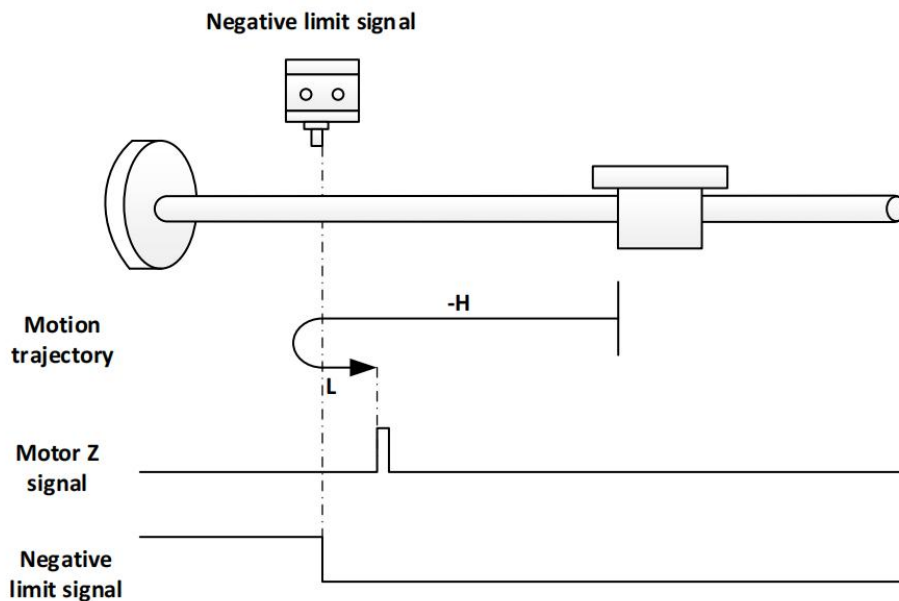
Appendix 1: Introduction to the method of returning to the origin

※ Method 1 (0031h = 1)

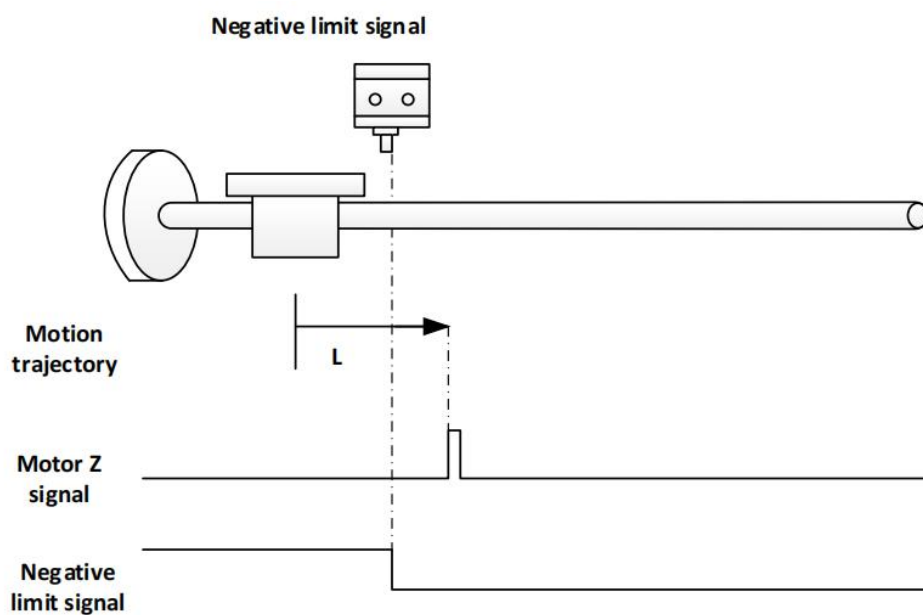
Origin: Motor Z signal

Deceleration point: negative limit

a) The deceleration point signal is invalid when returning to zero : when running in the reverse direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the forward direction at low speed, it will stop when encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is valid when starting from zero : the machine runs at a low speed in the forward direction and stops when it encounters the first Z signal after the falling edge of the deceleration point ;

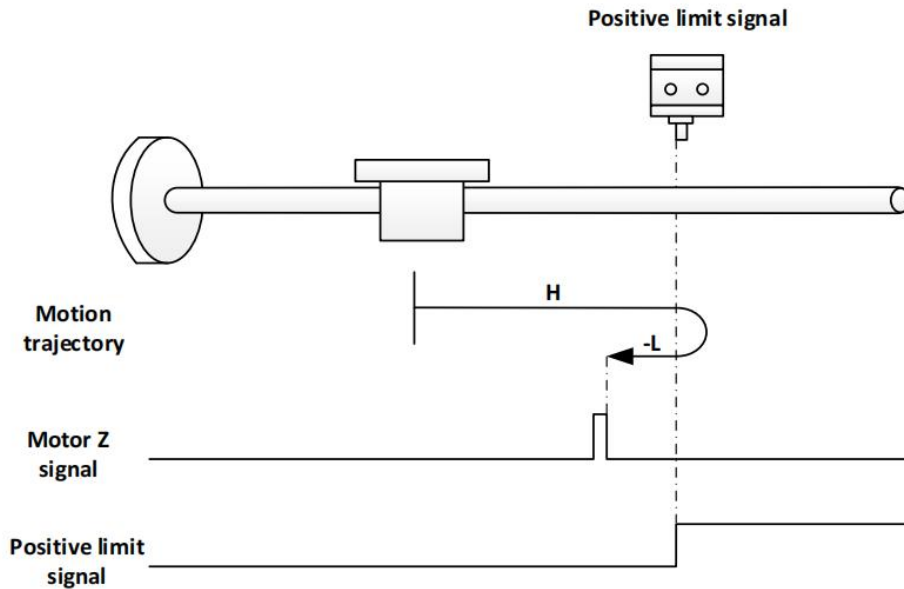


※ **Method 2 (0031h = 2)**

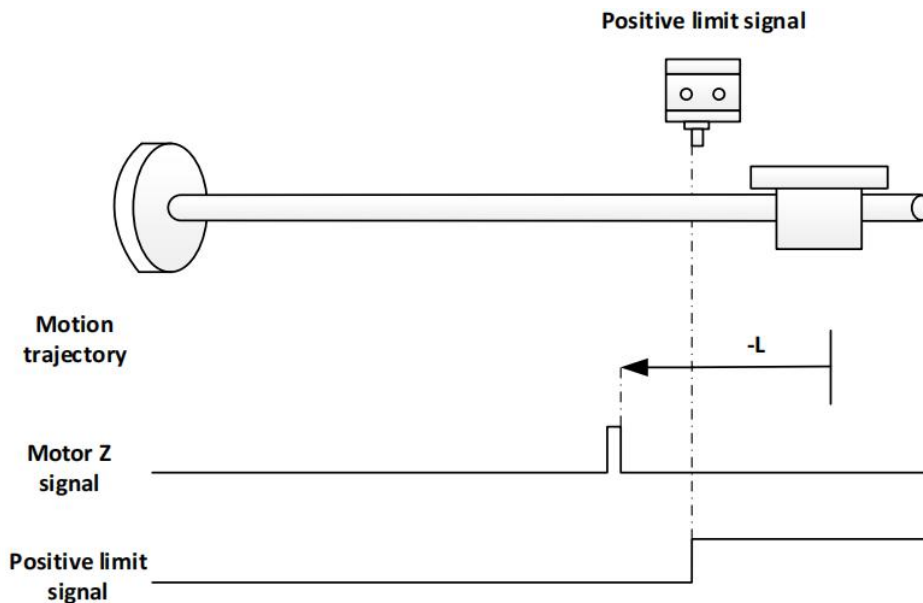
Origin: Motor Z signal

Deceleration point: positive limit

a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop when encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is valid when starting from zero : the machine runs in the reverse direction at a low speed and stops when encountering the first Z signal after the falling edge of the deceleration point ;

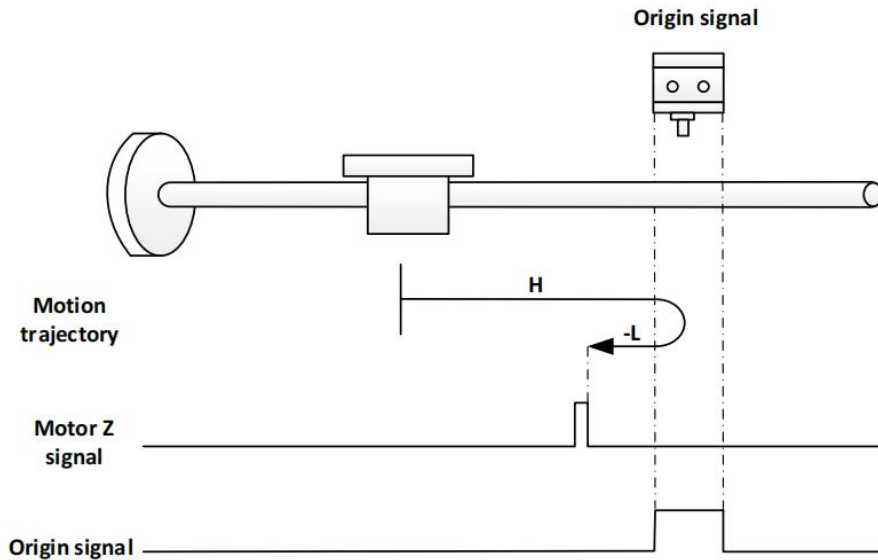


※ **Method 3 (0031h = 3)**

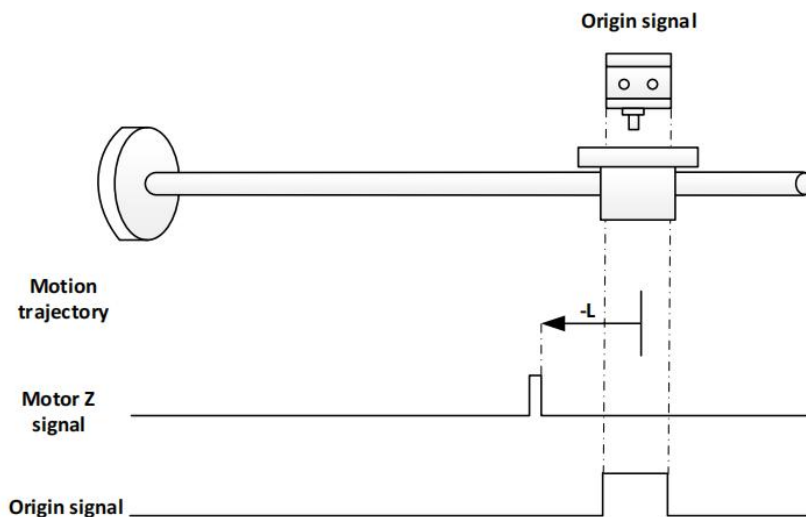
Origin: Motor Z signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop after encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is valid when starting from zero : the machine runs in the reverse direction at a low speed and stops when encountering the first Z signal after the falling edge of the deceleration point ;

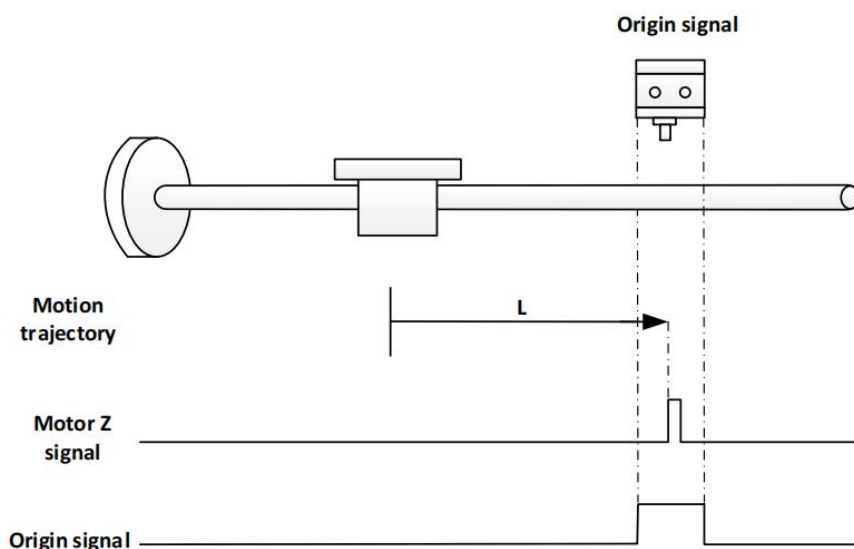


※ **Method 4 (0031h = 4)**

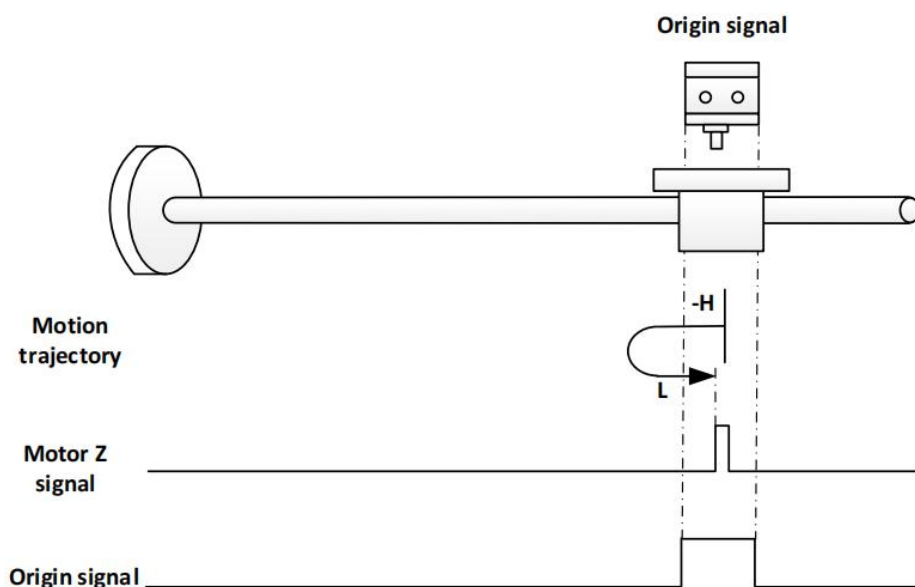
Origin: Motor Z signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when starting from zero : the machine runs at a low speed in the forward direction and stops when it encounters the first Z signal after the rising edge of the deceleration point;



b) The deceleration point signal is valid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop after encountering the falling edge of the deceleration point; when running in the forward direction at low speed, it will stop after encountering the first Z signal after the rising edge of the deceleration point;

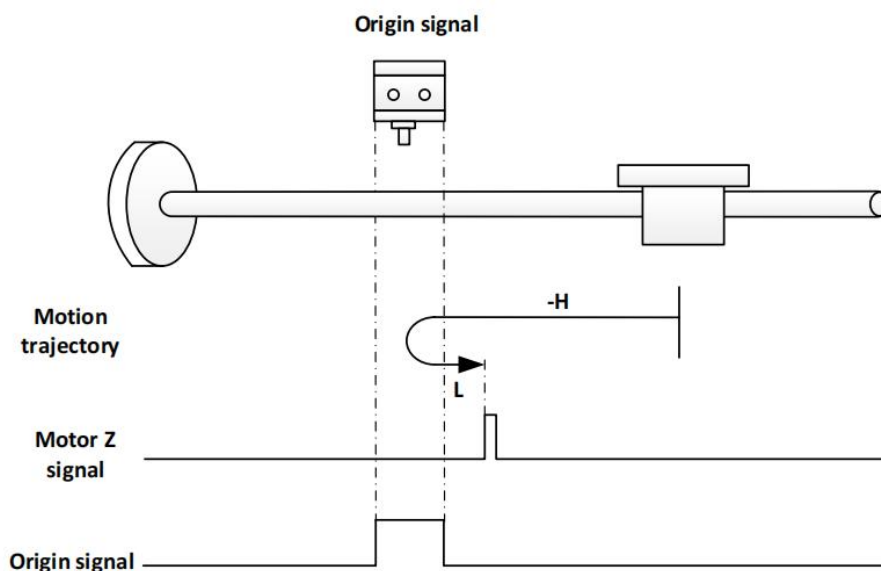


※ Method 5 (0031h = 5)

Origin: Motor Z signal

Deceleration point: origin signal

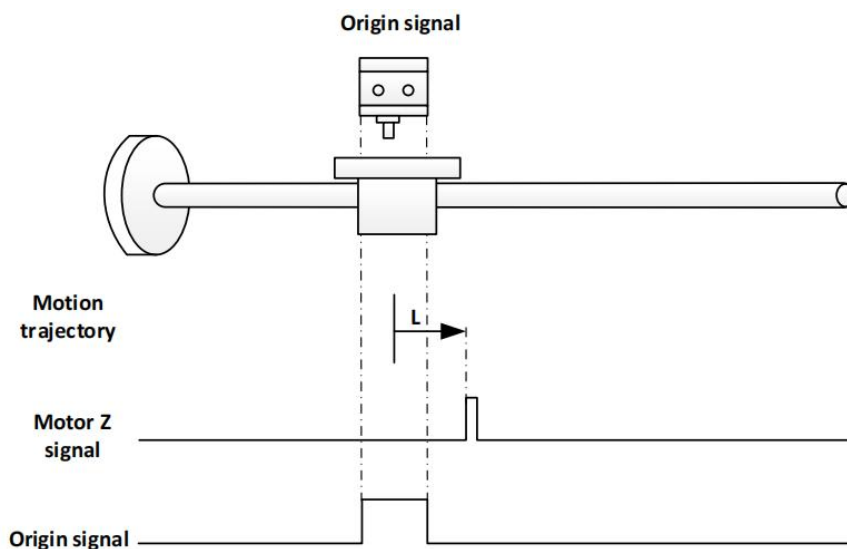
a) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop after encountering the rising edge of the deceleration point; when running in the forward direction at low speed, it will stop after encountering the first Z signal after the falling edge of the deceleration point ;



Origin: Motor Z signal

Deceleration point: origin signal

a) The deceleration point signal is valid when starting from zero : the machine runs at a low speed in the forward direction and stops when it encounters the first Z signal after the falling edge of the deceleration point ;

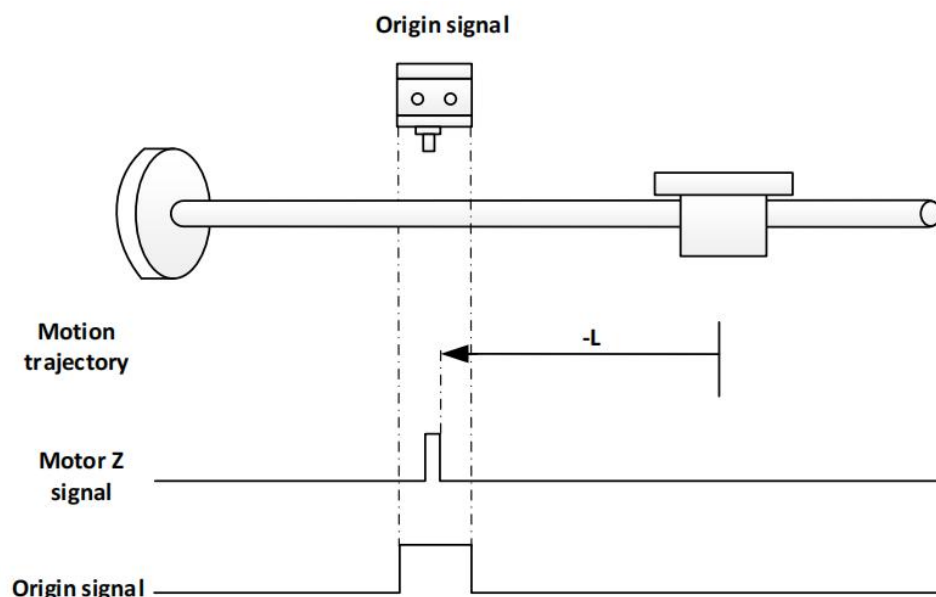


※ Method 6 (0031h = 6)

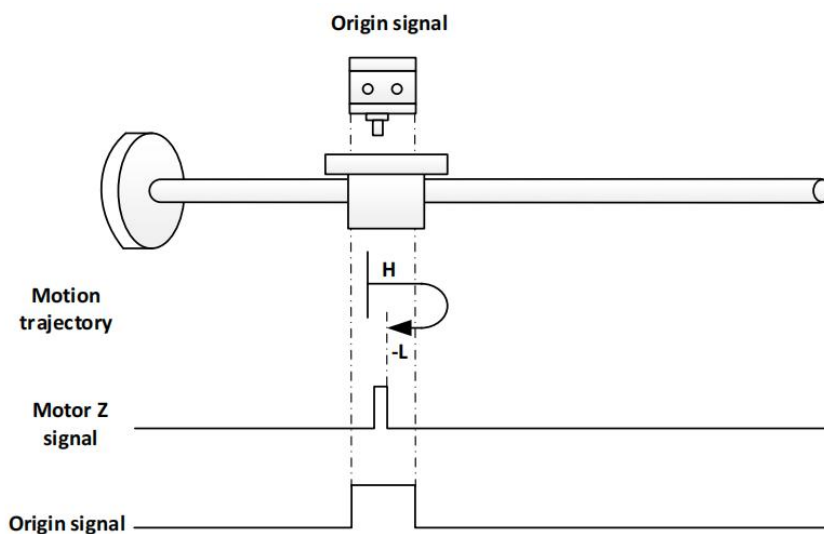
Origin: Motor Z signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when starting from zero return : the machine runs in the reverse direction at a low speed and stops when encountering the first Z signal after the rising edge of the deceleration point;



b) The deceleration point signal is valid when returning to zero and starting : when running at high speed in the forward direction , it will decelerate and stop after encountering the falling edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the first Z signal after the rising edge of the deceleration point;

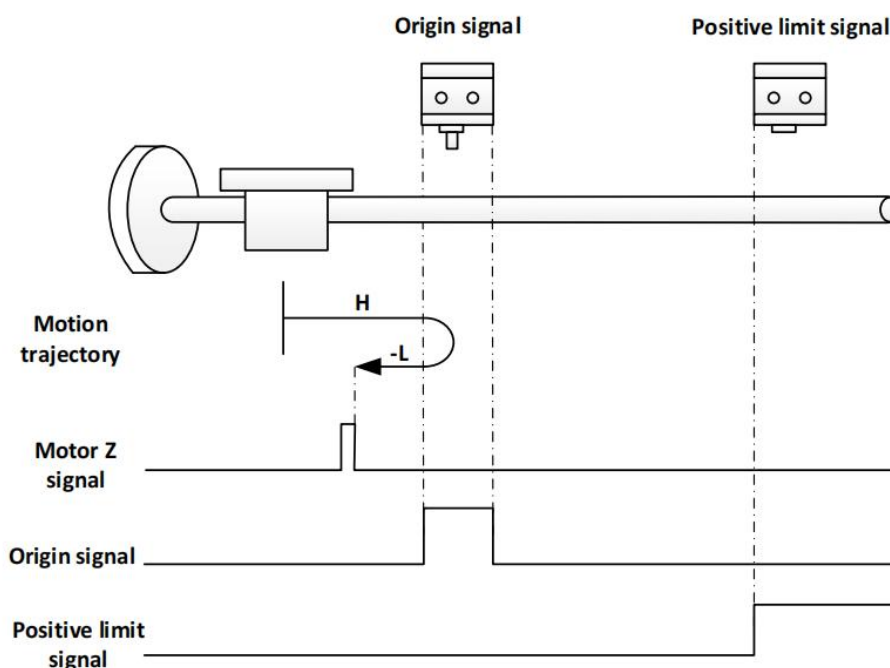


※ Method 7 (0031h = 7)

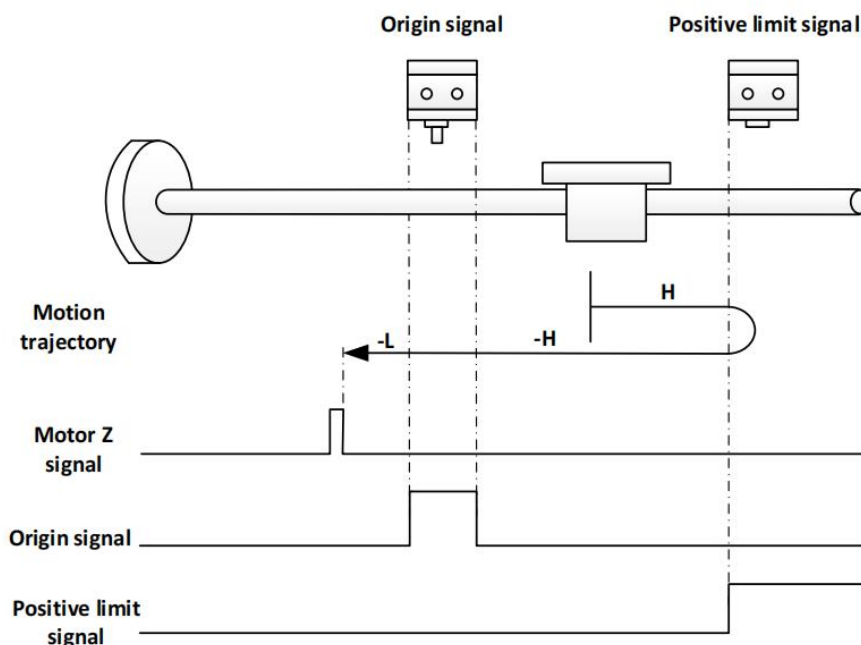
Origin: Motor Z signal

Deceleration point: origin signal

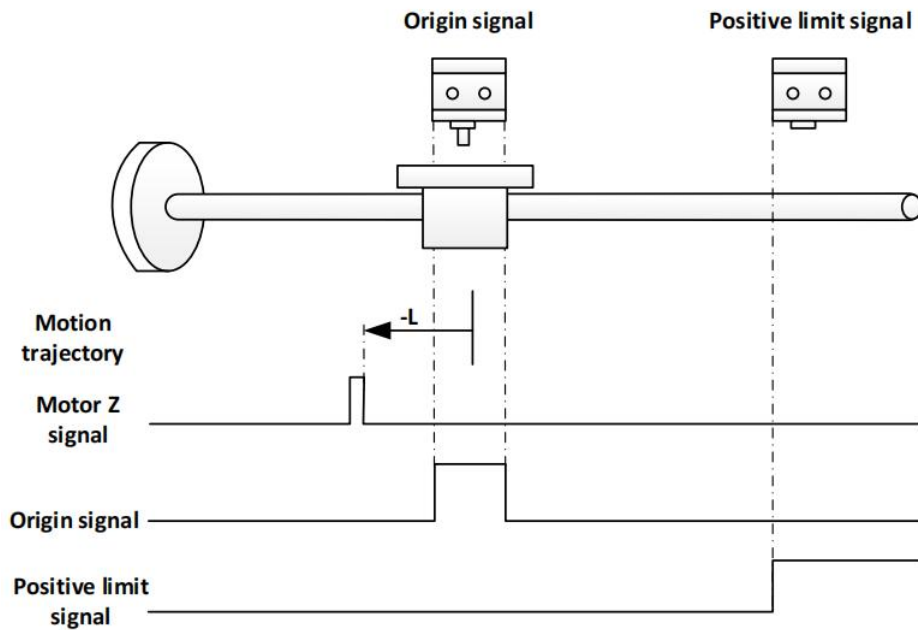
a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop after encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will run at low speed when encountering the rising edge of the deceleration point, and stop when encountering the first Z signal after the falling edge of the deceleration point ;



c) The deceleration point signal is valid when starting from zero : the machine runs in the reverse direction at a low speed and stops when encountering the first Z signal after the falling edge of the deceleration point ;



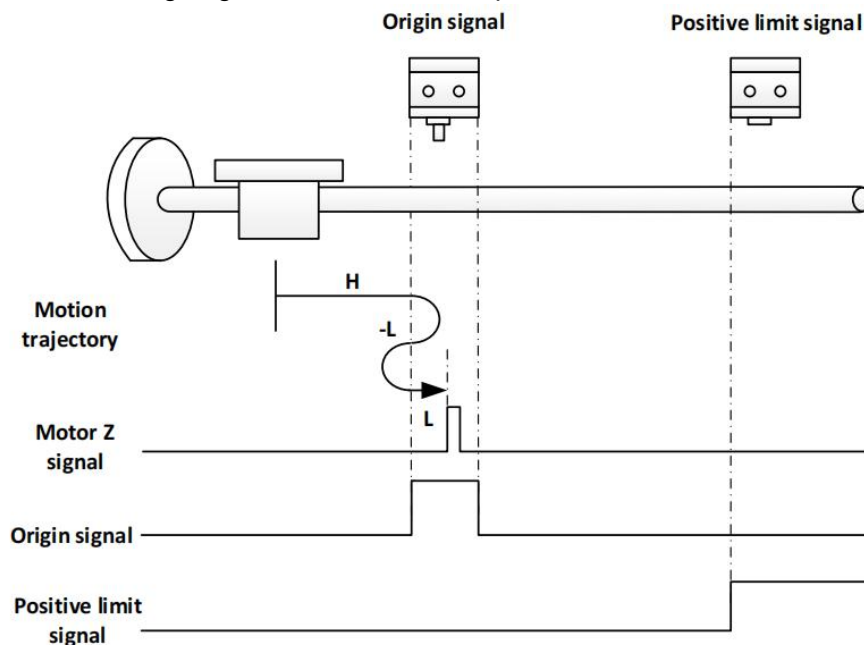
※ **Method 8 (0031h = 8)**

Origin: Motor Z signal

Deceleration point: origin signal

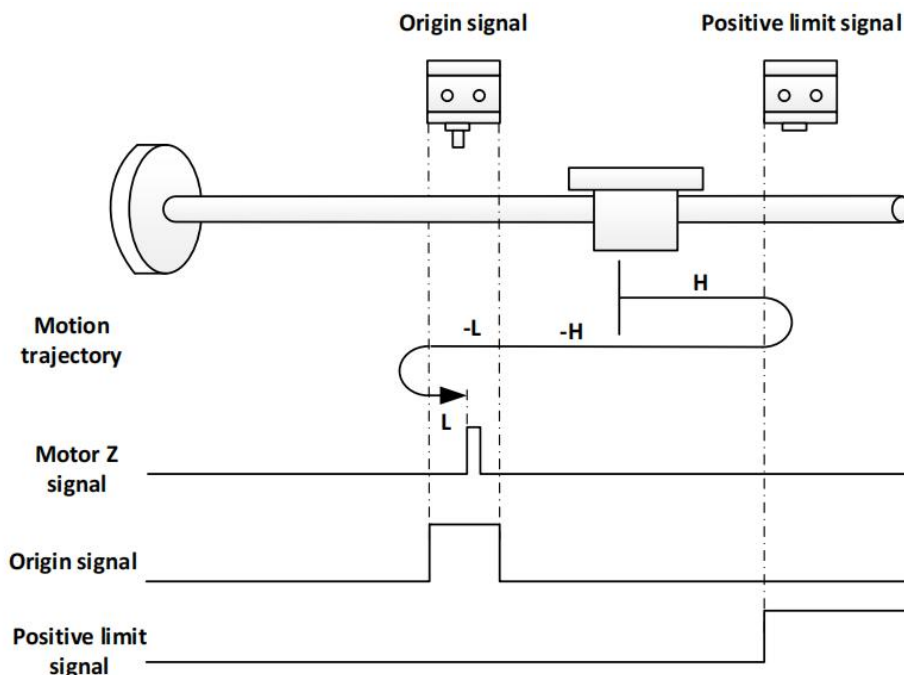
a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop after encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will decelerate and stop after encountering the falling edge of the deceleration point; when running at low speed in the forward direction, it will stop after encountering the first

Z signal after the rising edge of the deceleration point;

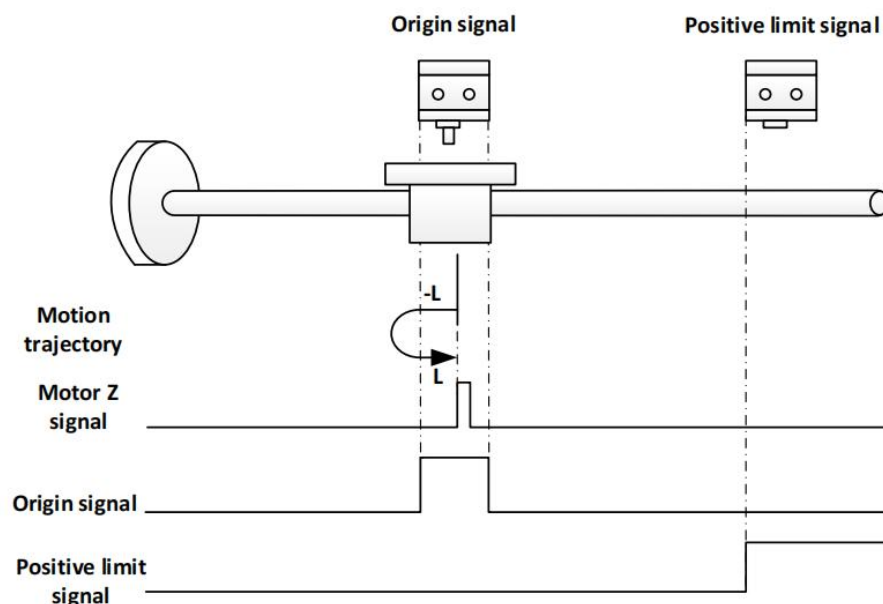


b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will run at low speed when encountering the rising edge of the deceleration point, and decelerate and stop when encountering the falling edge of

the deceleration point ; when running at low speed in the forward direction, it will stop when encountering the first Z signal after the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running in the reverse direction at a low speed , it will decelerate and stop when it encounters the falling edge of the deceleration point; when running in the forward direction at a low speed, it will stop when it encounters the first Z signal after the rising edge of the deceleration point;



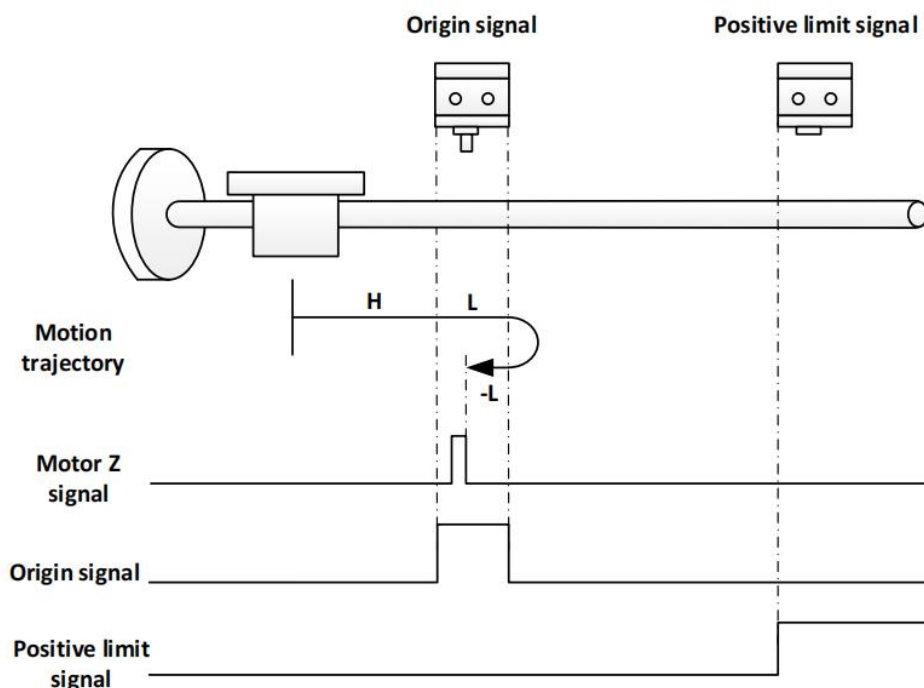
※ Method 9 (0031h = 9)

Origin: Motor Z signal

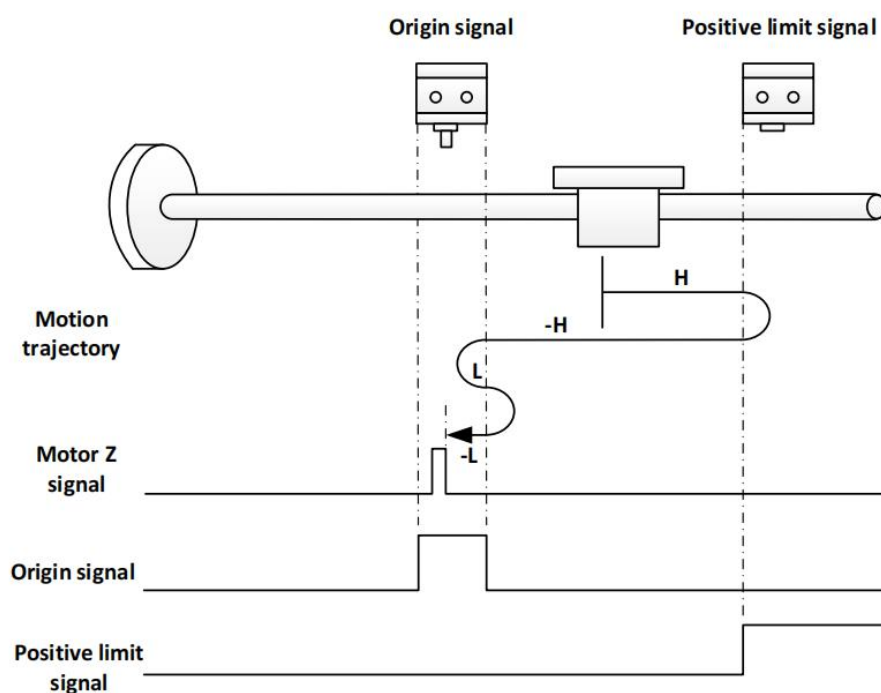
Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the forward direction, runs at low speed when encountering the rising edge of the deceleration point, and decelerates and stops when encountering the falling edge of the deceleration point ; it runs at

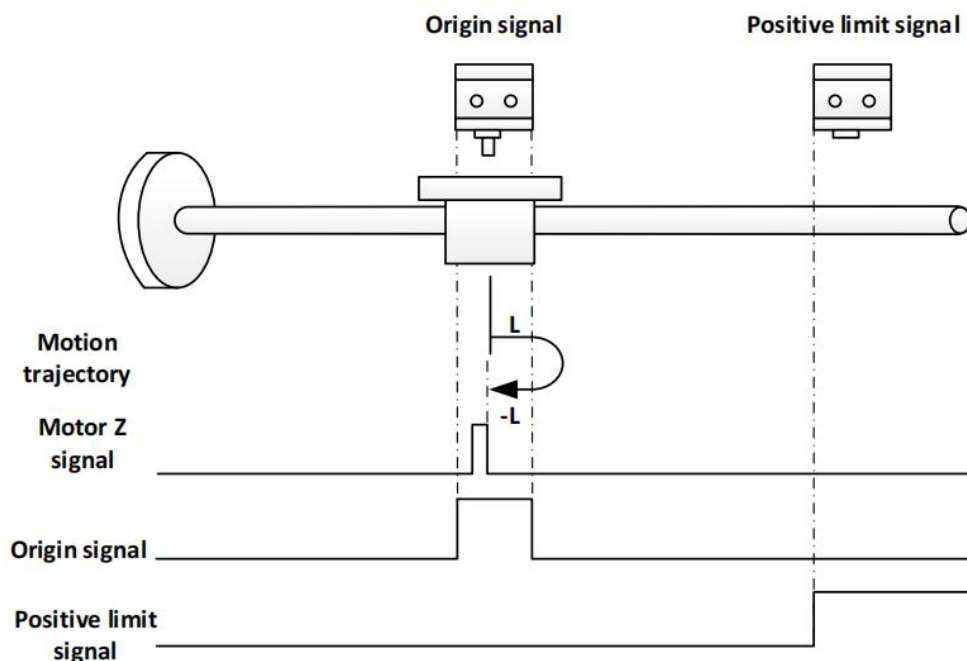
low speed in the reverse direction, and stops when encountering the first Z signal after the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the forward direction, it will decelerate and stop when encountering the falling edge of the deceleration point ; when running at low speed in the reverse direction, it will stop when encountering the first Z signal after the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running at low speed in the forward direction , it will decelerate and stop when it encounters the falling edge of the deceleration point; when running at low speed in the reverse direction, it will stop when it encounters the first Z signal after the rising edge of the deceleration point;

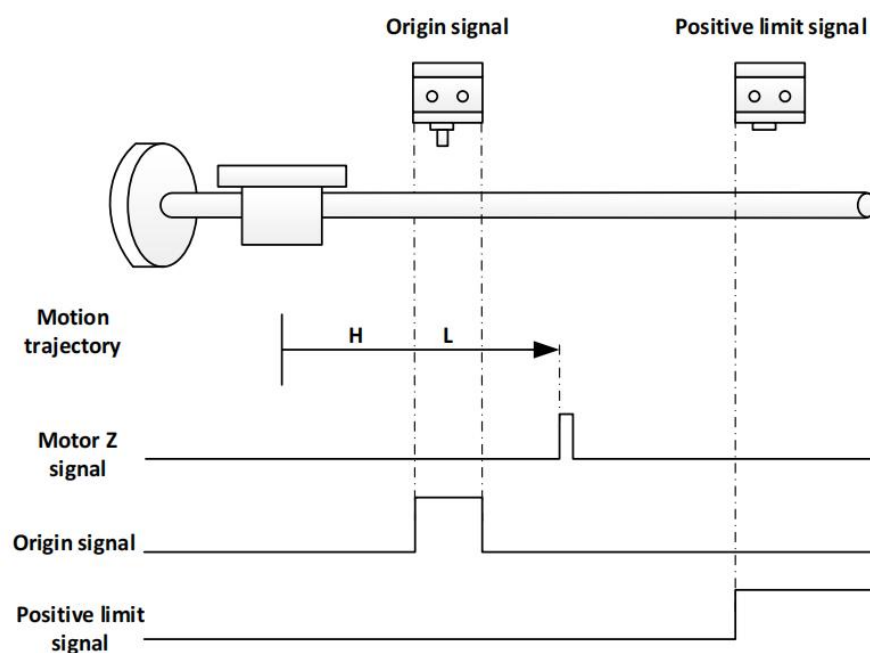


※ Method 10 (0031h = 10)

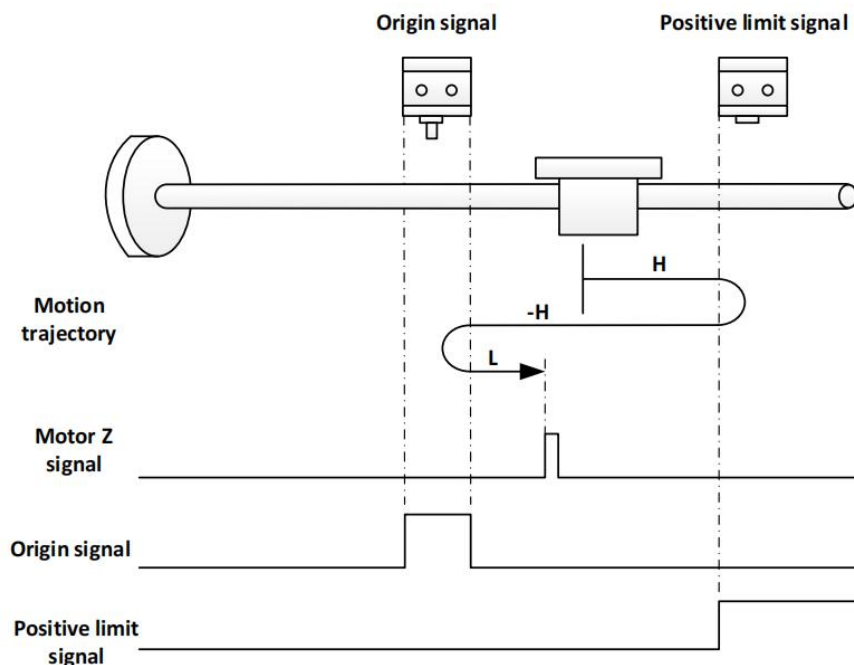
Origin: Motor Z signal

Deceleration point: origin signal

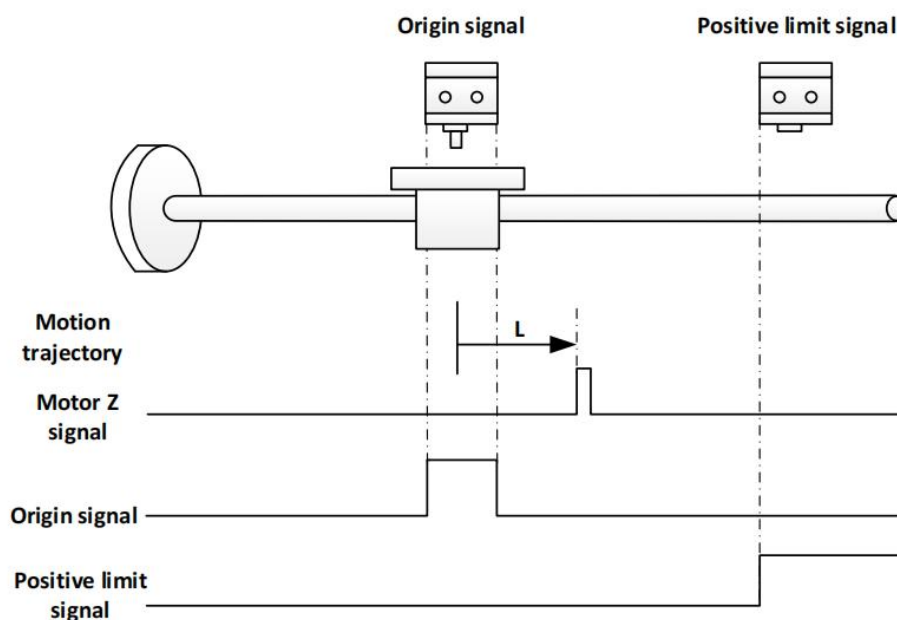
a) The deceleration point signal is invalid when starting from zero : it runs at high speed in the forward direction, runs at low speed when encountering the rising edge of the deceleration point, and stops when encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the forward direction, it will stop when encountering the first Z signal after the falling edge of the deceleration point ;



c) The deceleration point signal is invalid when starting from zero : it runs at a low speed in the forward direction and stops at the first Z signal after the falling edge of the deceleration point ;

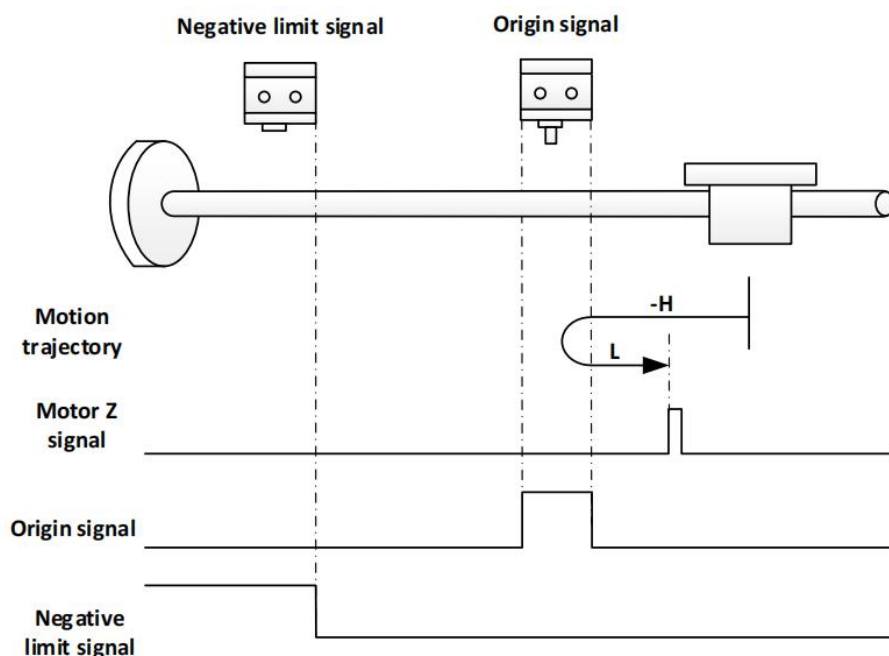


※ Method 11 (0031h = 11)

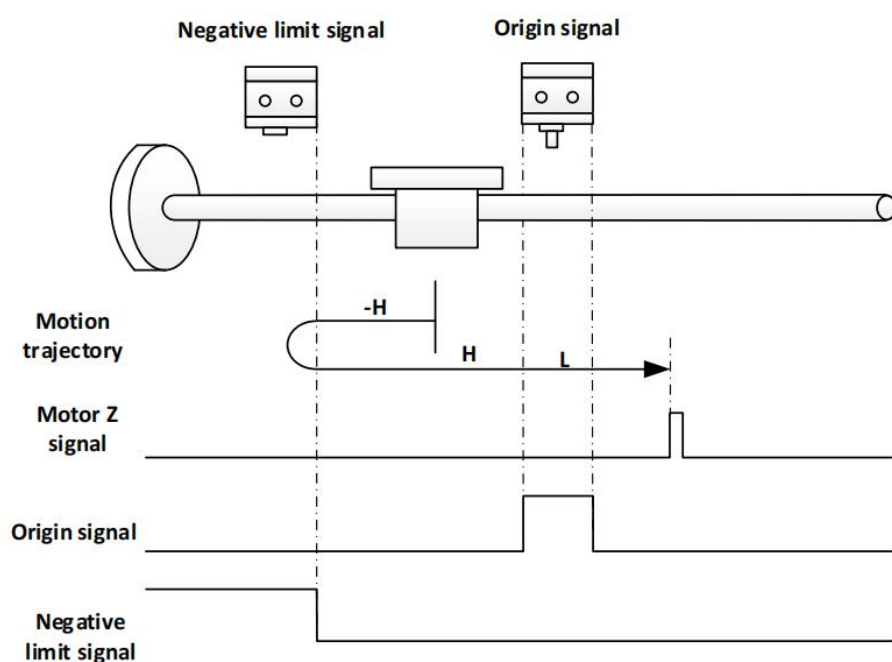
Origin: Motor Z signal

Deceleration point: origin signal

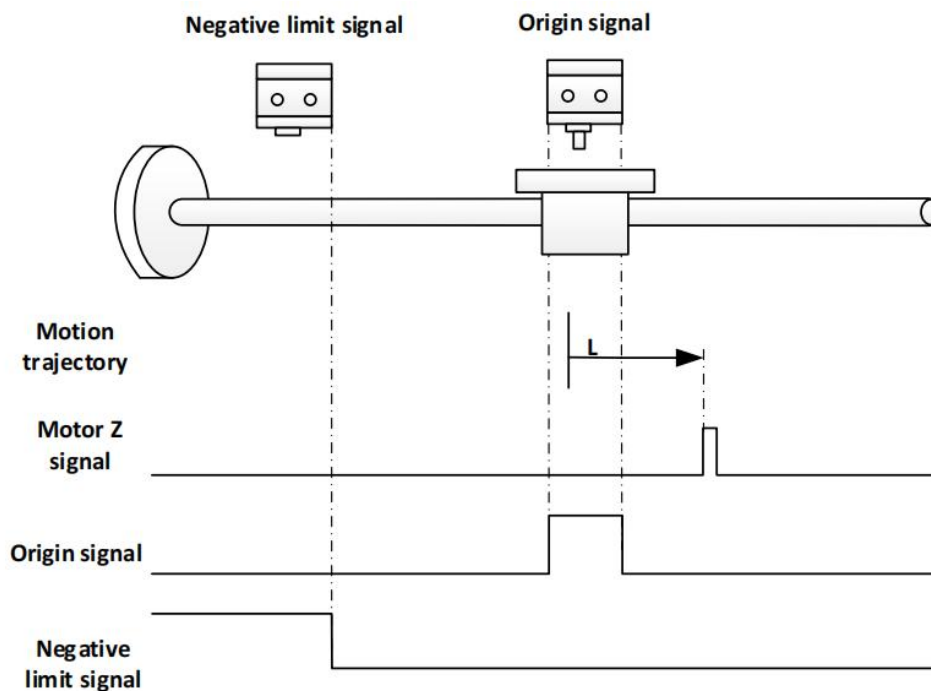
a) The deceleration point signal is invalid when returning to zero : when running in the reverse direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the forward direction at low speed, it will stop when encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will run at low speed when encountering the rising edge of the deceleration point, and stop when encountering the first Z signal after the falling edge of the deceleration point



c) The deceleration point signal is valid when starting from zero : the machine runs at a low speed in the forward direction and stops when it encounters the first Z signal after the falling edge of the deceleration point ;

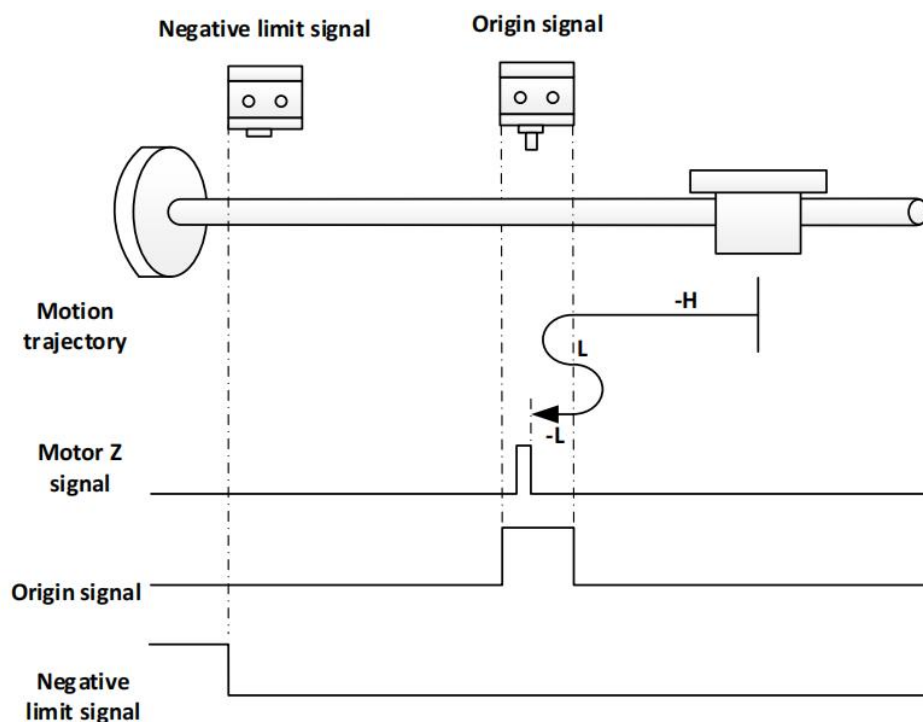


※ **Method 12 (0031h = 12)**

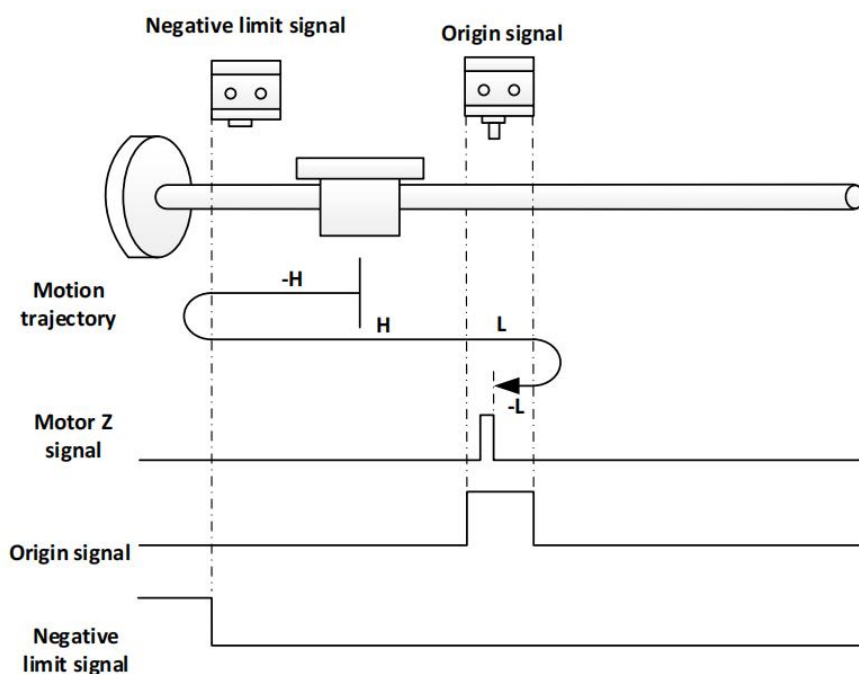
Origin: Motor Z signal

Deceleration point: origin signal

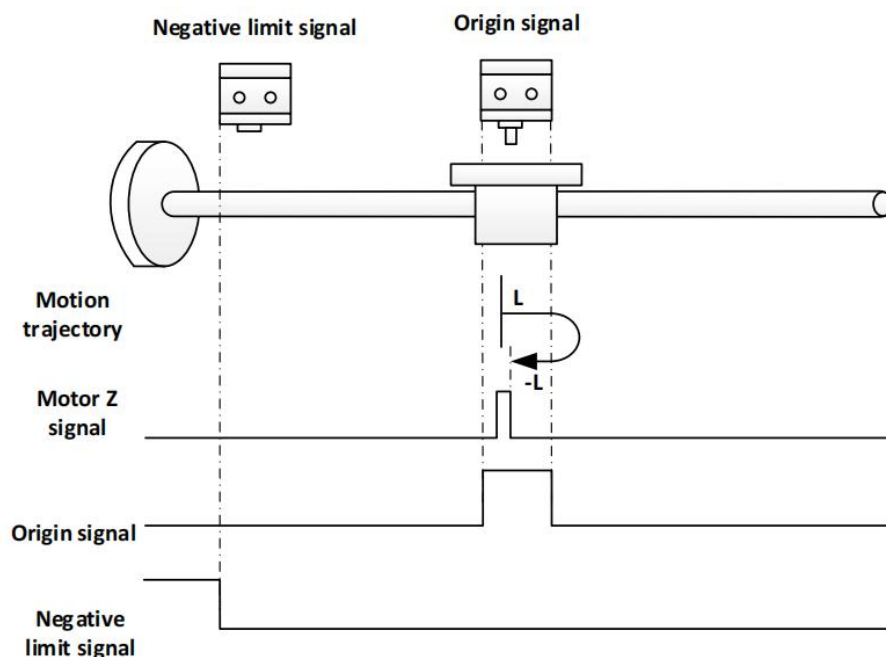
a) The deceleration point signal is invalid when returning to zero : when running in the reverse direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the forward direction at low speed, it will decelerate and stop when encountering the falling edge of the deceleration point; when running in the reverse direction at low speed, it will stop when encountering the first Z signal after the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit ; when running in the forward direction at high speed, it will run at low speed when encountering the rising edge of the deceleration point, and decelerate and stop when encountering the falling edge of the deceleration point ; when running in the reverse direction at low speed, it will stop when encountering the first Z signal after the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running at low speed in the forward direction , it will decelerate and stop when it encounters the falling edge of the deceleration point; when running at low speed in the reverse direction, it will stop when it encounters the first Z signal after the rising edge of the deceleration point;

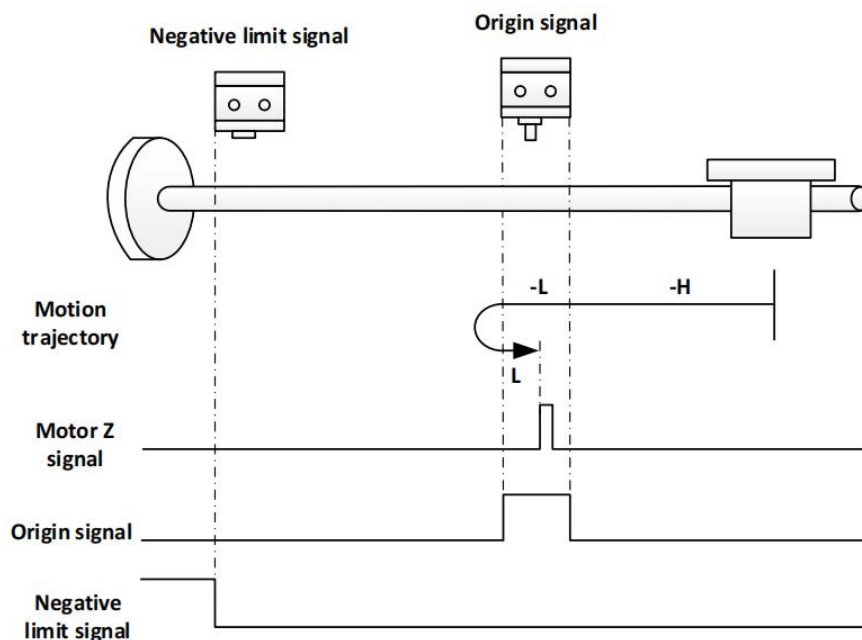


※ Method 13 (0031h = 13)

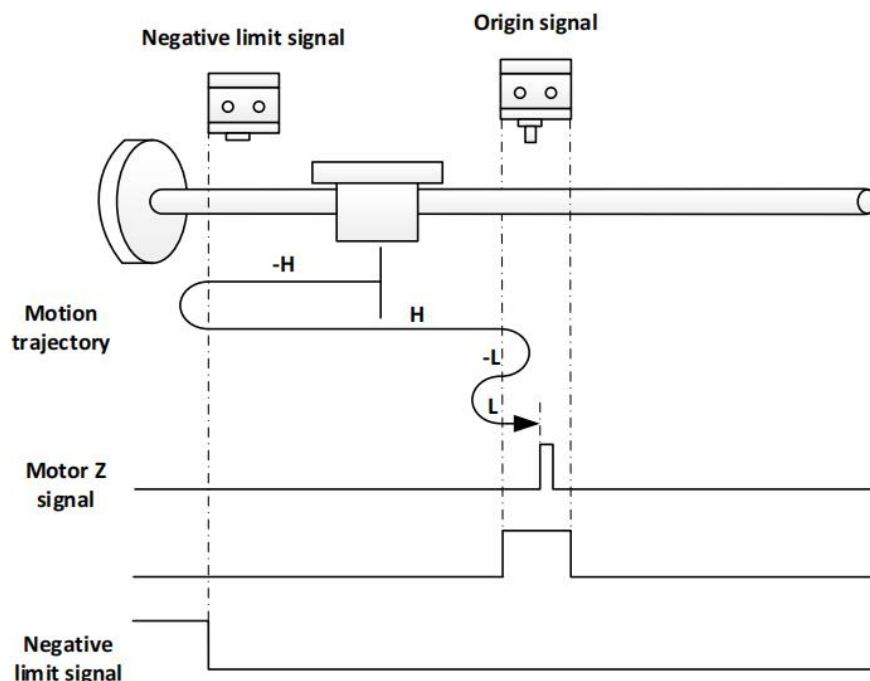
Origin: Motor Z signal

Deceleration point: origin signal

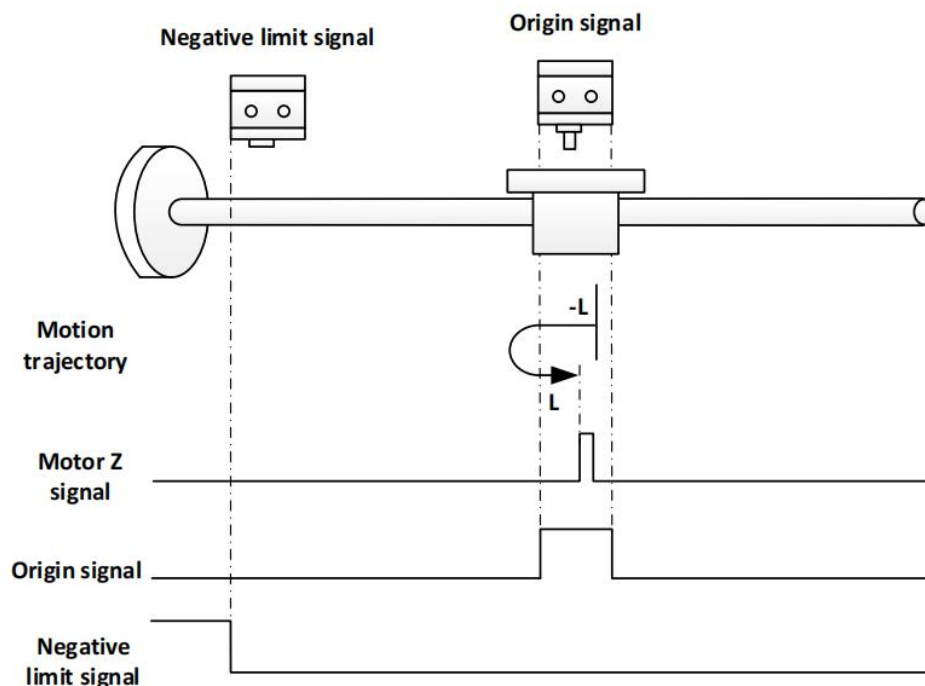
a) The deceleration point signal is invalid when returning to zero : it runs at high speed in the reverse direction, runs at low speed when encountering the rising edge of the deceleration point, and decelerates and stops when encountering the falling edge of the deceleration point; it runs at low speed in the forward direction, and stops when encountering the first Z signal after the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the reverse direction at low speed, it will decelerate and stop when encountering the falling edge of the deceleration point ; when running in the forward direction at low speed, it will stop when encountering the first Z signal after the rising edge of the deceleration point;



c) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at a low speed , it will decelerate and stop when it encounters the falling edge of the deceleration point; when running in the forward direction at a low speed, it will stop when it encounters the first Z signal after the rising edge of the deceleration point;

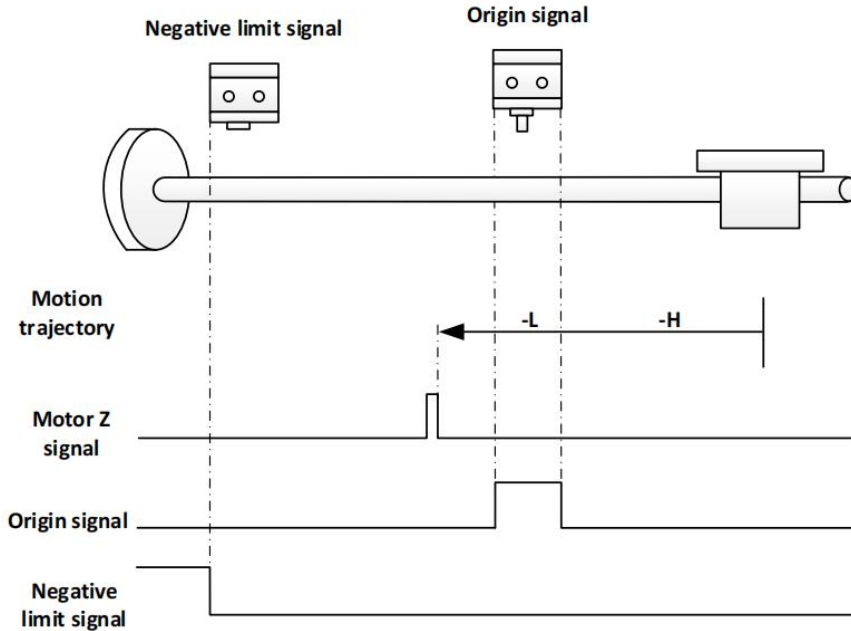


※ Method 14 (0031h = 14)

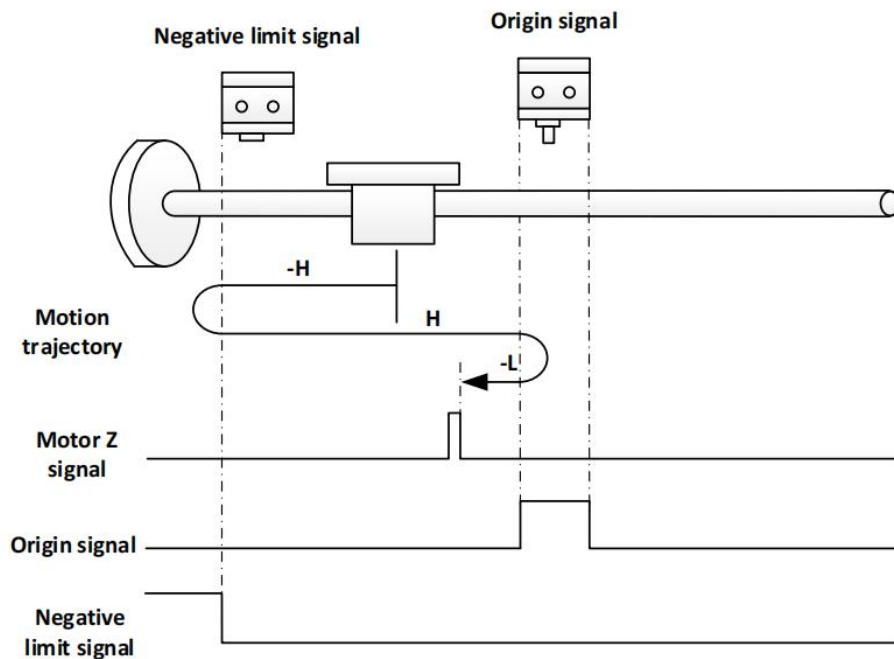
Origin: Motor Z signal

Deceleration point: origin signal

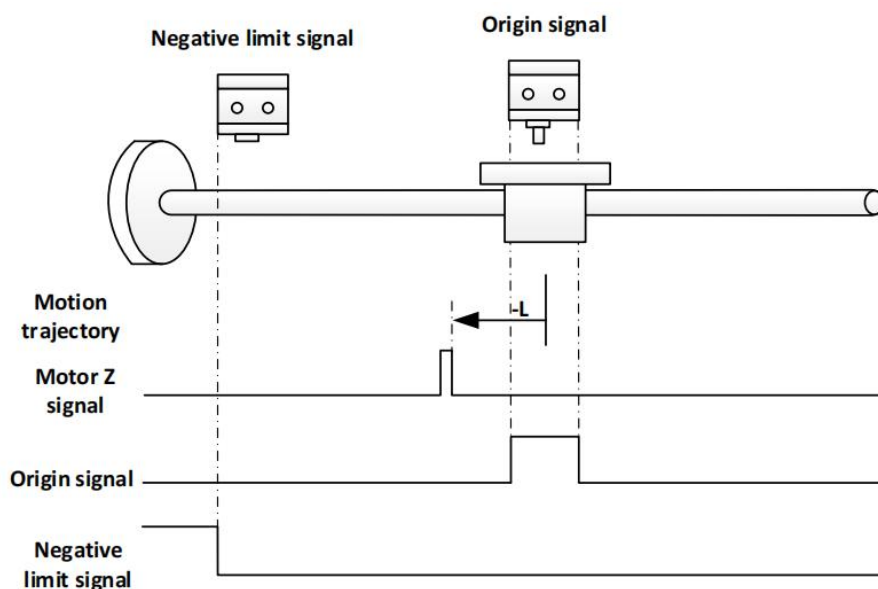
a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, runs at low speed when encountering the rising edge of the deceleration point, and stops when encountering the first Z signal after the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero : when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the reverse direction at low speed, it will stop when encountering the first Z signal after the falling edge of the deceleration point ;



c) The deceleration point signal is invalid when returning to zero : the machine runs in the reverse direction at a low speed and stops when encountering the first Z signal after the falling edge of the deceleration point ;

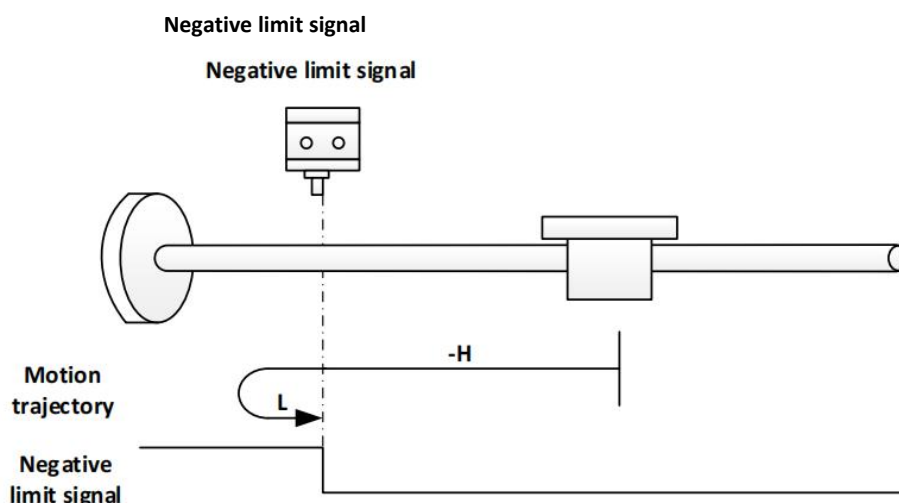


※ Method 17 (0031h = 17)

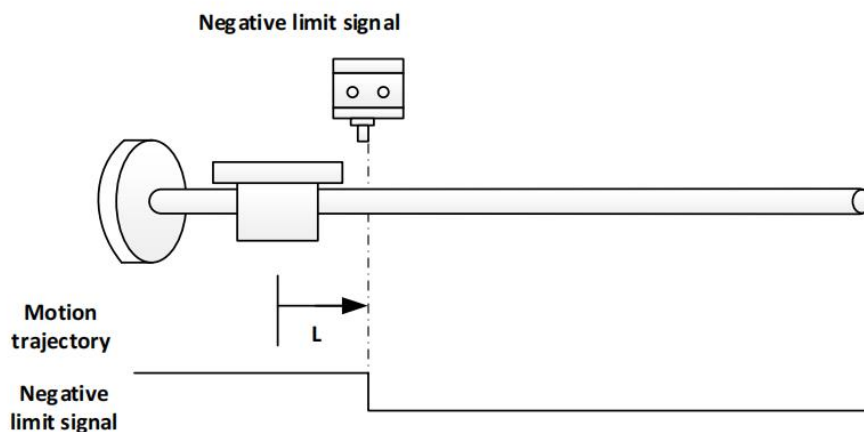
Origin: Negative limit

Deceleration point: negative limit

a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, and stops when it encounters the rising edge of the deceleration point; it runs at low speed in the forward direction, and stops when it encounters the falling edge of the deceleration point ;



b) The deceleration point signal is valid when returning to zero and starting : the machine runs at a low speed in the forward direction and stops after encountering the falling edge of the deceleration point ;

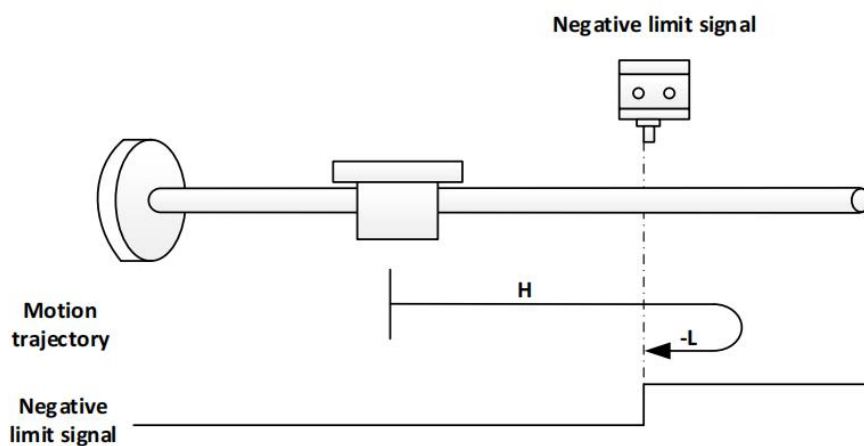


※ Method 18 (0031h = 18)

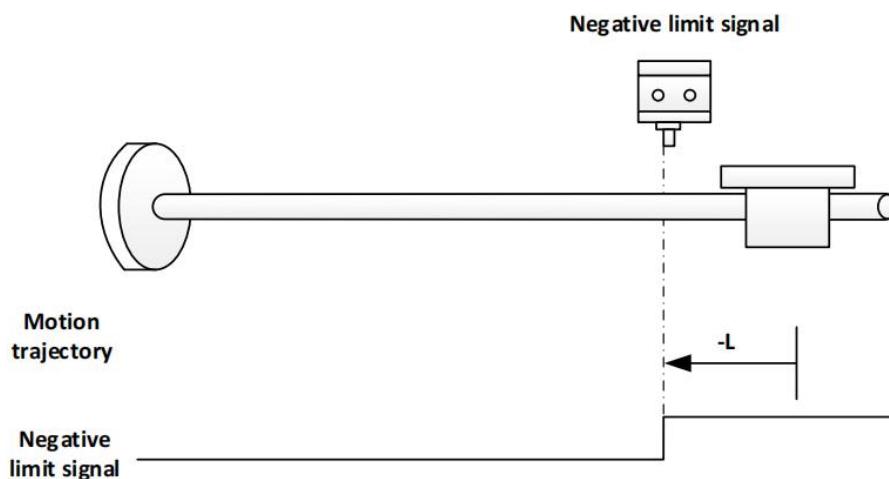
Origin: Positive limit

Deceleration point: positive limit

a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when it encounters the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop when it encounters the falling edge of the deceleration point ;



b) The deceleration point signal is valid when returning to zero and starting : the machine will run in the reverse direction at a low speed and stop when it encounters the falling edge of the deceleration point ;

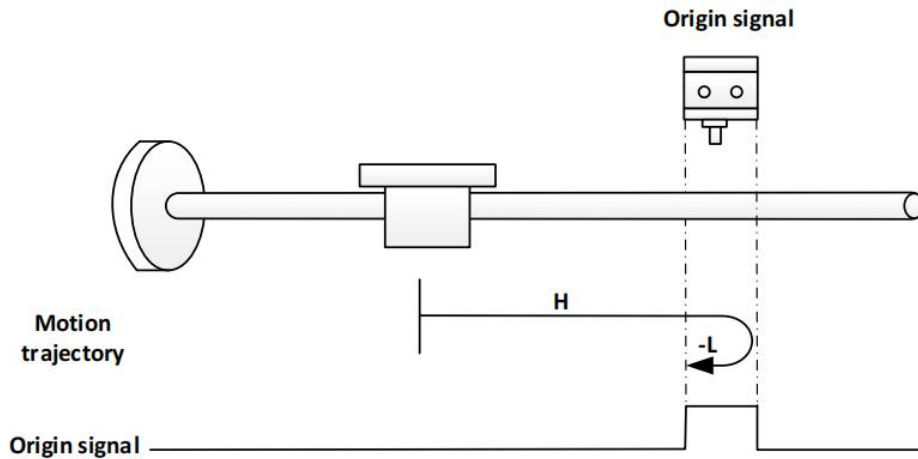


※ **Method 19 (0031h = 19)**

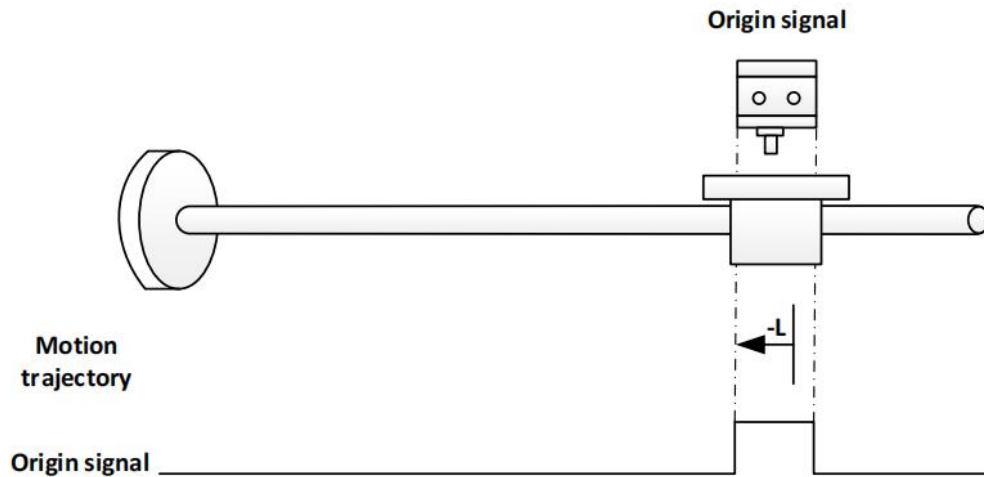
Origin: Origin signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop after encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the falling edge of the deceleration point ;



b) The deceleration point signal is valid when returning to zero and starting : the machine will run in the reverse direction at a low speed and stop when it encounters the falling edge of the deceleration point ;

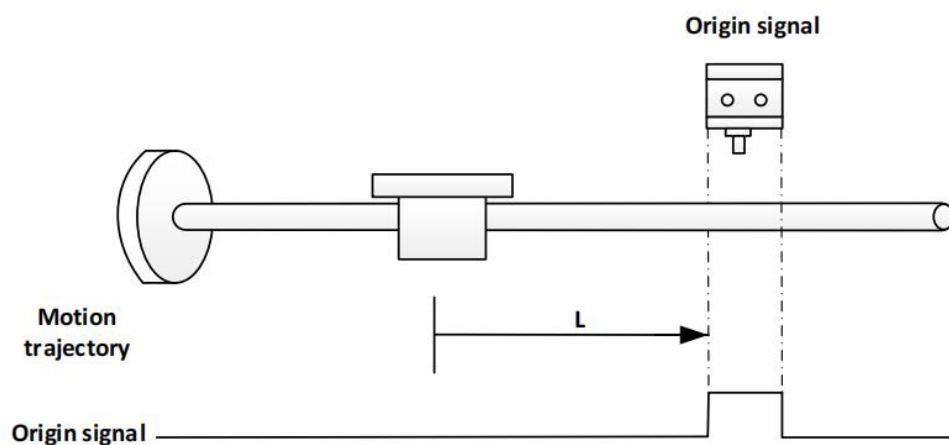


※ **Method 20 (0031h = 20)**

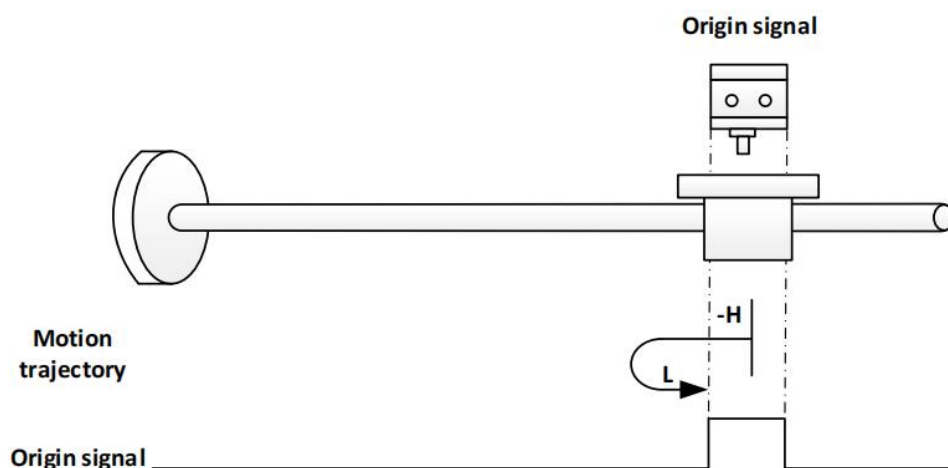
Origin: Origin signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : the machine runs at a low speed in the forward direction and stops after encountering the rising edge of the deceleration point;



b) The deceleration point signal is valid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop after encountering the falling edge of the deceleration point; when running in the forward direction at low speed, it will stop after encountering the rising edge of the deceleration point;

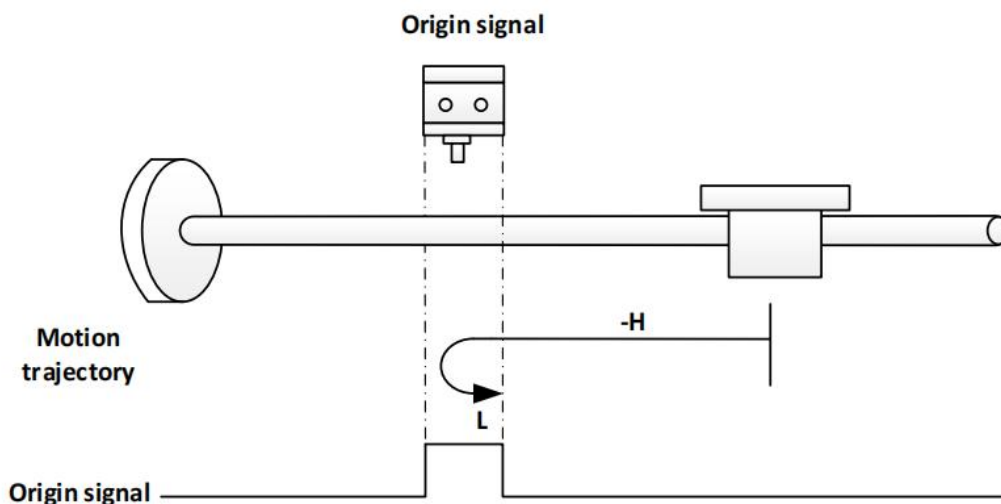


※ **Method 21 (0031h = 21)**

Origin: Origin signal

Deceleration point: origin signal

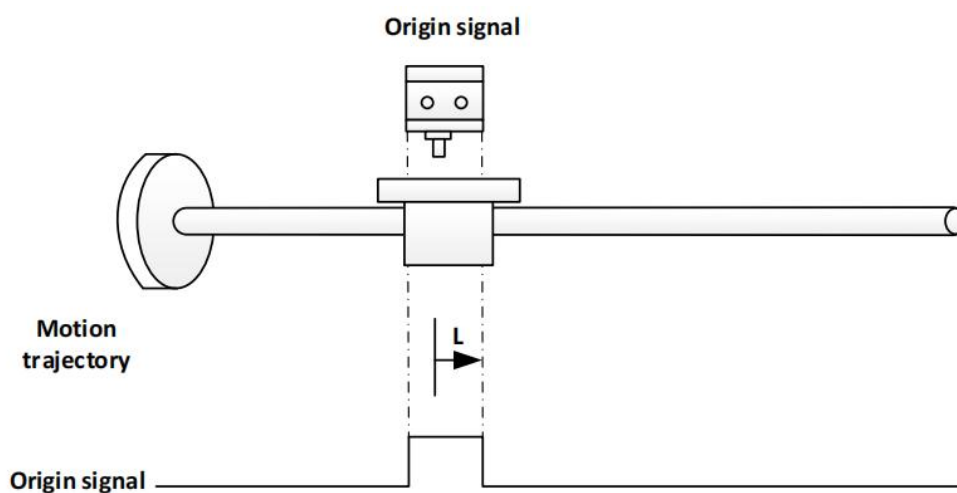
a) The deceleration point signal is invalid when returning to zero and starting : the machine runs at high speed in the reverse direction, and stops after encountering the rising edge of the deceleration point; the machine runs at low speed in the forward direction, and stops after encountering the falling edge of the deceleration point ;



Origin: Origin signal

Deceleration point: origin signal

a) The deceleration point signal is valid when starting from zero : the machine runs at a low speed in the forward direction and stops after encountering the falling edge of the deceleration point ;

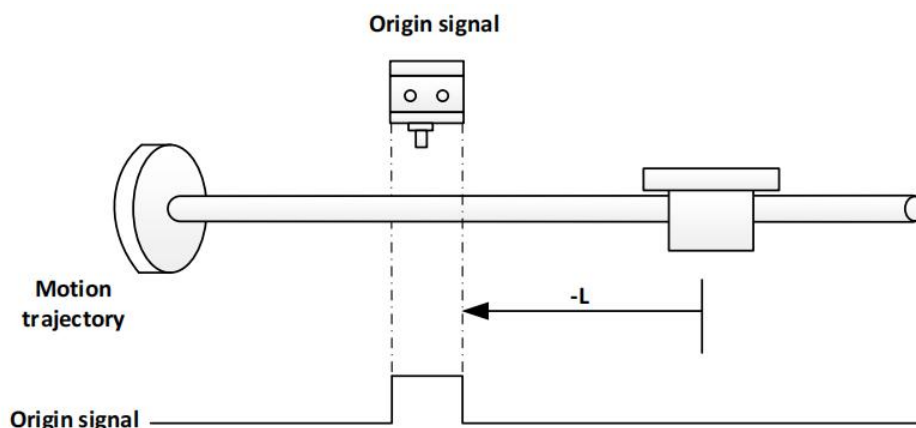


※ Method 22 (0031h = 22)

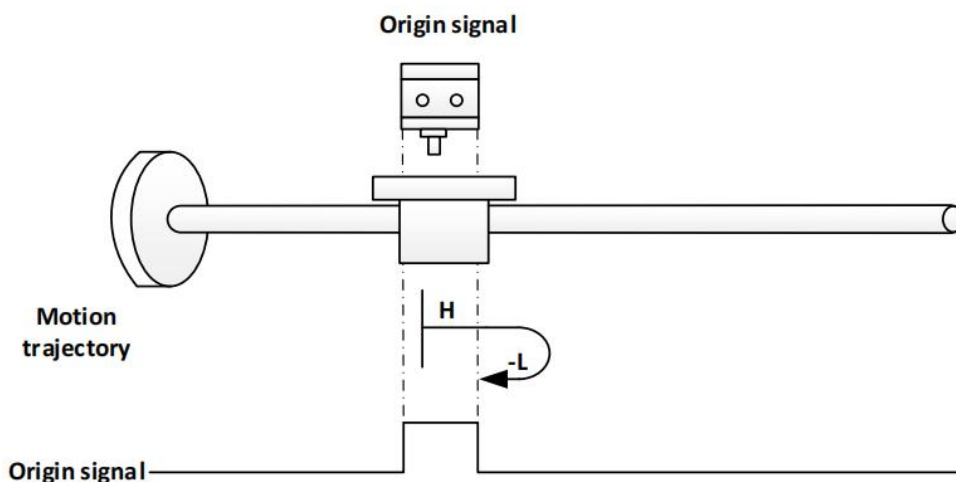
Origin: Origin signal

Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : the machine will run in the reverse direction at a low speed and stop after encountering the rising edge of the deceleration point;



b) The deceleration point signal is valid when returning to zero and starting : when running at high speed in the forward direction , it will decelerate and stop after encountering the falling edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the rising edge of the deceleration point;

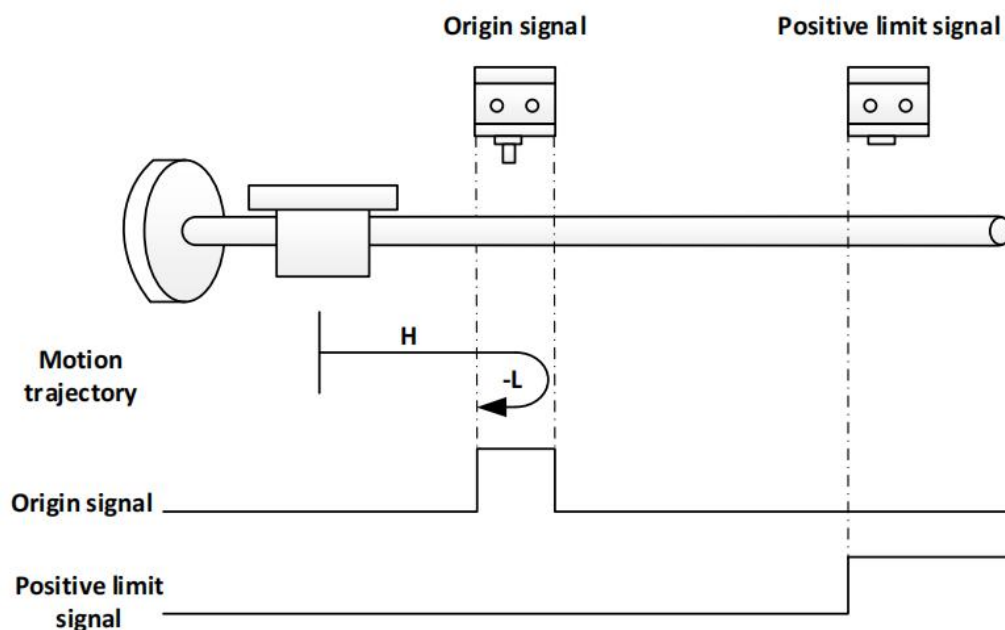


※ Method 23 (0031h = 23)

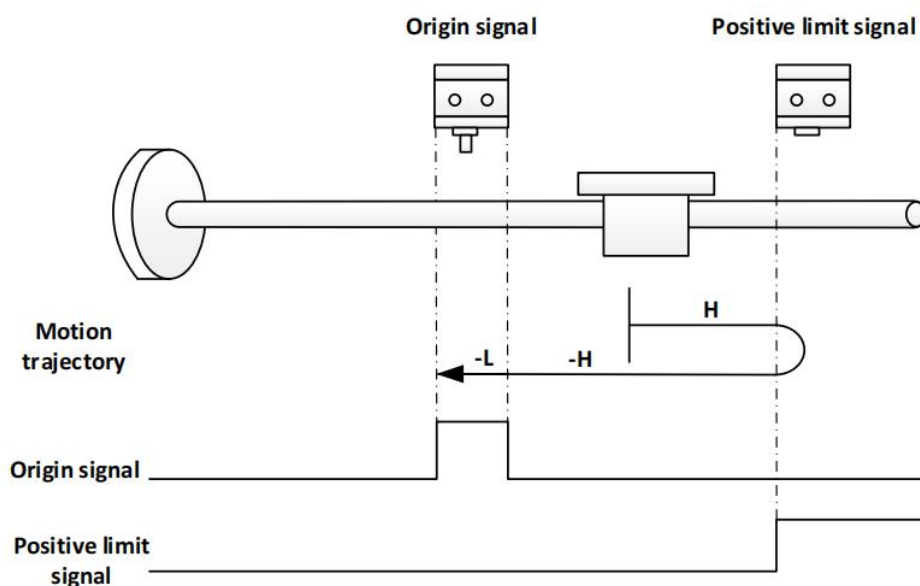
Origin: Origin signal

Deceleration point: origin signal

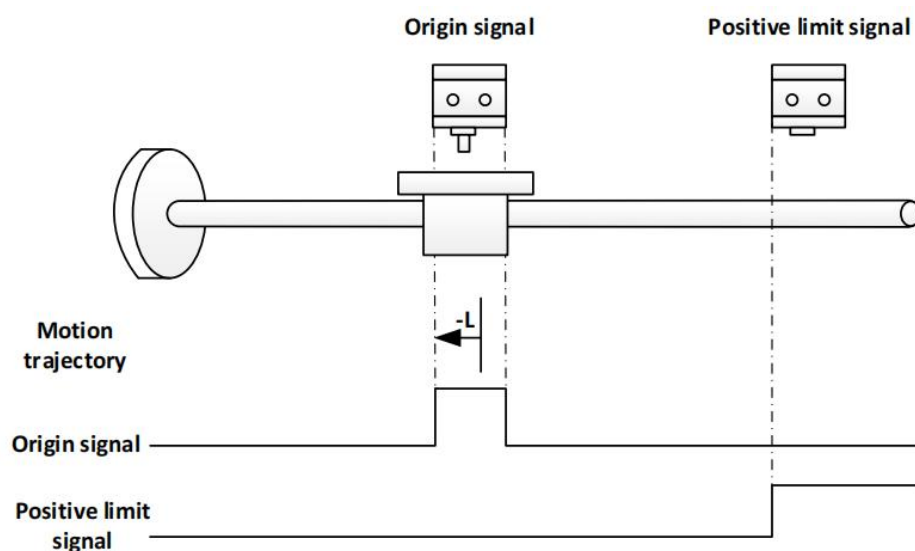
a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop after encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will stop after encountering the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the forward direction, and decelerates and stops when encountering the positive limit signal; it runs at high speed in the reverse direction, and runs at low speed when encountering the rising edge of the deceleration point, and stops when encountering the falling edge of the deceleration point ;



c) The deceleration point signal is valid when returning to zero and starting : the machine will run in the reverse direction at a low speed and stop after encountering the falling edge of the deceleration point ;

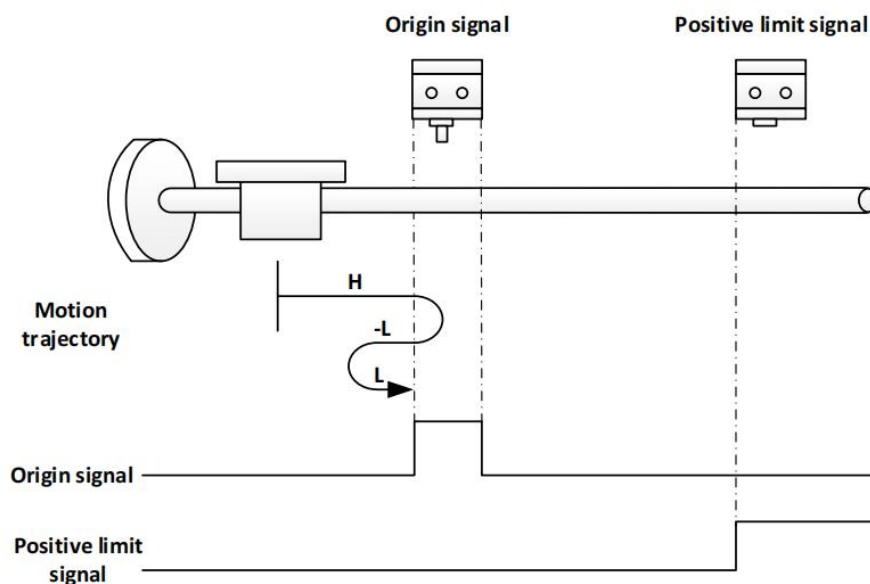


※ Method 24 (0031h = 24)

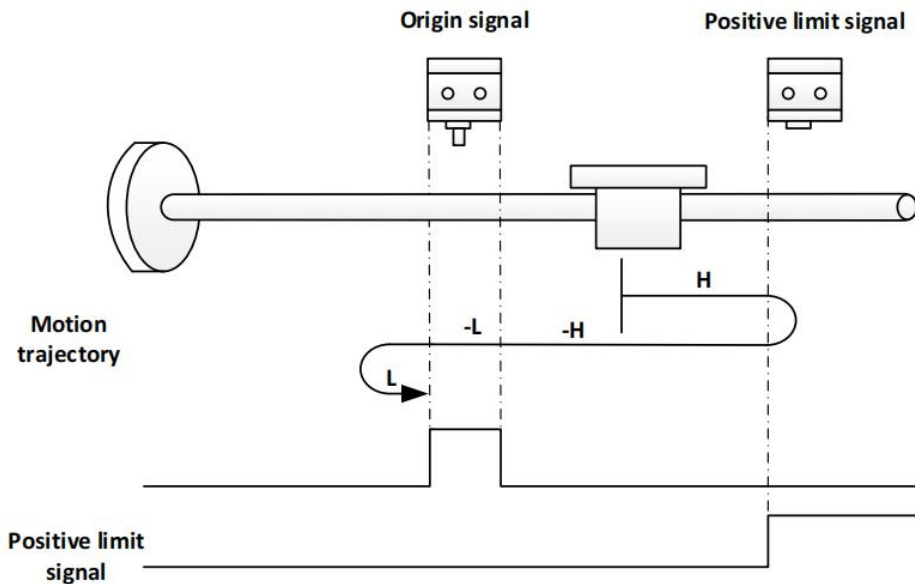
Origin: Origin signal

Deceleration point: origin signal

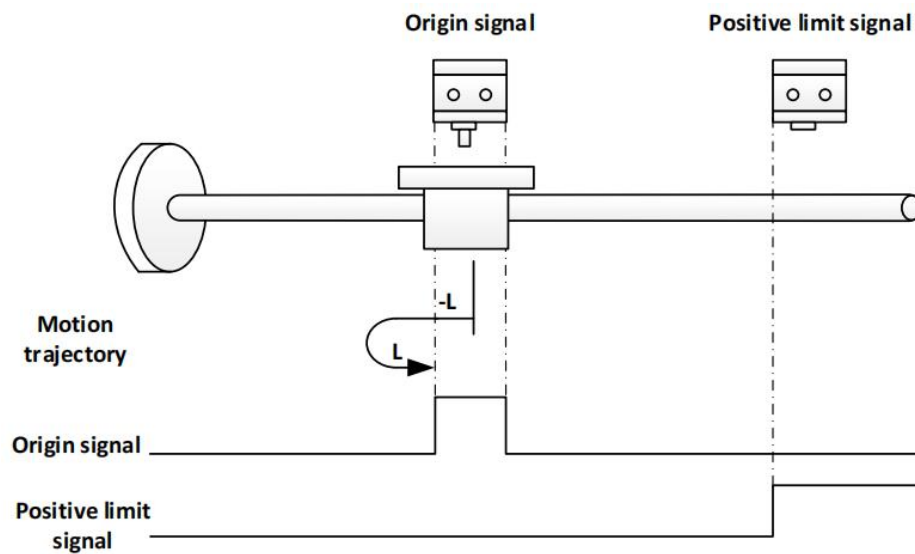
a) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the reverse direction, it will decelerate and stop when encountering the falling edge of the deceleration point; when running at low speed in the forward direction, it will stop when encountering the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will run at low speed when encountering the rising edge of the deceleration point, and decelerate and stop when encountering the falling edge of the deceleration point ; when running at low speed in the forward direction, it will stop after encountering the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running in the reverse direction at a low speed , the machine will decelerate and stop when encountering the falling edge of the deceleration point; when running in the forward direction at a low speed, the machine will stop when encountering the rising edge of the deceleration point;

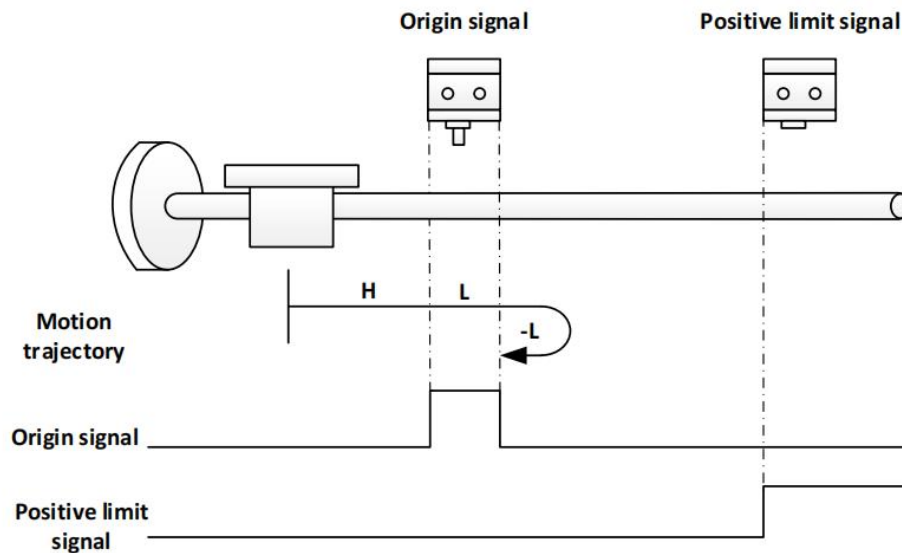


※ Method 25 (0031h = 25)

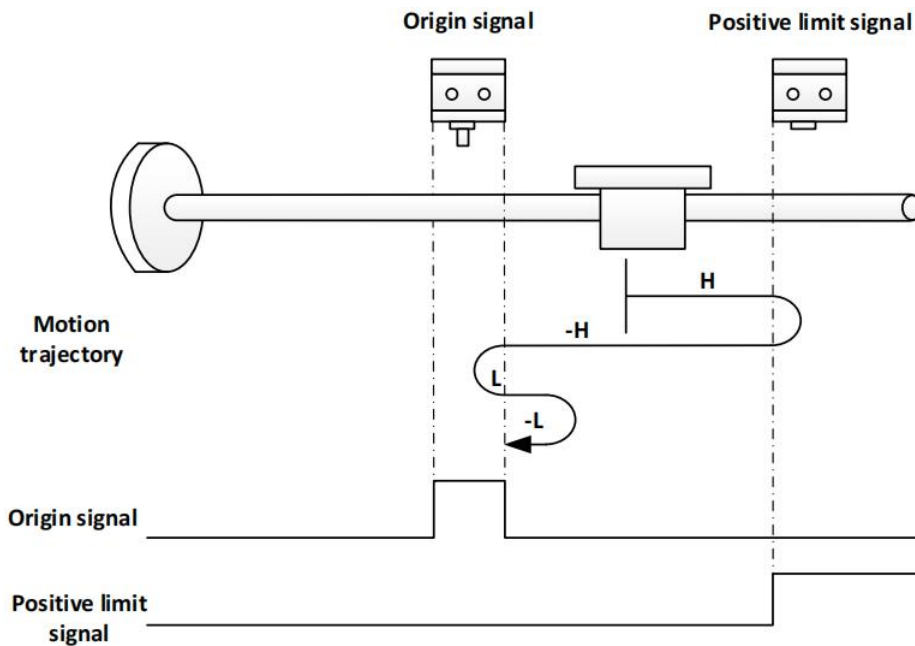
Origin: Origin signal

Deceleration point: origin signal

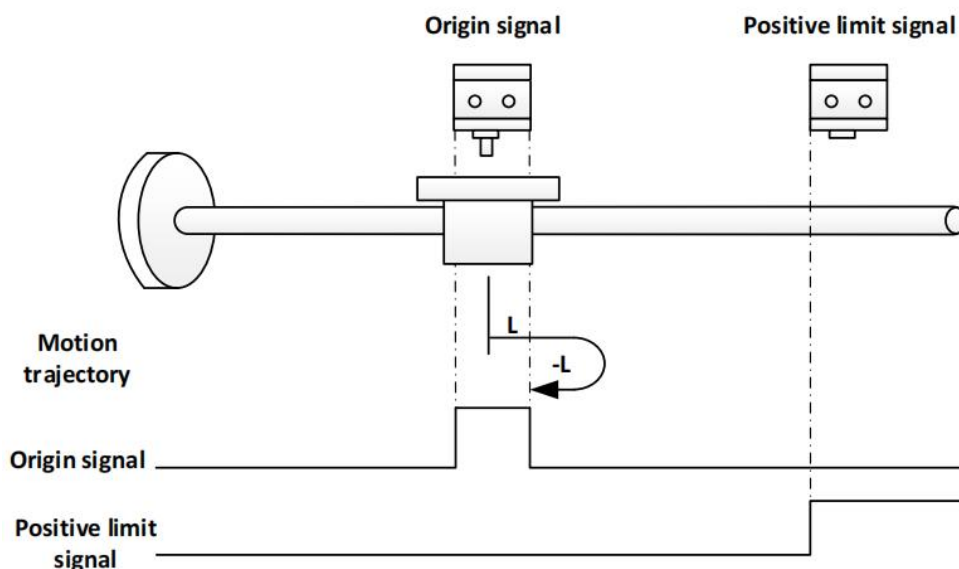
a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the forward direction, runs at low speed when encountering the rising edge of the deceleration point, and decelerates and stops when encountering the falling edge of the deceleration point ; it runs at low speed in the reverse direction, and stops when encountering the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the forward direction, it will decelerate and stop when encountering the falling edge of the deceleration point ; when running at low speed in the reverse direction, it will stop after encountering the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running at low speed in the forward direction , the machine will decelerate and stop when encountering the falling edge of the deceleration point; when running at low speed in the reverse direction, the machine will stop when encountering the rising edge of the deceleration point;

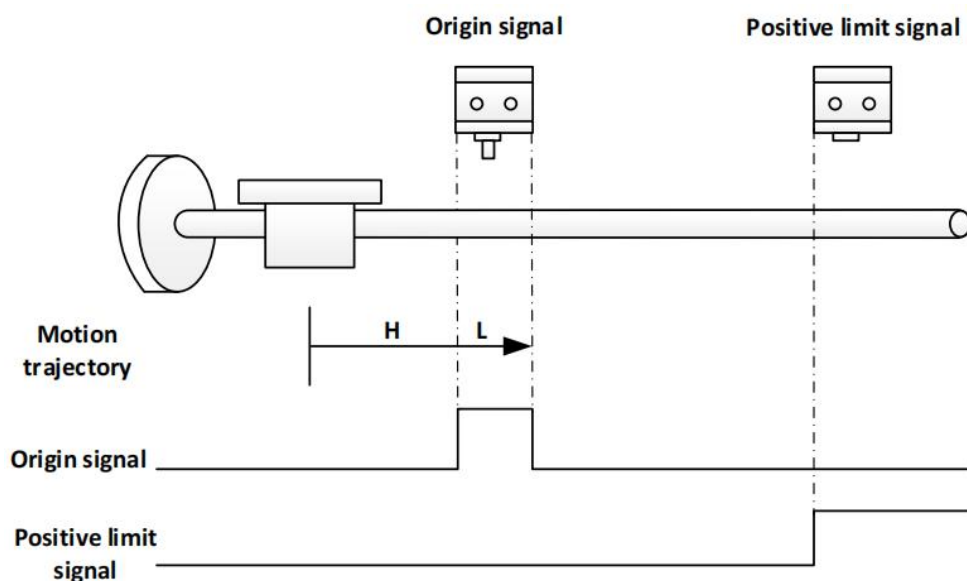


※ **Method 26 (0031h = 26)**

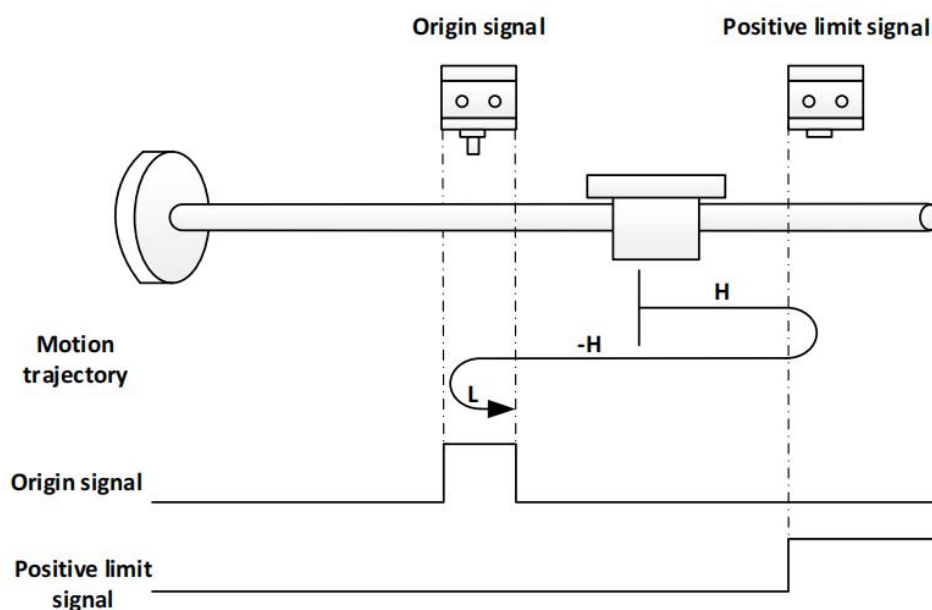
Origin: Origin signal

Deceleration point: origin signal

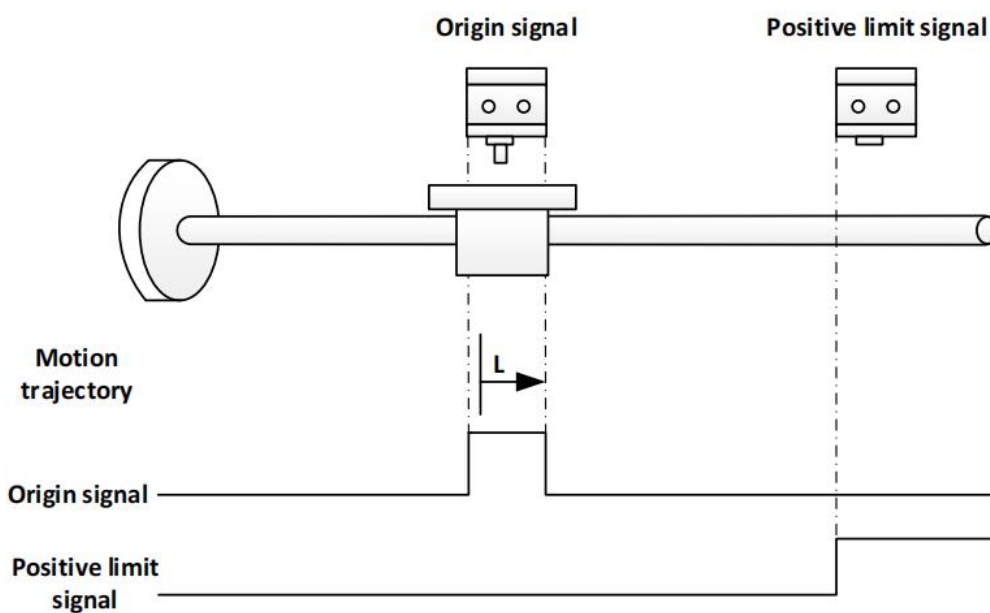
a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the forward direction, runs at low speed when encountering the rising edge of the deceleration point, and stops when encountering the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : when running at high speed in the forward direction, it will decelerate and stop when encountering the positive limit signal; when running at high speed in the reverse direction, it will decelerate and stop when encountering the rising edge of the deceleration point; when running at low speed in the forward direction, it will stop after encountering the falling edge of the deceleration point ;



c) The deceleration point signal is invalid when returning to zero and starting : the machine runs at a low speed in the forward direction and stops after encountering the falling edge of the deceleration point ;

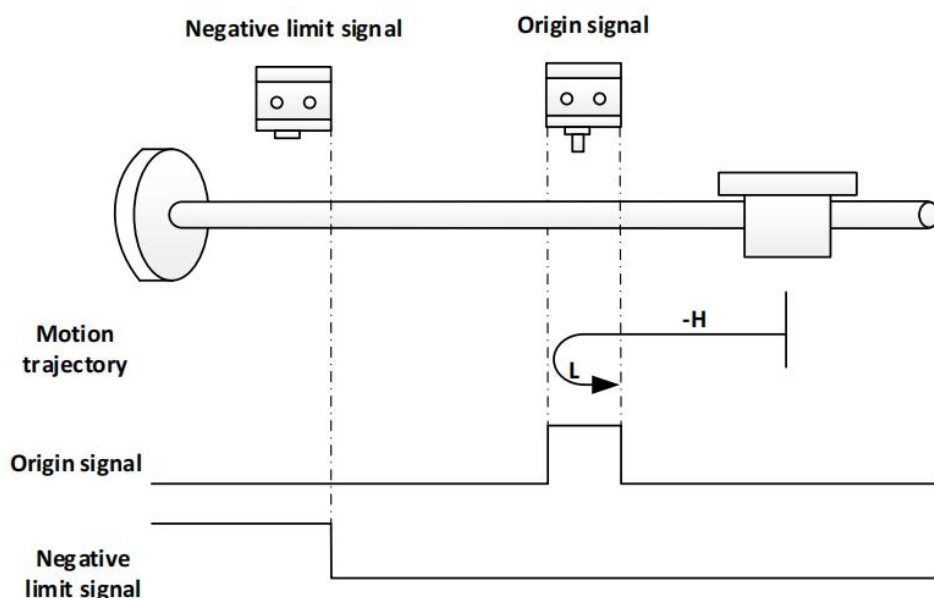


※ **Method 27 (0031h = 27)**

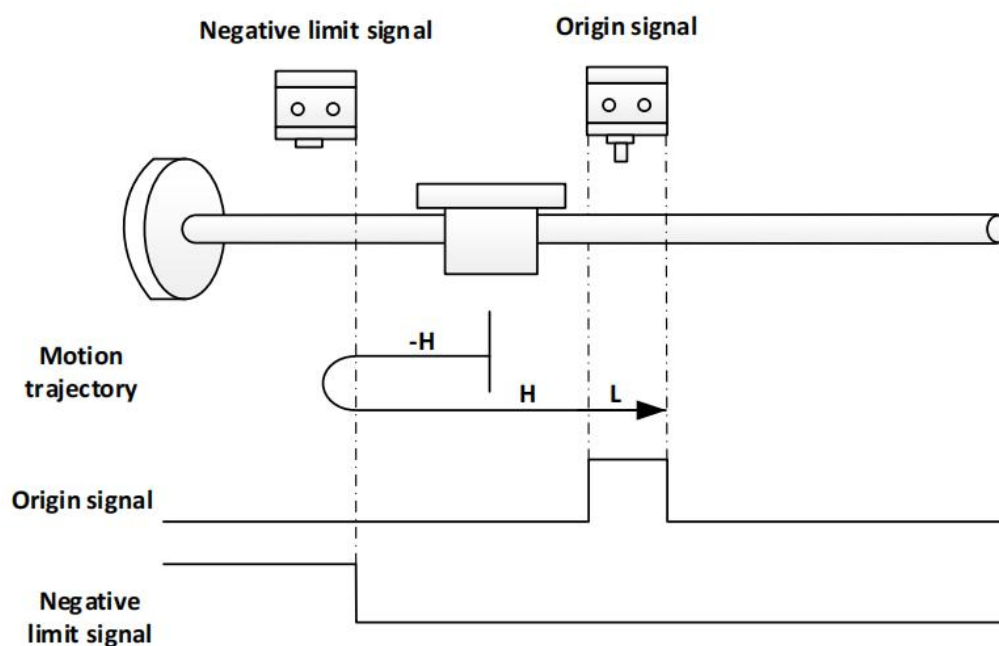
Origin: Origin signal

Deceleration point: origin signal

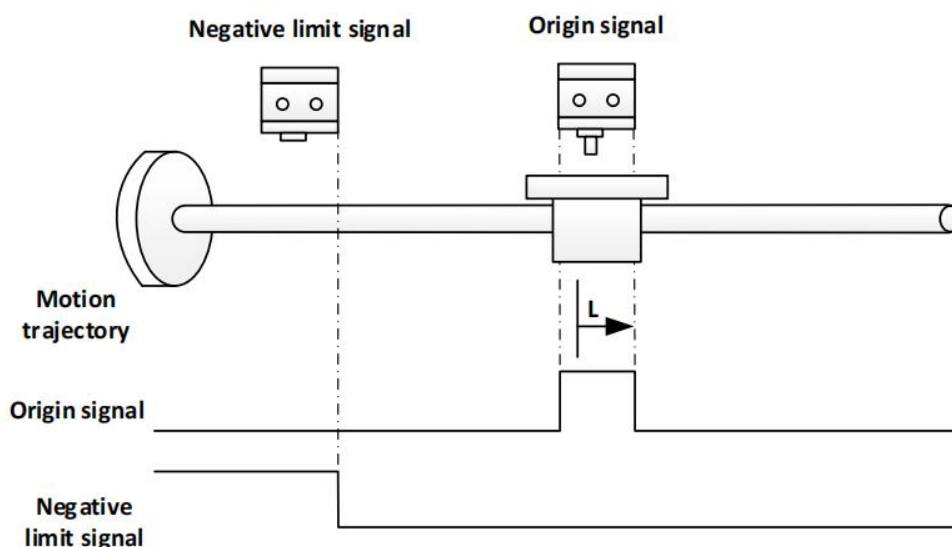
a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, and stops when it encounters the rising edge of the deceleration point; it runs at low speed in the forward direction, and stops when it encounters the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, and stops when it encounters the negative limit ; it runs at high speed in the forward direction, and runs at low speed when it encounters the rising edge of the deceleration point, and stops when it encounters the falling edge of the deceleration point ;



c) The deceleration point signal is valid when returning to zero and starting : the machine runs at a low speed in the forward direction and stops after encountering the falling edge of the deceleration point ;

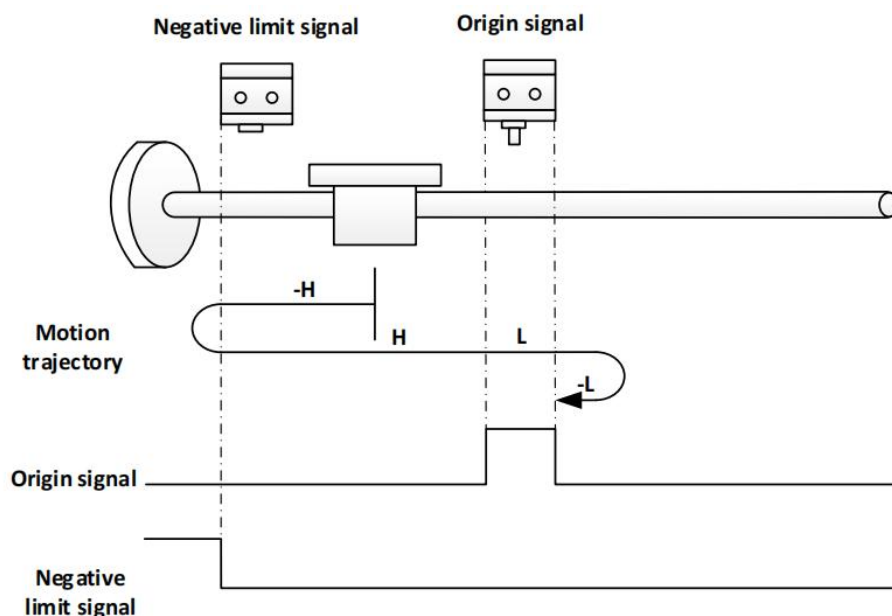


※ Method 28 (0031h = 28)

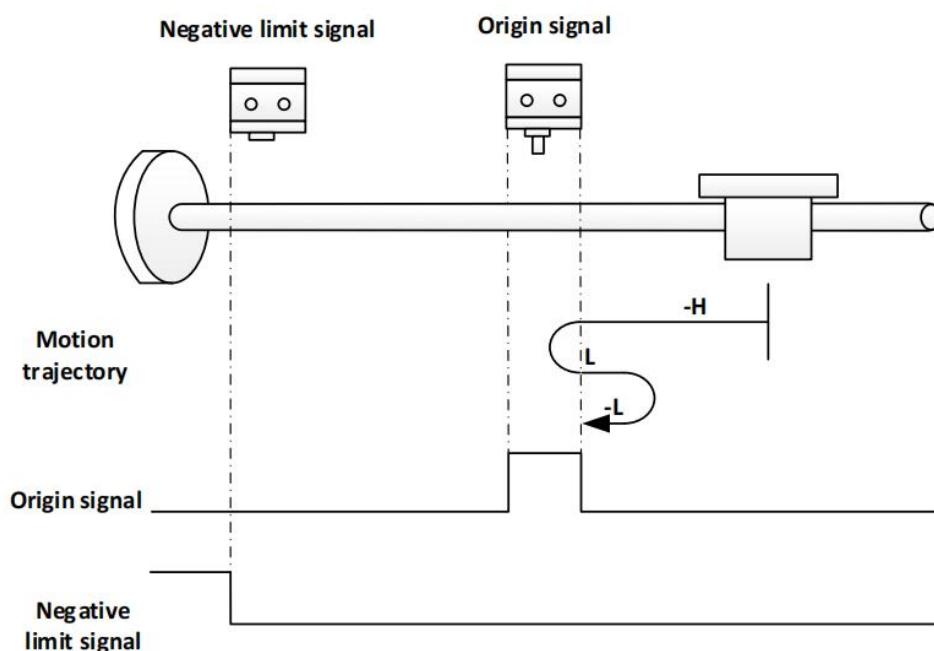
Origin: Origin signal

Deceleration point: origin signal

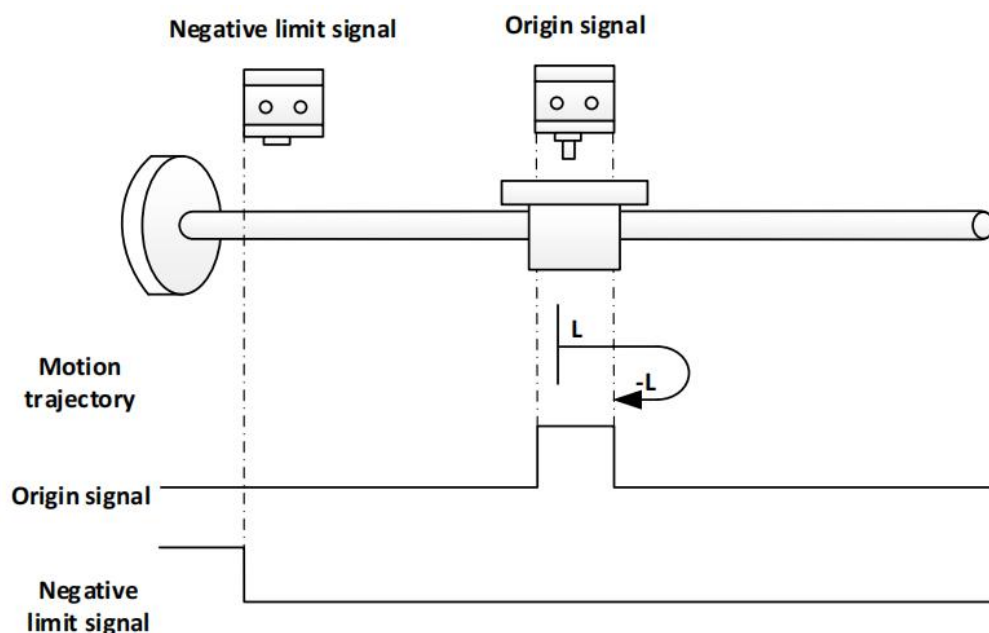
a) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the forward direction at low speed, it will decelerate and stop when encountering the falling edge of the deceleration point; when running in the reverse direction at low speed, it will stop after encountering the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting: when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will run at low speed when encountering the rising edge of the deceleration point, and decelerate and stop when encountering the falling edge of the deceleration point; when running in the reverse direction at low speed, it will stop after encountering the rising edge of the deceleration point;



c) The deceleration point signal is valid when returning to zero and starting : when running at low speed in the forward direction , the machine will decelerate and stop when encountering the falling edge of the deceleration point; when running at low speed in the reverse direction, the machine will stop when encountering the rising edge of the deceleration point;



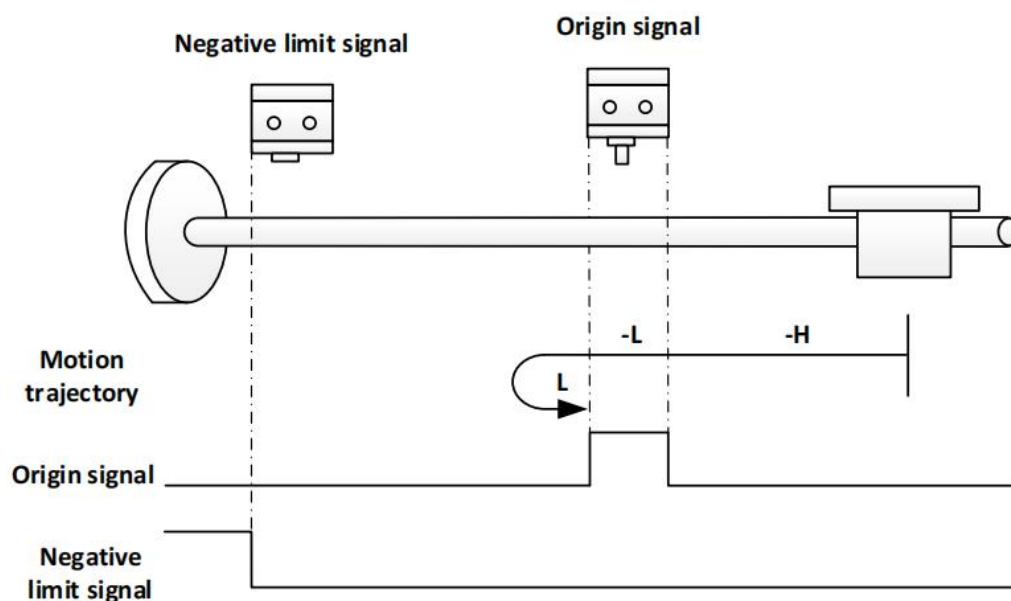
※ Method 29 (0031h = 29)

Origin: Origin signal

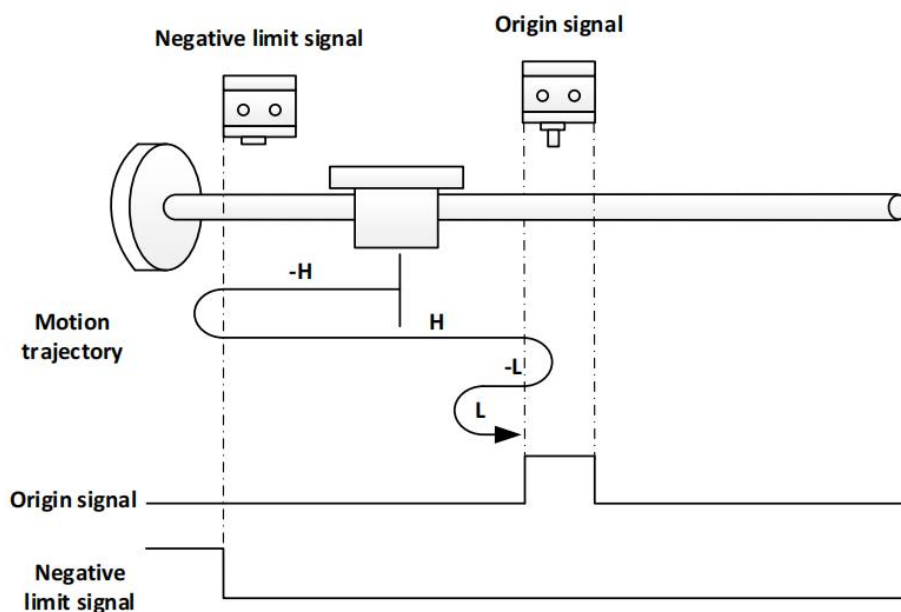
Deceleration point: origin signal

a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, runs at low speed when encountering the rising edge of the deceleration point, and decelerates and stops when encountering the falling edge of the deceleration point ; it runs at

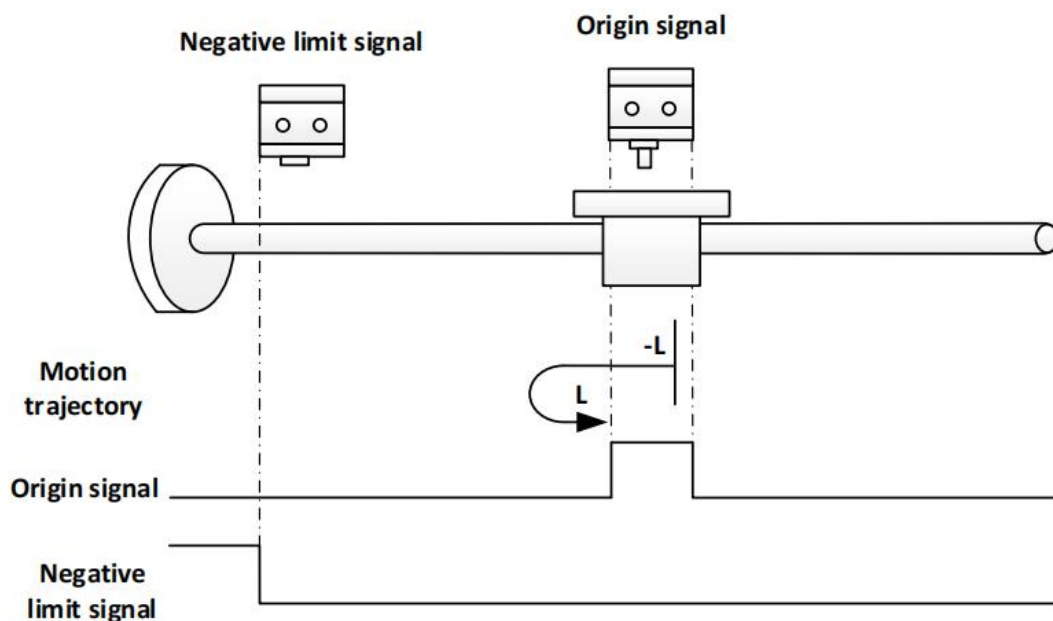
low speed in the forward direction, and stops after encountering the rising edge of the deceleration point;



b) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed, it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the reverse direction at low speed, it will decelerate and stop when encountering the falling edge of the deceleration point ; when running in the forward direction at low speed, it will stop after encountering the rising edge of the deceleration point;



c) The deceleration point signal is invalid when returning to zero and starting : it runs at a low speed in the reverse direction , and stops when it encounters the falling edge of the deceleration point; it runs at a low speed in the forward direction, and stops when it encounters the rising edge of the deceleration point;

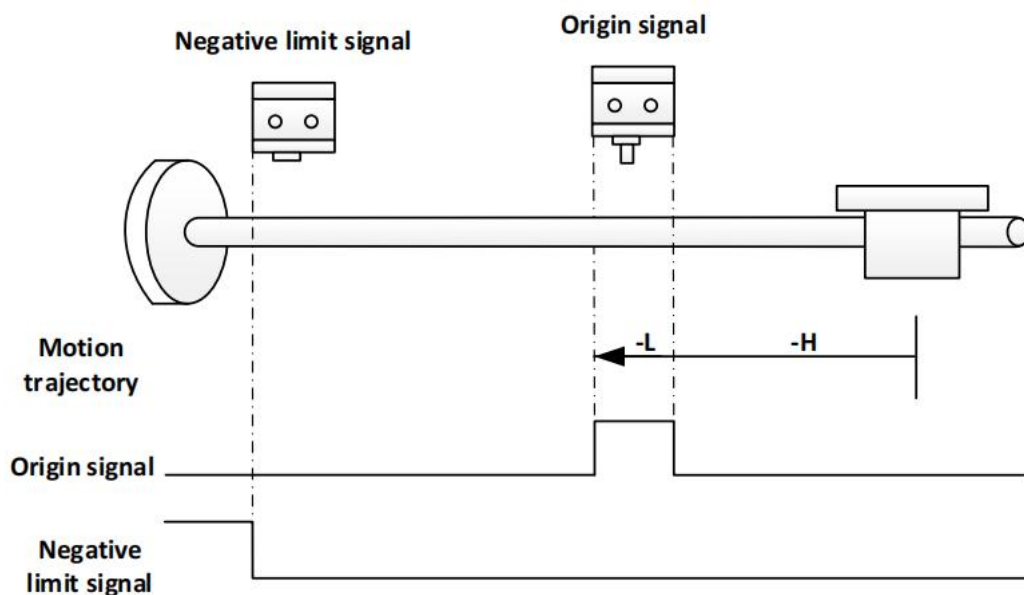


※ **Method 30 (0031h = 30)**

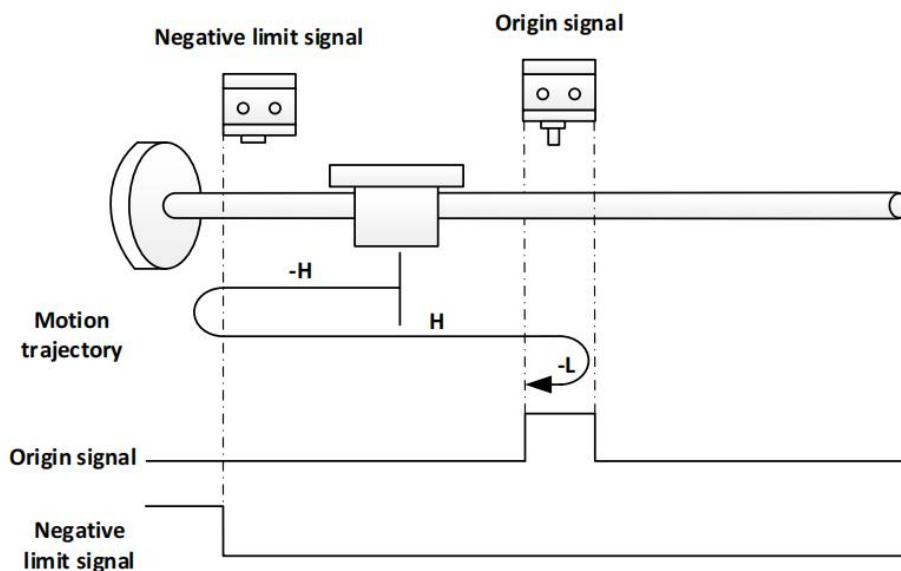
Origin: Origin signal

Deceleration point: origin signal

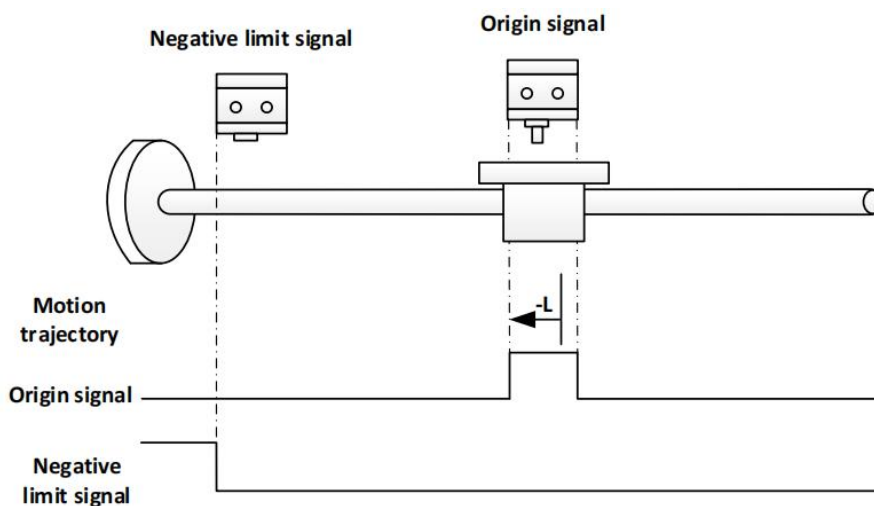
a) The deceleration point signal is invalid when returning to zero and starting : it runs at high speed in the reverse direction, runs at low speed when encountering the rising edge of the deceleration point, and stops when encountering the falling edge of the deceleration point ;



b) The deceleration point signal is invalid when returning to zero and starting : when running in the reverse direction at high speed , it will decelerate and stop when encountering the negative limit; when running in the forward direction at high speed, it will decelerate and stop when encountering the rising edge of the deceleration point; when running in the reverse direction at low speed, it will stop after encountering the falling edge of the deceleration point ;



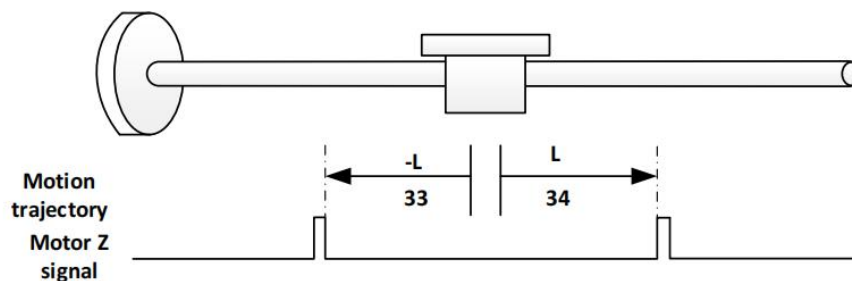
c) The deceleration point signal is invalid when returning to zero and starting : the machine will run in the reverse direction at a low speed and stop after encountering the falling edge of the deceleration point ;



※ Method 33 and Method 34 (0031h = 33 and 34)

Origin: Motor Z signal

Slowdown point: None



※ Method 35 (0031h = 35)

Return to zero mode 35, with the current position as the mechanical origin;

Appendix 2: Modbus Register Parameter Table - ESS-RS Series

Register Address	project	illustrate	Setting range Note: Other values are invalid.	default value
Status parameter group (read-only)				
0x0000	Driver Model	Drive model code	(RO)	0x0305
0x0001	Driver version	Driver version	(RO)	0x010 0
0x0002	Drive Node Number	MODBUS current communication slave node number	(RO)	-
0x0003	DIP status code	Bit0~Bit6: SW1~SW7 status; 0: OFF; 1:ON;	(RO)	-
0x0006	Current error code	0 : Normal; 1~5 : Error;	(RO)	-
0x0007	Motion status bit	Bit0: In place flag; 0: not in place, 1: in place; Bit1: Return to origin completed bit; 0: not completed, 1: completed; Bit2: Motor running position; 0: stationary, 1: running; Bit3: alarm bit; 0: normal, 1: alarm; Bit4: Motor enable bit; 0: enable, 1: release; Bit5: Forward soft limit overtravel flag ; 0: invalid; 1: valid; Bit6: Negative soft limit overtravel flag ; 0: invalid; 1: valid;	(RO)	-
0x0008	Input terminal status flag	Bit 0 : X0 terminal input status; Bit 1 : X1 terminal input status; Bit 2 : X2 terminal input status; Bit 3 : X3 terminal input status; Bit 4 ~Bit15 : Reserved;	(RO)	-

		0 : Input level is invalid; 1 : Input level is valid;		
0x0009	Output terminal status flag	Bit0: Y0 terminal output status; Bit1: Y1 terminal output status; Bit 2 ~Bit15: reserved; 0 : Output level is invalid; 1: Output level is valid;	(RO)	-
0x000A	Current position high	The current position value of the motor. For an open-loop motor, this register value is the position given value. For a closed-loop motor, this register value is the subdivision equivalent of the encoder feedback value.	(RO)	-
0x000B	Current position low		(RO)	-
0x000C	Current speed	Current motor running speed	(RO)	-
Driver basic control parameter group 1				
0x0010	Default movement direction	0 : default direction ; 1 : Reverse direction ;	0~ 1 (RW)	0
0x0011	Segment settings	Address - Segment 400~51200;	400~51200 (RW)	1000
0x0013	Custom drive node number	0 ~ 255 : custom slave address; This register takes effect when all the drive address DIP switches are OFF;	0~ 255 (RW)	0
0x0014	Customize communication baud rate	0:115200 1 : 38400 2 : 19200 3 : 9600 Note: After modification, power must be turned on again to take effect;	0~ 3 (RW)	0
0x0015	Serial port data format	0 : 8 -bit data, no parity, 1 stop bit; 1 : 8 -bit data, no parity, 2 stop bits; 2 : 8 -bit data, even parity, 1 stop bit; 3 : 8 -bit data, odd parity, 1 stop bit; Note: After modification, power must be turned on again to take effect;	0~ 3 (RW)	0

0x0017	Over-limit parking method	0 : free parking; 1 : Emergency stop;	0~ 1 (RW)	0
0x0018	Internal software limit switches	0: invalid; 1: Take effect after returning to zero;	0~ 1 (RW)	0
0x0019	32-bit register endianness configuration	0: high position first, low position last; 1: High position at the back, low position at the front; Adapt to different PLC or touch screen usage habits;	0~ 1 (RW)	0
Basic motion control parameters of the drive				
0x001D	Jog mode operation speed	Jog mode running speed, unit: r/min;	-3000 -300 0 r/min (RW)	120 (5r/min)
0x001E	Jog mode acceleration time	Jog mode acceleration time, in ms;	0-2000ms (RW)	50 (100ms)
0x001F	Jog mode deceleration time	Jog mode deceleration time, unit: ms;	0-2000ms (RW)	50 (100ms)
0x0020	Starting speed	The starting speed of the positioning movement , in r/min ;	0 ~3000r/min (RW)	30 (60r/min)
0x0021	Acceleration time	Acceleration time , in ms;	0-2000ms (RW)	50 (100ms)
0x0022	Deceleration time	Deceleration time , in ms;	0-2000ms (RW)	50 (100ms)
0x0023	Positioning movement speed	Movement speed during positioning movement , unit: r/min ;	0 ~3000r/min (RW)	60 (60r/min)
0x0024	Total pulse count high	The total number of pulses for positioning motion operation is a 32-bit register. If 100000 pulses are set, the high bit is 0x0001 , the low bit is 0x86A0 , and the value written to the register is 0x000186A0;	- 0xFFFFFFFF ~ 0xFFFFFFFF (RW)	5000
0x0025	Total pulse count low			
0x0027	Movement Control Order	Bit0: Position mode start command bit; 0; invalid; 1: Valid; Bit1: Speed mode start command bit; 0; invalid; 1: Valid;	0~65535 (WO)	-

		Bit2: Position mode positioning method; 0: relative positioning; 1: absolute positioning; Bit3: Sports mode switching method; 0: Ignore motion commands; 1: interrupt the current motion and execute it immediately; Bit4: Return to origin start command bit; 0: invalid; 1: Valid; Bit8: stop command bit; 0: invalid; 1: Valid; Bit9: Emergency stop command bit; 0: invalid; 1: Valid;		
0x002D	Auxiliary control instructions	0x0000: invalid; 0x0011: Motor release; 0x0012: Motor enable; 0x0021: drive alarm cleared; 0x0031: Clear the current position of the motor; 0x0041: Restore parameters to factory settings; 0x0042: save all parameters; Note: When performing factory recovery and parameter saving operations on the drive, it is necessary to ensure that the motor is in the stopped state, otherwise the relevant instructions will be ignored;	0~ 65535 (WO)	-
Drive return to zero motion control parameters				

0x0030	Zero return auxiliary setting	0: Run the offset, and after completion, the current position is the offset value; 1: Run the offset, and the current position is 0 after completion; 2 : Run the offset, and after completion the current position is a negative offset value; 3: Run the offset, and after completion, the current position is the actual value; 4: Run the offset. After completion, the current position is the actual value plus the offset value. 5: Run the offset. After completion, the current position is the actual value minus the offset value. 6: Do not run the offset, and the current position after completion is the offset value; 7 : Do not run the offset, the current position is 0 after completion; 8: Do not run the offset, and the current position will be a negative offset value after completion; 9: Do not run the offset, and the current position is the actual value after completion; 1 0 : Do not run the offset. After completion, the current position is the actual value plus the offset value. 1 1 : Do not run the offset. After completion, the current position is the actual value minus the offset value.	0 ~11 (RW)	0
0x0031	Return to origin mode	Support -1~-4 , 1~14, 17~30, 33~35 modes;	0~ 65535 (RW)	twenty four
0x0032	Return to origin speed	Running speed when querying the origin position;	5-3000r/min (RW)	120 (60r/min)

0x0033	Return to origin query speed	The return speed after querying the origin;	5-300 r/min (RW)	60 (60r/min)
0x0034	Acceleration and deceleration time when returning to origin	The acceleration and deceleration time when querying the origin position;	30-2000ms (RW)	50 (100ms)
0x0035	Origin offset value high	Origin offset value: after finding the origin sensor, it moves to the correct position according to the value set in this register .	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0036	Origin offset value low			
0x0037	Soft limit positive high position	Software positive limit setting point, this limit point will take effect only after zero return is completed ;	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0038	Soft limit positive low position			
0x0039	Soft limit negative high position	Software negative limit setting point, this limit point will take effect only after zero return is completed ;	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x003A	Soft limit negative low position			
0x003B	Collision return threshold	the zero return mode is set to -1 ~-4 , if the position error of the stall is greater than this value, it is judged as a valid collision;	5 0~4000	200
0x003C	Collision return current percentage	the zero return mode is set to -1 ~-4 and the zero return motion is started, the motion current will be reduced to the set percentage;	2 0~100	50
Input and output terminal parameter group				
0x0040	Input terminal effective level	Bit 0 : Input terminal X0 control bit; Bit 1 : Input terminal X1 control bit; Bit 2 : Input terminal X2 control bit; Bit 3 : Input terminal X3 control bit; Bit 4 ~Bit15 : Reserved; 0 : default; 1 : Level inversion;	0~65535 (RW)	0
0x0041	Input terminal X0 terminal function	0 : undefined; 1 : origin signal;	0~17 (RW)	1

	selection	2 : Positive limit signal; 3 : Anti-limit signal;		
0x004 2	Input terminal X1 terminal function selection	4 : Motor MF signal; 5 : Stop signal; 6 : Emergency stop signal;	0~17 (RW)	2
0x004 3	Input terminal X2 terminal function selection	7 : Position mode movement; 8: Speed mode movement; 9 : JOG+ point motion;	0~17 (RW)	3
0x004 4	Input terminal X3 terminal function selection	10 : JOG -point movement; 1 1 : Return to origin enable signal; 1 2 : PT trigger signal ; 13: PV trigger signal; 1 4 : PIN0 ; 1 5 : PIN1 ; 1 6 : PIN2 ; 1 7 : PIN3 ;	0~17 (RW)	0
0x004B	Output terminal effective level	Bit0 : Output terminal Y0 control bit; Bit1 : Output terminal Y1 control bit; Bit 2 ~Bit15 : Reserved; 0 : default; 1 : Level inversion;	0~65535 (RW)	0
0x004C	Output terminal Y0 terminal function selection	0: undefined ; 1: Alarm signal; 2: Driver status signal;	0~11 (RW)	0
0x004D	Output terminal Y1 terminal function selection	3: Return to origin completion signal; 4: Arrival signal; 5: Braking signal; 9: User defined 0; 10: User defined 1; 11: User defined 2;	0~11 (RW)	0
0x004F	Y group terminal custom output	Bit0 : Y0 terminal output status; Bit1 : Y1 terminal output status; Bit 2 ~Bit15 : Reserved; 0 : Output is invalid; 1 : Output is valid;	0~65535 (RW)	0
Multi-stage positioning/speed control parameter group				

0x0050	Relative position / absolute position	When using IO to control positioning motion / multisegment position control , this bit is valid: 0 : relative position; 1 : absolute position;	0~1 (RW)	0
0x0051	PV trigger signal level selection	0: level is valid; 1: rising edge is valid;	0~1 (RW)	0
0x0060	The first segment motion pulse command high	The first section of motion pulse instruction is a 32-bit register. If 100000 pulses are set, the high bit is 0x0001, the low bit is 0x86A0, and the value written into the register is 0x000186A0.	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0061	The first segment motion pulse command low			500 0
0x0062	Positioning speed of the first stage	the first position positioning, unit: r/min;	0 -3000 r/min (RW/S)	120 (0/min)
0x0063	1st stage positioning motion acceleration	The first segment position positioning motion acceleration, in ms;	0 -2000ms (RW/S)	5 0 (0ms)
0x0064	1st stage position positioning motion deceleration	The first stage of the positioning motion deceleration, unit: ms;	0 -2000ms (RW/S)	5 0 (0ms)
0x0065	reserve	reserve	0 (RW/S)	0
0x0066 ~0x006B	Parameters of Section 2	Refer to 0x0060~0x0065;	-	-
0x006C~0x0071	Parameters of Section 3	Refer to 0x0060~0x0065;	-	-
0x0072~0x0077	Parameters of Section 4	Refer to 0x0060~0x0065;	-	-
0x0078~0x007D	Parameters of Section 5	Refer to 0x0060~0x0065;	-	-
0x007E~0x0083	Parameters of Section 6	Refer to 0x0060~0x0065;	-	-
0x0084~0x0089	Parameters of Section 7	Refer to 0x0060~0x0065;	-	-
0x008A~0x008F	Parameters of Section 8	Refer to 0x0060~0x0065;	-	-

0x0090~0x0095	Parameters of Section 9	Refer to 0x0060~0x0065;	-	-
0x0096~0x009B	Parameters of Section 10	Refer to 0x0060~0x0065;	-	-
0x009C~0x00A1	Parameters of Section 11	Refer to 0x0060~0x0065;	-	-
0x00A2~0x00A7	Parameters of Section 12	Refer to 0x0060~0x0065;	-	-
0x00A8~0x00AD	Parameters of Section 13	Refer to 0x0060~0x0065;	-	-
0x00AE~0x00B3	Parameters of Section 14	Refer to 0x0060~0x0065;	-	-
0x00B4~0x00B9	Parameters of Section 15	Refer to 0x0060~0x0065;	-	-
0x00BA~0x00BF	Parameters of Section 16	Refer to 0x0060~0x0065;	-	-
0x00 C0	1st speed segment running speed	running speed of the first speed segment is in r/min ;	-3000 -3000 r/min (RW/S)	10 0 (0/min)
0x00C1	1st speed segment acceleration time	The first speed segment motion acceleration, unit: ms;	0 -2000ms (RW/S)	5 0 (0ms)
0x00C2	1st speed stage deceleration time	The first speed segment is the deceleration of the movement, in ms;	0 -2000ms (RW/S)	5 0 (0ms)
0x00C3~0x00C5	Parameters of Section 2	Reference 0x00C0 ~0x00C2	-	-
0x00C6~0x00C8	Parameters of Section 3	Reference 0x00C0 ~0x00C2	-	-
0x00C9~0x00CB	Parameters of Section 4	Reference 0x00C0 ~0x00C2	-	-
0x00CC~0x00CE	Parameters of Section 5	Reference 0x00C0 ~0x00C2	-	-
0x00CF~0x00D1	Parameters of Section 6	Reference 0x00C0 ~0x00C2	-	-
0x00D2~0x00D4	Parameters of Section 7	Reference 0x00C0 ~0x00C2	-	-
0x00D5~0x00D7	Parameters of Section 8	Reference 0x00C0 ~0x00C2	-	-

0x00D8~0x00DA	Parameters of Section 9	Reference 0x00C0 ~0x00C2	-	-
0x00DB~0x00DD	Parameters of Section 10	Reference 0x00C0 ~0x00C2	-	-
0x00DE~0x00E0	Parameters of Section 11	Reference 0x00C0 ~0x00C2	-	-
0x00E1~0x00E3	Parameters of Section 12	Reference 0x00C0 ~0x00C2	-	-
0x00E4~0x00E6	Parameters of Section 13	Reference 0x00C0 ~0x00C2	-	-
0x00E7~0x00E9	Parameters of Section 14	Reference 0x00C0 ~0x00C2	-	-
0x00EA~0x00EC	Parameters of Section 15	Reference 0x00C0 ~0x00C2	-	-
0x00ED~0x00EF	Parameters of Section 16	Reference 0x00C0 ~0x00C2	-	-
0x0130~0x013F	Multi-segment motion starting speed	The starting speed of each segment of movement (shared by PT mode and PV mode)	-180~180 rpm (RW /S)	Or pm
Performance parameter group				
0x0100	Closed-loop algorithm	Drive operating mode; 1: open loop; 2 : Algorithm 1;	1 ~ 2 (RW/S)	2
0x0101	Encoder resolution	Encoder resolution, which is four times the encoder value, and the default is a 1000-line encoder;	0~ 65535 (RW/S)	4000
0x0102	Maximum effective current	The maximum current value output by the driver. The register value is given according to the rated current of the motor, in mA.	0~ 5600 (RW/S)	5600/220 0
0x0103	Closed loop maximum current percentage	The maximum current percentage in closed-loop mode , in % , when the driver is in closed-loop mode, the maximum current that can be output under rated load is the product of the 2042h register value and this register value;	0~150 (RW/S)	100

0x0104	Base current percentage	Registered current percentage , unit % , when the driver is running at no load, the output current value is the product of the 2042h register value and this register value;	0~ 75 (RW/S)	40
0x0105	Open loop maximum current percentage	The maximum open-loop current percentage, in % , when the driver is in open-loop mode, the maximum output current during operation is the product of the 2042h register value and this register value;	0~100 (RW/S)	100
0x0106	Lock current percentage	Lock current percentage, unit % . When the driver is in the locked state , the lock current value is the product of the 2042h register value and the register value.	0 - 100 (RW/S)	100
0x0107	Lock time	for the motor to switch to the lock state after it stops moving , in ms;	0~20000 (RW/S)	4000
0x0108	IO terminal filter coefficient	Terminal filter coefficient;	0 ~ 65535 (RW/S)	2
0x0109	Pulse low-pass filter coefficient	Pulse control command filter coefficient; Low-pass filter coefficient, the smaller the value, the more obvious the filtering effect;	0~1024 (RW/S)	5
0x010A	Position deviation alarm threshold	Position deviation alarm threshold ;	1~ 65535 (RW/S)	4000
0x010B	Positioning error range	Positioning error range ;	1~256 (RW/S)	5
0x010C	End time	End time of arrival ;	0~200 (RW/S)	1 0
0x010D	Pulse mean filter coefficient	Mean filter coefficient, the larger the value, the more obvious the filtering effect;	0~ 512 (RW/S)	512
0x010E	Current loop Kp gain multiple	Current loop Kp gain multiple ;	0~ 65535 (RW/S)	4096

0x010F	Current loop Kp gain	Current loop Kp gain ;	0~ 65535 (RW/S)	1024
0x0110	Current loop Ki gain	Current loop Ki gain ;	0~ 65535 (RW/S)	28
0x0111	Current loop Kc gain	Current loop Kc gain ;	0~ 65535 (RW/S)	1228
0x0112	LA speed Kp1	LA speed Kp1 ;	0~65535 (RW/S)	10
0x0113	LA Speed Kv1	LA speed Kv1 ;	0~ 65535 (RW/S)	32
0x0114	LA Speed Node 1	LA speed node 1 ;	0~ 65535 (RW/S)	320
0x0115	LA speed Kp2	LA speed Kp2 ;	0~65535 (RW/S)	15
0x0116	LA Speed Kv2	LA speed Kv2 ;	0~65535 (RW/S)	33
0x0117	LA Speed Node 2	LA speed node 2 ;	0~ 65535 (RW/S)	320
0x0118	LA speed feedforward Kvf gain	LA speed feedforward Kvf gain ;	0~65535 (RW/S)	20
0x0119	LA position loop Ki gain	LA position loop Ki gain ;	0~65535 (RW/S)	35
0x0122	Collision return to zero threshold	Position error value after collision;	200~4000 (RW/S)	2 00
0x0123	Collision return to zero current percentage	The ratio of the current magnitude when the collision returns to zero to that during normal motion;	2 0~100 (RW /S)	50

Appendix 3: Modbus register parameter table – DM-PR series

Register Address	project	illustrate	Setting range Note: Other values are invalid.	default value
Status parameter group (read-only)				
0x0000	Driver Model	Drive model code	(RO)	0x0305
0x0001	Driver version	Driver version	(RO)	0x0100
0x0002	Drive Node Number	MODBUS current communication slave node number	(RO)	-
0x0003	DIP status code	Bit0~Bit6: SW1~SW7 status; 0: OFF; 1:ON;	(RO)	-
0x0006	Current error code	0 : Normal; 1~5 : Error;	(RO)	-
0x0007	Motion status bit	Bit0: In place flag; 0: not in place, 1: in place; Bit1: Return to origin completed bit; 0: not completed, 1: completed; Bit2: Motor running position; 0: stationary, 1: running; Bit3: alarm bit; 0: normal, 1: alarm; Bit4: Motor enable bit; 0: enable, 1: release; Bit5: Forward soft limit overtravel flag ; 0: invalid; 1: valid; Bit6: Negative soft limit overtravel flag ; 0: invalid; 1: valid;	(RO)	-
0x0008	Input terminal status flag	Bit 0 : X0 terminal input status; Bit 1 : X1 terminal input status; Bit 2 : X2 terminal input status; Bit 3 : X3 terminal input status; Bit 4 : X4 terminal input status; Bit 5 : X5 terminal input status;	(RO)	-

		Bit 6 : X6 terminal input status; Bit 7 ~Bit15 : Reserved; 0 : Input level is invalid; 1 : Input level is valid;		
0x0009	Output terminal status flag	Bit0: Y0 terminal output status; Bit1: Y1 terminal output status; Bit 2 : Y 2 terminal output status; Bit 2 ~Bit15: reserved; 0: output level is invalid; 1: Output level is valid;	(RO)	-
0x000A	Current position high	The current position value of the motor. For an open-loop motor, this register value is the position given value. For a closed-loop motor, this register value is the subdivision equivalent of the encoder feedback value.	(RO)	-
0x000B	Current position low		(RO)	-
0x000C	Current speed	Current motor running speed	(RO)	-
Driver basic control parameter group 1				
0x0010	Default movement direction	0 : default direction ; 1 : Reverse direction ;	0~ 1 (RW)	0
0x0011	Segment settings	Address - Segment 400~51200;	400~51200 (RW)	1000
0x0013	Custom drive node number	0 ~ 255 : custom slave address; This register takes effect when all the drive address DIP switches are OFF;	0~ 255 (RW)	0
0x0014	Customize communication baud rate	0:115200 1 : 38400 2 : 19200 3 : 9600 Note: After modification, power must be turned on again to take effect;	0~ 3 (RW)	0
0x0015	Serial port data format	0 : 8 -bit data, no parity, 1 stop bit; 1 : 8 -bit data, no parity, 2 stop bits; 2 : 8 -bit data, even parity, 1 stop bit; 3 : 8 -bit data, odd parity, 1 stop	0~ 3 (RW)	0

		bit; Note: After modification, power must be turned on again to take effect;		
0x0017	Over-limit parking method	0 : free parking; 1 : Emergency stop;	0~ 1 (RW)	0
0x0018	Internal software limit switches	0: invalid; 1: Take effect after returning to zero;	0~ 1 (RW)	0
0x0019	32-bit register endianness configuration	0: high position first, low position last; 1: High position at the back, low position at the front; Adapt to different PLC or touch screen usage habits;	0~ 1 (RW)	0
motion control parameters of the drive				
0x001D	Jog mode operation speed	Jog mode running speed, unit: r/min;	-3000 -300 0 r/min (RW)	120 (5r/min)
0x001E	Jog mode acceleration time	Jog mode acceleration time, in ms;	0-2000ms (RW)	50 (100ms)
0x001F	Jog mode deceleration time	Jog mode deceleration time, unit: ms;	0-2000ms (RW)	50 (100ms)
0x0020	Starting speed	The starting speed of the positioning movement , in r/min ;	0 ~3000r/min (RW)	30 (60r/min)
0x0021	Acceleration time	Acceleration time , in ms;	0-2000ms (RW)	50 (100ms)
0x0022	Deceleration time	Deceleration time , in ms;	0-2000ms (RW)	50 (100ms)
0x0023	Positioning movement speed	Movement speed during positioning movement , unit: r/min ;	0 ~3000r/min (RW)	60 (60r/min)
0x0024	Total pulse count high	The total number of pulses for positioning motion operation is a 32-bit register. If 100000 pulses are set, the high bit is 0x0001 , the low bit is 0x86A0 , and the value written to the register is 0x000186A0;	- 0xFFFFFFFF ~ 0xFFFFFFFF (RW)	5000
0x0025	Total pulse count low			
0x0027	Movement Control Order	Bit0: Position mode start command bit; 0; invalid;	0~6 5535 (WO)	-

		1: Valid; Bit1: Speed mode start command bit; 0: invalid; 1: Valid; Bit2: Position mode positioning method; 0: relative positioning; 1: absolute positioning; Bit3: Sports mode switching method; 0: Ignore motion commands; 1: interrupt the current motion and execute it immediately; Bit4: Return to origin start command bit; 0: invalid; 1: Valid; Bit8: stop command bit; 0: invalid; 1: Valid; Bit9: Emergency stop command bit; 0: invalid; 1: Valid;		
0x002D	Auxiliary control instructions	0x0000: invalid; 0x0011: Motor release; 0x0012: Motor enable; 0x0021: drive alarm cleared; 0x0031: Clear the current position of the motor; 0x0041: Restore parameters to factory settings; 0x0042: save all parameters; Note: When performing factory recovery and parameter saving operations on the drive, it is necessary to ensure that the motor is in the stopped state,	0~ 65535 (WO)	-

		otherwise the relevant instructions will be ignored;		
Drive return to zero motion control parameters				
0x0030	Zero return auxiliary setting	0: Run the offset, and after completion, the current position is the offset value; 1: Run the offset, and the current position is 0 after completion; 2 : Run the offset, and after completion the current position is a negative offset value; 3: Run the offset, and after completion, the current position is the actual value; 4: Run the offset. After completion, the current position is the actual value plus the offset value. 5: Run the offset. After completion, the current position is the actual value minus the offset value. 6: Do not run the offset, and the current position after completion is the offset value; 7 : Do not run the offset, the current position is 0 after completion; 8: Do not run the offset, and the current position will be a negative offset value after completion; 9: Do not run the offset, and the current position is the actual value after completion; 1 0 : Do not run the offset. After completion, the current position is the actual value plus the offset value. 1 1 : Do not run the offset. After completion, the current position is the actual value minus the offset value.	0 ~11 (RW)	0

0x0031	Return to origin mode	Support 17~30, 35 modes;	0~ 65535 (RW)	twenty four
0x0032	Return to origin speed	Running speed when querying the origin position;	5-3000r/min (RW)	120 (60r/min)
0x0033	Return to origin query speed	The return speed after querying the origin;	5-300 r/min (RW)	60 (60r/min)
0x0034	Acceleration and deceleration time when returning to origin	The acceleration and deceleration time when querying the origin position;	30-2000ms (RW)	50 (100ms)
0x0035	Origin offset value high	Origin offset value: after finding the origin sensor, it moves to the correct position according to the value set in this register .	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0036	Origin offset value low			
0x0037	Soft limit positive high position	Software positive limit setting point, this limit point will take effect only after zero return is completed ;	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0038	Soft limit positive low position			
0x0039	Soft limit negative high position	Software negative limit setting point, this limit point will take effect only after zero return is completed ;	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x003A	Soft limit negative low position			
Input and output terminal parameter group				
0x0040	Input terminal effective level	Bit 0 : Input terminal X0 control bit; Bit 1 : Input terminal X1 control bit; Bit 2 : Input terminal X2 control bit; Bit 3 : Input terminal X3 control bit; Bit 4 : Input terminal X 4 control bit; Bit 5 : Input terminal X 5 control bit; Bit 6 : Input terminal X 6 control bit; Bit 7 ~Bit15 : Reserved; 0 : default; 1 : Level inversion;	0~65535 (RW)	0

0x0041	Input terminal X0 terminal function selection	0 : undefined; 1 : origin signal; 2 : Positive limit signal; 3 : Anti-limit signal; 4 : Motor MF signal; 5 : Stop signal; 6 : Emergency stop signal; 7 : Position mode movement; 8: Speed mode movement; 9 : JOG+ point motion; 10 : JOG -point movement; 11 : Return to origin enable signal; 12 : PT trigger signal ; 13: PV trigger signal; 14 : PIN0 ; 15 : PIN1 ; 16 : PIN2 ; 17 : PIN3 ; 18 : PV direction signal;	0~17 (RW)	1
0x0042	Input terminal X1 terminal function selection		0~17 (RW)	2
0x0043	Input terminal X2 terminal function selection		0~17 (RW)	3
0x0044	Input terminal X3 terminal function selection		0~17 (RW)	0
0x0045	Input terminal X 4 terminal function selection		0~17 (RW)	0
0x0046	Input terminal X 5 terminal function selection		0~17 (RW)	0
0x0047	Input terminal X 6 terminal function selection		0~18 (RW)	0
0x004B	Output terminal effective level	Bit0 : Output terminal Y0 control bit; Bit1 : Output terminal Y1 control bit; Bit 2 : Output terminal Y 2 control bit; Bit 2 ~Bit15 : Reserved; 0 : default; 1 : Level inversion;	0~65535 (RW)	0
0x004C	Output terminal Y0 terminal function selection	0: undefined ; 1: Alarm signal; 2: Driver status signal; 3: Return to origin completion signal; 4: Arrival signal; 5: Braking signal; 9: User defined 0; 10: User defined 1; 11: User defined 2;	0~11 (RW)	0
0x004D	Output terminal Y1 terminal function selection		0~11 (RW)	0
0x004E	Output terminal Y2 terminal function selection		0~11 (RW)	0
0x004F	Y group terminal custom output	Bit0 : Y0 terminal output status; Bit1 : Y1 terminal output status; Bit 2 ~Bit15 : Reserved;	0~65535 (RW)	0

		0 : Output is invalid; 1 : Output is valid;		
Multi-stage positioning/speed control parameter group				
0x0050	Relative position / absolute position	When using IO to control positioning motion , this bit is valid: 0 : relative position; 1 : absolute position;	0~1 (RW)	0
0x0051	PV trigger signal level selection	0: level is valid; 1: rising edge is valid;	0~1 (RW)	0
0x0060	The first segment motion pulse command high	The first section of motion pulse instruction is a 32-bit register. If 100000 pulses are set, the high bit is 0x0001 , the low bit is 0x86A0 , and the value written into the register is 0x000186A0.	-0xFFFFFFFF ~ 0xFFFFFFFF (RW)	0
0x0061	The first segment motion pulse command low			5000
0x0062	Positioning speed of the first stage	the first position positioning, unit: r/min;	0 -3000 r/min (RW/S)	120 (0/min)
0x0063	1st stage positioning motion acceleration	The first segment position positioning motion acceleration, in ms;	0 -2000ms (RW/S)	50 (0ms)
0x0064	1st stage position positioning motion deceleration	The first stage of the positioning motion deceleration, unit: ms;	0 -2000ms (RW/S)	50 (0ms)
0x0065	1st stage positioning motion mode	When using IO to control multisegment position motion , this bit is valid: 0: relative position; 1: absolute position;	0 ~1 (RW/S)	0
0x0066~0x006B	Parameters of Section 2	Reference 0x0060~0x0065;	-	-
0x006C~0x0071	Parameters of Section 3	Refer to 0x0060~0x0065;	-	-
0x0072~0x0077	Parameters of Section 4	Reference 0x0060~0x0065;	-	-
0x0078~0x007D	Parameters of Section 5	Reference 0x0060~0x0065;	-	-

0x007E~0x0083	Parameters of Section 6	Reference 0x0060~0x0065;	-	-
0x0084~0x0089	Parameters of Section 7	Reference 0x0060~0x0065;	-	-
0x008A~0x008F	Parameters of Section 8	Reference 0x0060~0x0065;	-	-
0x0090~0x0095	Parameters of Section 9	Reference 0x0060~0x0065;	-	-
0x0096~0x009B	Parameters of Section 10	Reference 0x0060~0x0065;	-	-
0x009C~0x00A1	Parameters of Section 11	Reference 0x0060~0x0065;	-	-
0x00A2~0x00A7	Parameters of Section 12	Reference 0x0060~0x0065;	-	-
0x00A8~0x00AD	Parameters of Section 13	Refer to 0x0060~0x0065;	-	-
0x00AE~0x00B3	Parameters of Section 14	Reference 0x0060~0x0065;	-	-
0x00B4~0x00B9	Parameters of Section 15	Reference 0x0060~0x0065;	-	-
0x00BA~0x00BF	Parameters of Section 16	Refer to 0x0060~0x0065;	-	-
0x00C0	1st speed segment running speed	running speed of the first speed segment is in r/min ;	-3000 -3000 r/min (RW/S)	100 (0/min)
0x00C1	1st speed segment acceleration time	The first speed segment motion acceleration, unit: ms;	0 -2000ms (RW/S)	50 (0ms)
0x00C2	1st speed stage deceleration time	The first speed segment is the deceleration of the movement, in ms;	0 -2000ms (RW/S)	50 (0ms)
0x00C3~0x00C5	Parameters of Section 2	Reference 0x00C0 ~0x00C2	-	-
0x00C6~0x00C8	Parameters of Section 3	Reference 0x00C0 ~0x00C2	-	-
0x00C9~0x00CB	Parameters of Section 4	Reference 0x00C0 ~0x00C2	-	-
0x00CC~0x00CE	Parameters of Section 5	Reference 0x00C0 ~0x00C2	-	-

0x00CF~0x00D1	Parameters of Section 6	Reference 0x00C0 ~0x00C2	-	-
0x00D2~0x00D4	Parameters of Section 7	Reference 0x00C0 ~0x00C2	-	-
0x00D5~0x00D7	Parameters of Section 8	Reference 0x00C0 ~0x00C2	-	-
0x00D8~0x00DA	Parameters of Section 9	Reference 0x00C0 ~0x00C2	-	-
0x00DB~0x00DD	Parameters of Section 10	Reference 0x00C0 ~0x00C2	-	-
0x00DE~0x00E0	Parameters of Section 11	Reference 0x00C0 ~0x00C2	-	-
0x00E1~0x00E3	Parameters of Section 12	Reference 0x00C0 ~0x00C2	-	-
0x00E4~0x00E6	Parameters of Section 13	Reference 0x00C0 ~0x00C2	-	-
0x00E7~0x00E9	Parameters of Section 14	Reference 0x00C0 ~0x00C2	-	-
0x00EA~0x00EC	Parameters of Section 15	Reference 0x00C0 ~0x00C2	-	-
0x00ED~0x00EF	Parameters of Section 16	Reference 0x00C0 ~0x00C2	-	-
0x0130~0x013F	Multi-segment motion starting speed	The starting speed of each segment of movement (shared by PT mode and PV mode)	-180~180 rpm (RW /S)	0rpm
Performance parameter group				
0x0102	Maximum effective current	The maximum current value output by the driver. The register value is given according to the rated current of the motor, in mA.	0~ 7700 (RW/S)	5600/7700
0x0103	Closed loop maximum current percentage	The maximum current percentage in closed-loop mode , in % , when the driver is in closed-loop mode, the maximum current that can be output under rated load is the product of the 2042h register value and this register value;	0~1 0 0 (RW/S)	100

0x0104	Base current percentage	Registered current percentage , unit % , when the driver is running at no load, the output current value is the product of the 2042h register value and this register value;	0~ 100 (RW/S)	40
0x0105	Open loop maximum current percentage	The maximum open-loop current percentage, in % , when the driver is in open-loop mode, the maximum output current during operation is the product of the 2042h register value and this register value;	0~100 (RW/S)	100
0x0106	Lock current percentage	Lock current percentage, unit % . When the driver is in the locked state , the lock current value is the product of the 2042h register value and the register value.	0 - 1 00 (RW/S)	100
0x0107	Lock time	for the motor to switch to the lock state after it stops moving , in ms;	0~2000 (RW/S)	200
0x0108	IO terminal filter coefficient	Terminal filter coefficient;	0 ~ 65535 (RW/S)	2
0x0109	Pulse low-pass filter coefficient	Pulse control command filter coefficient; Low-pass filter coefficient, the smaller the value, the more obvious the filtering effect;	0~1024 (RW/S)	5
0x010C	End time	End time of arrival ;	0~200 (RW/S)	10
0x010D	Pulse mean filter coefficient	Mean filter coefficient, the larger the value, the more obvious the filtering effect;	0~ 512 (RW/S)	512
0x010E	Current loop Kp gain multiple	Current loop Kp gain multiple ;	0~ 65535 (RW/S)	4096
0x010F	Current loop Kp gain	Current loop Kp gain ;	0~ 65535 (RW/S)	1024
0x0110	Current loop Ki gain	Current loop Ki gain ;	0~ 65535 (RW/S)	28
0x0111	Current loop Kc gain	Current loop Kc gain ;	0~ 65535 (RW/S)	1228